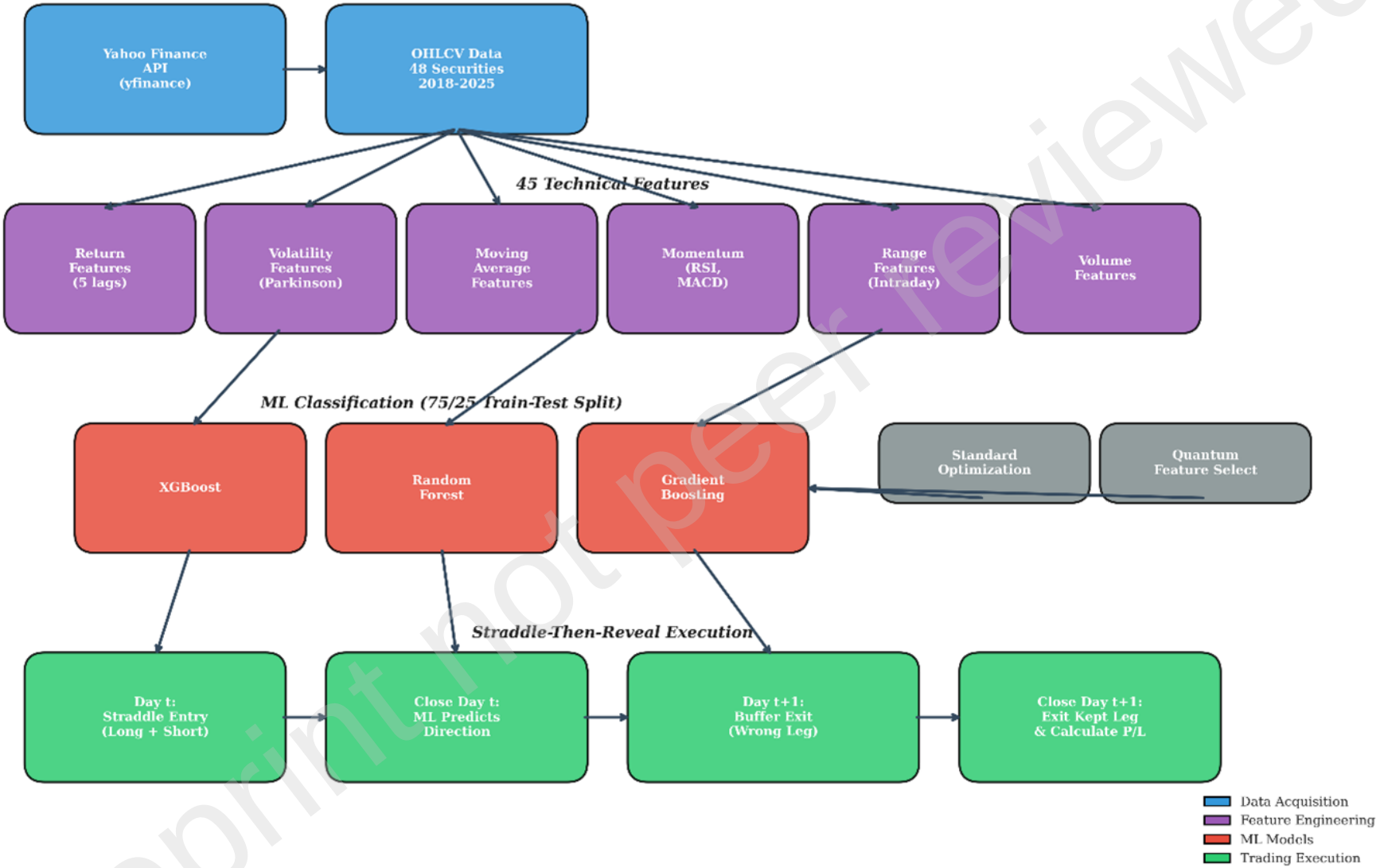
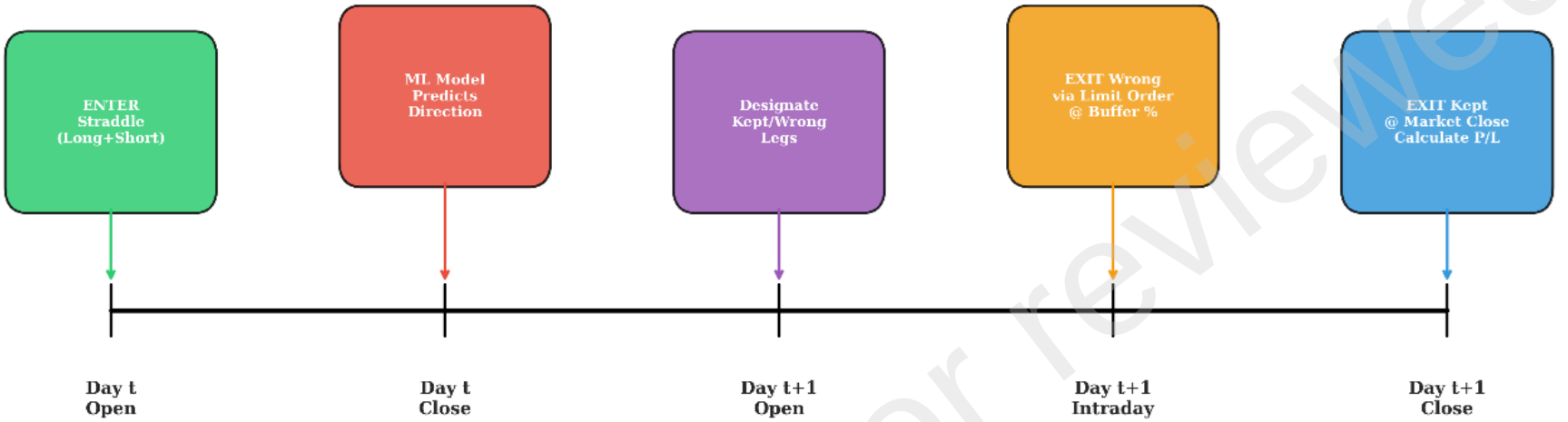
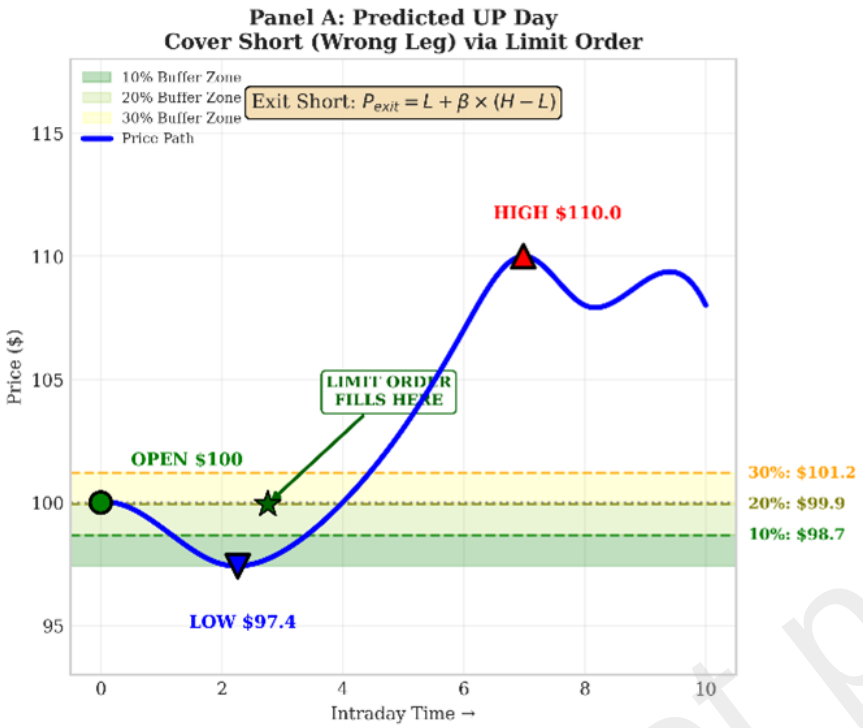
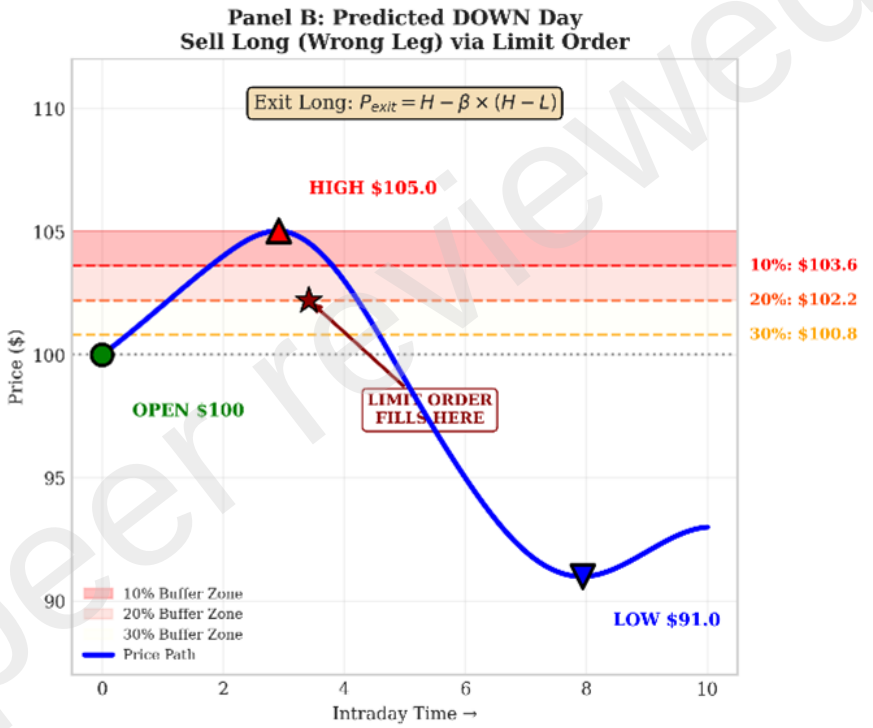


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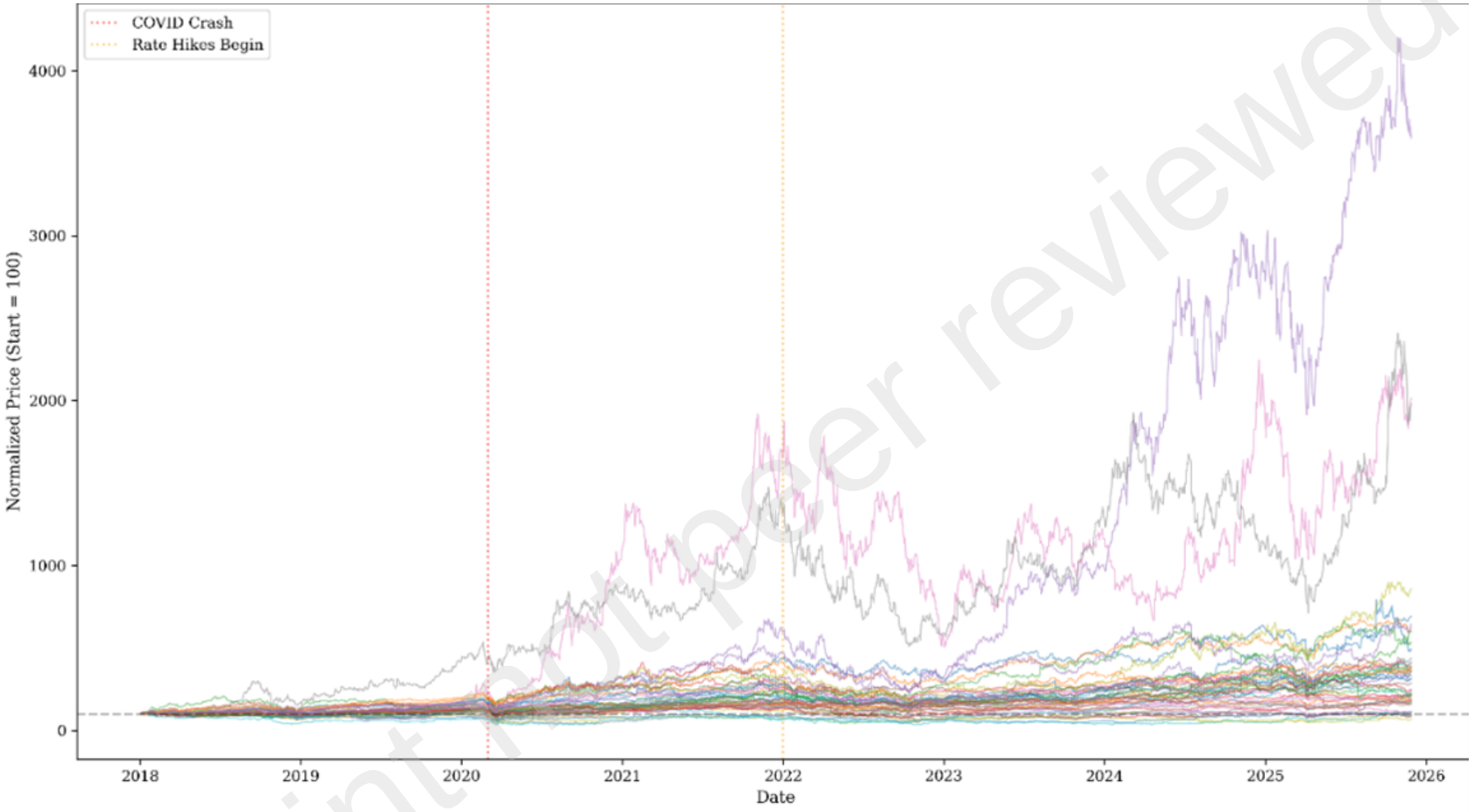


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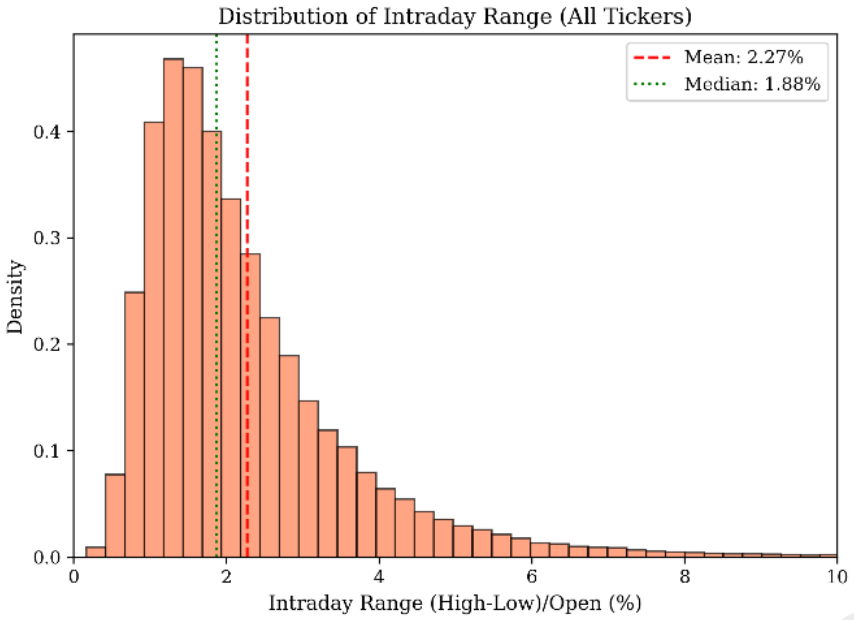


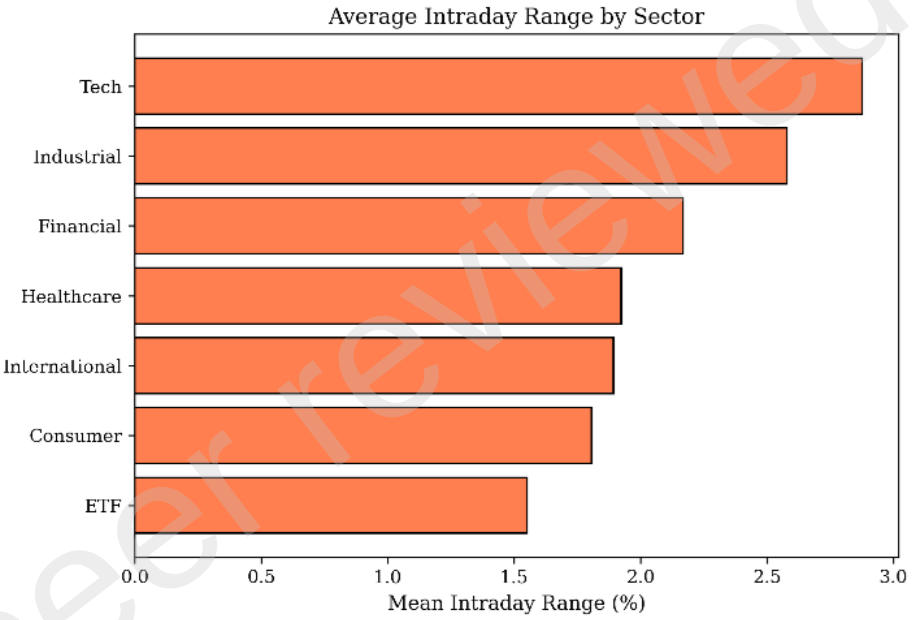


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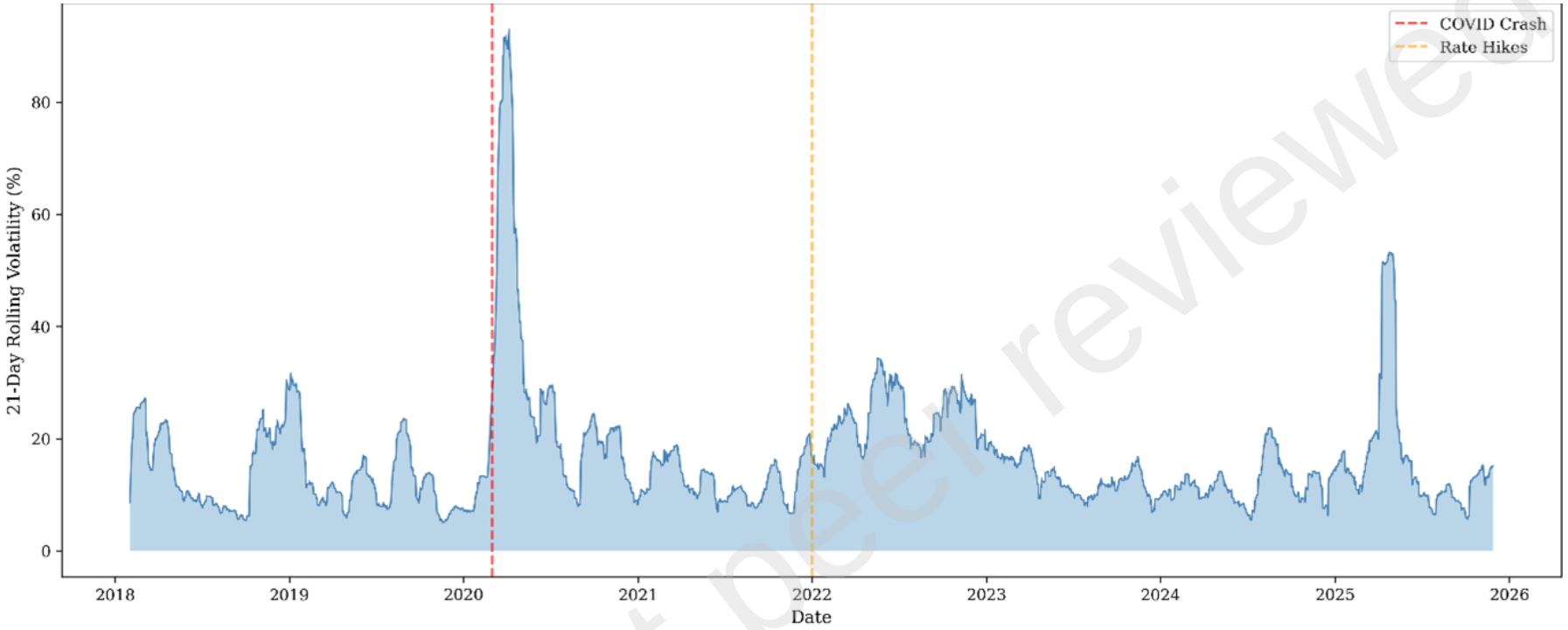


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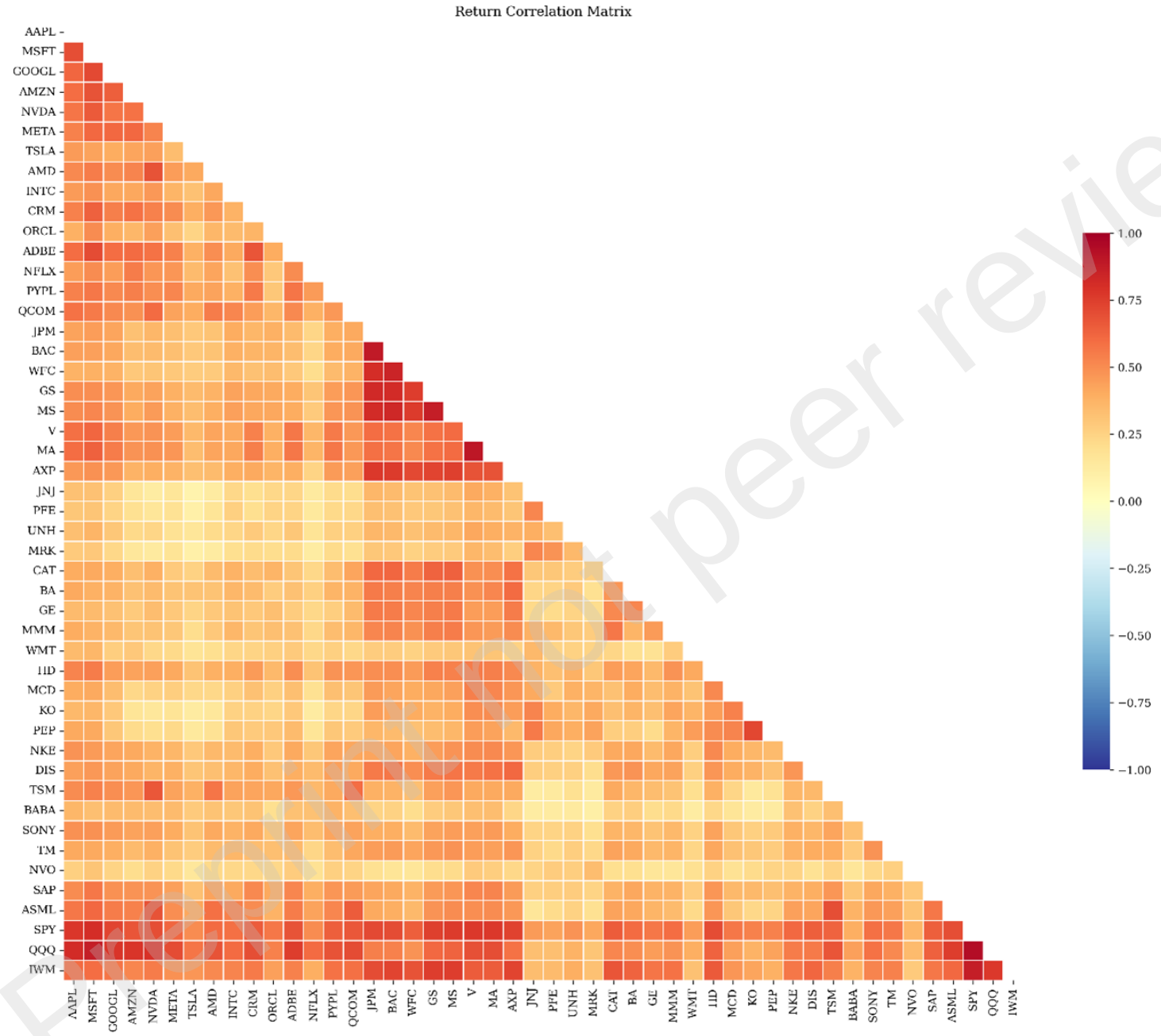




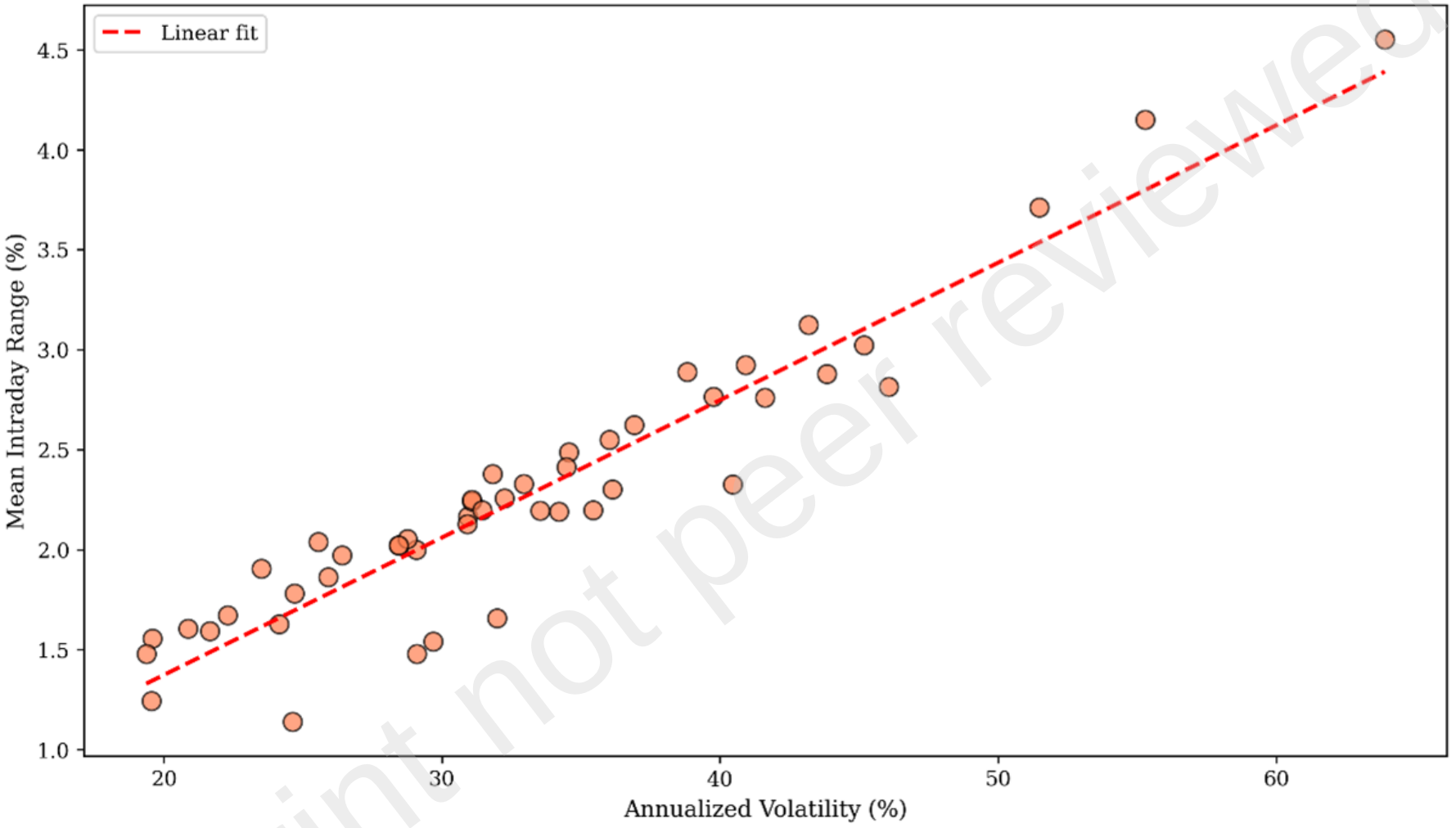
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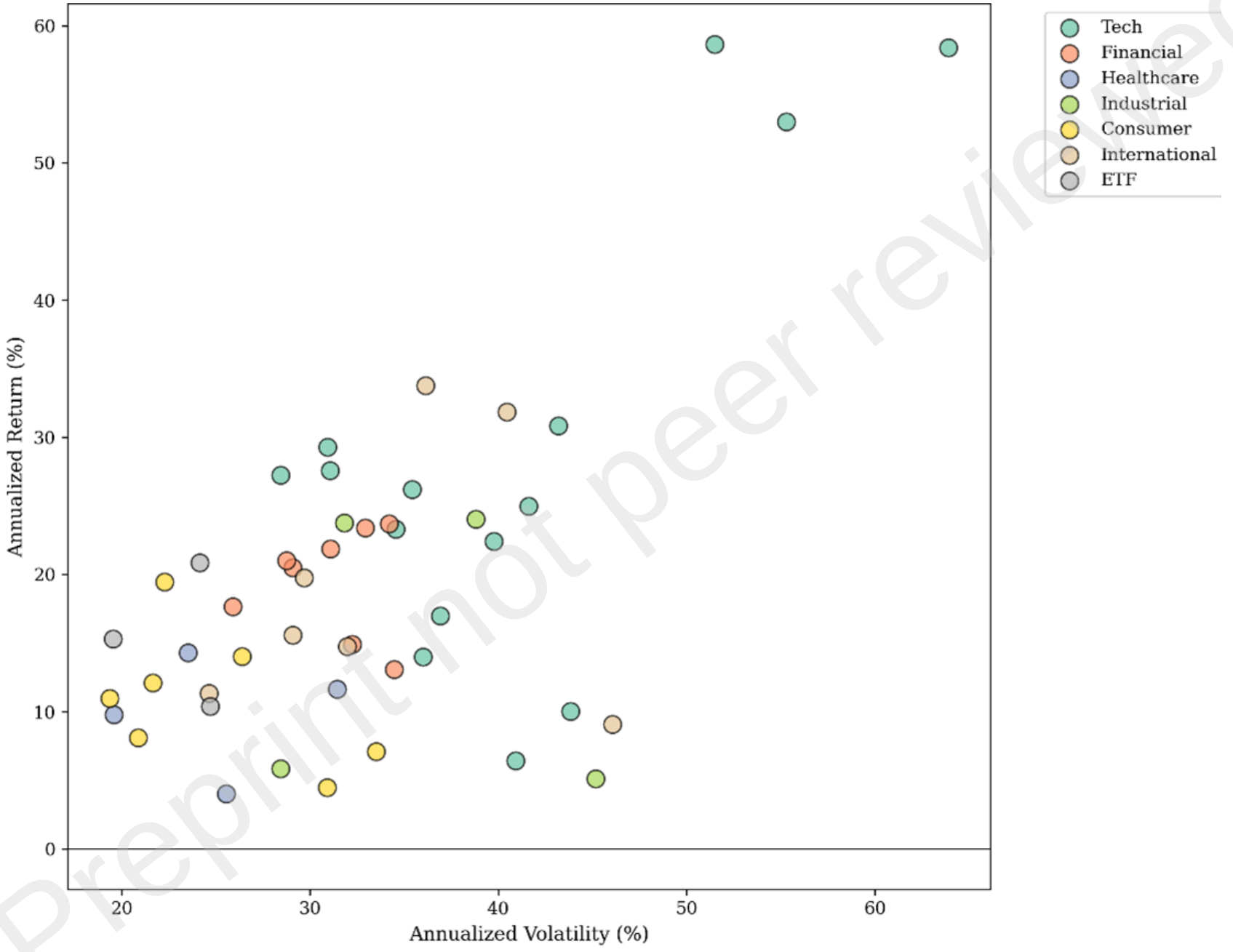
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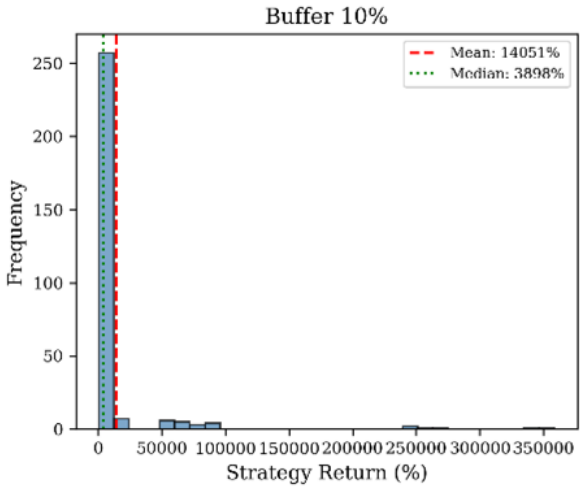
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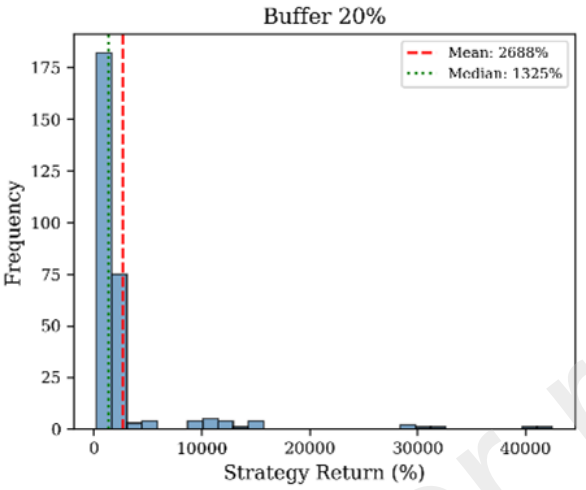


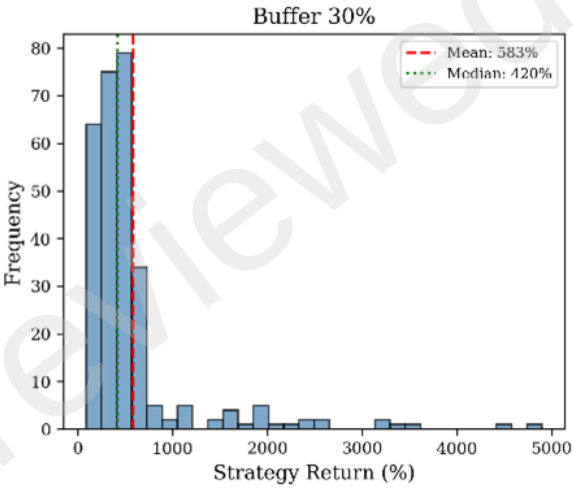
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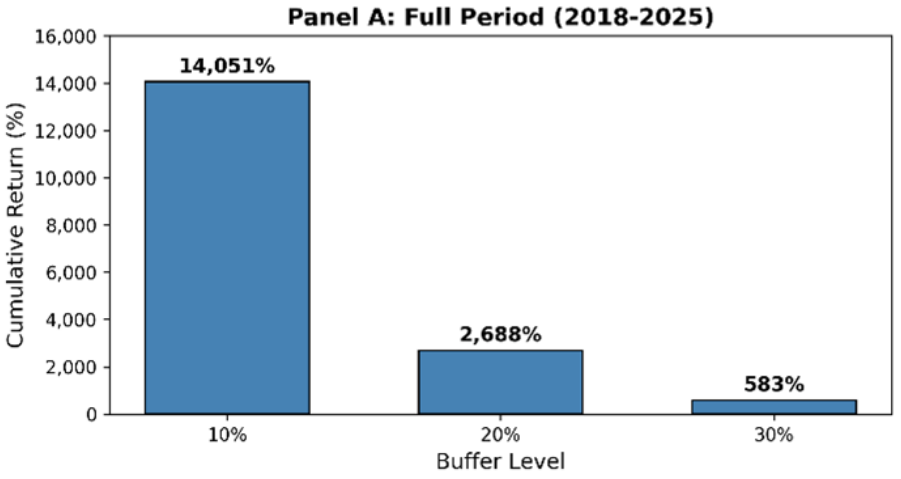
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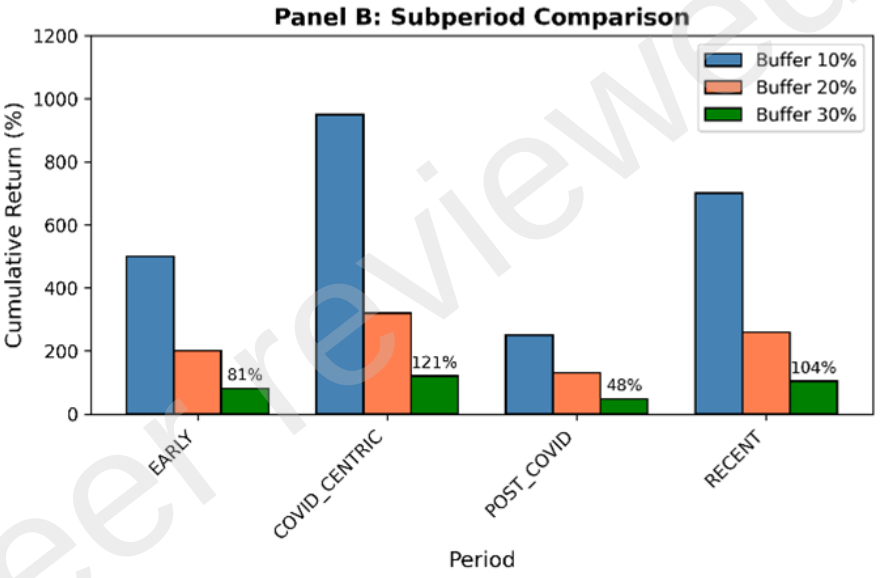




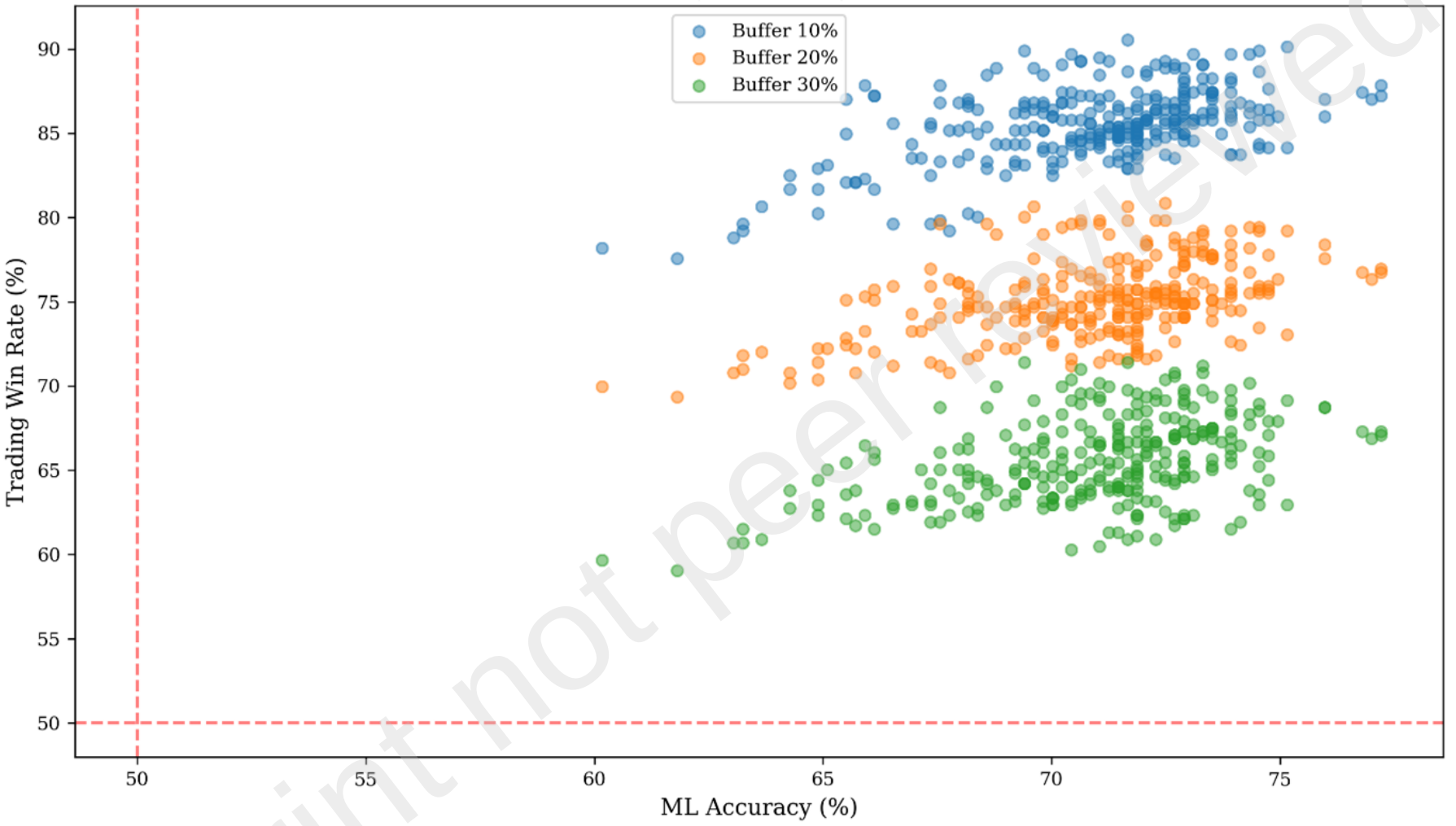


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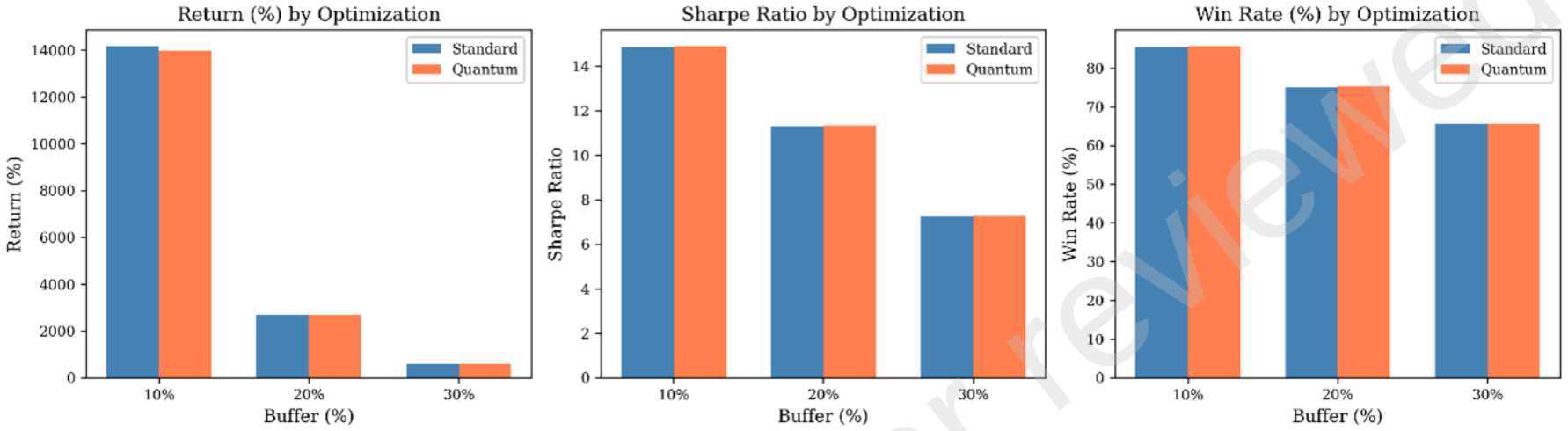




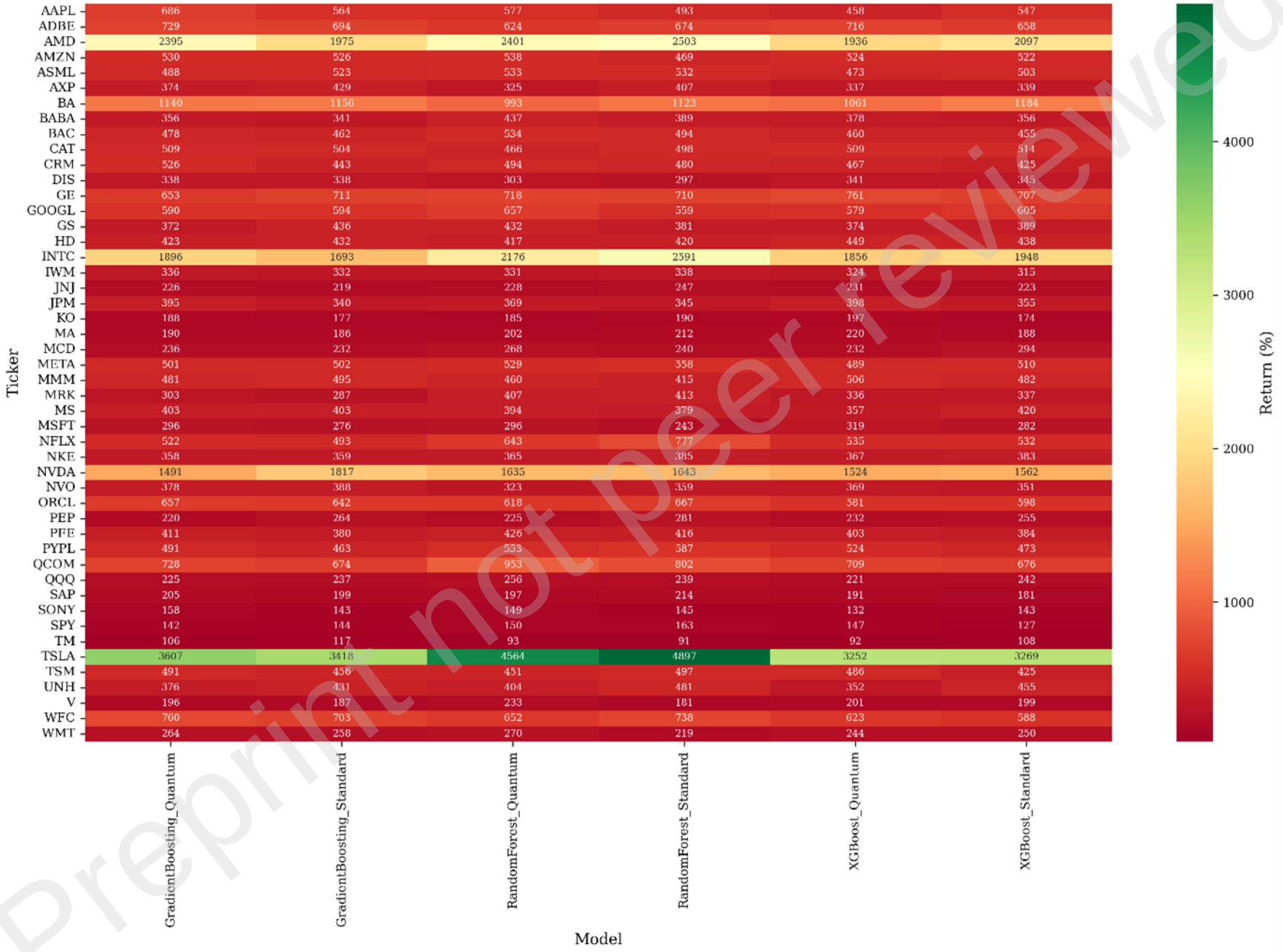
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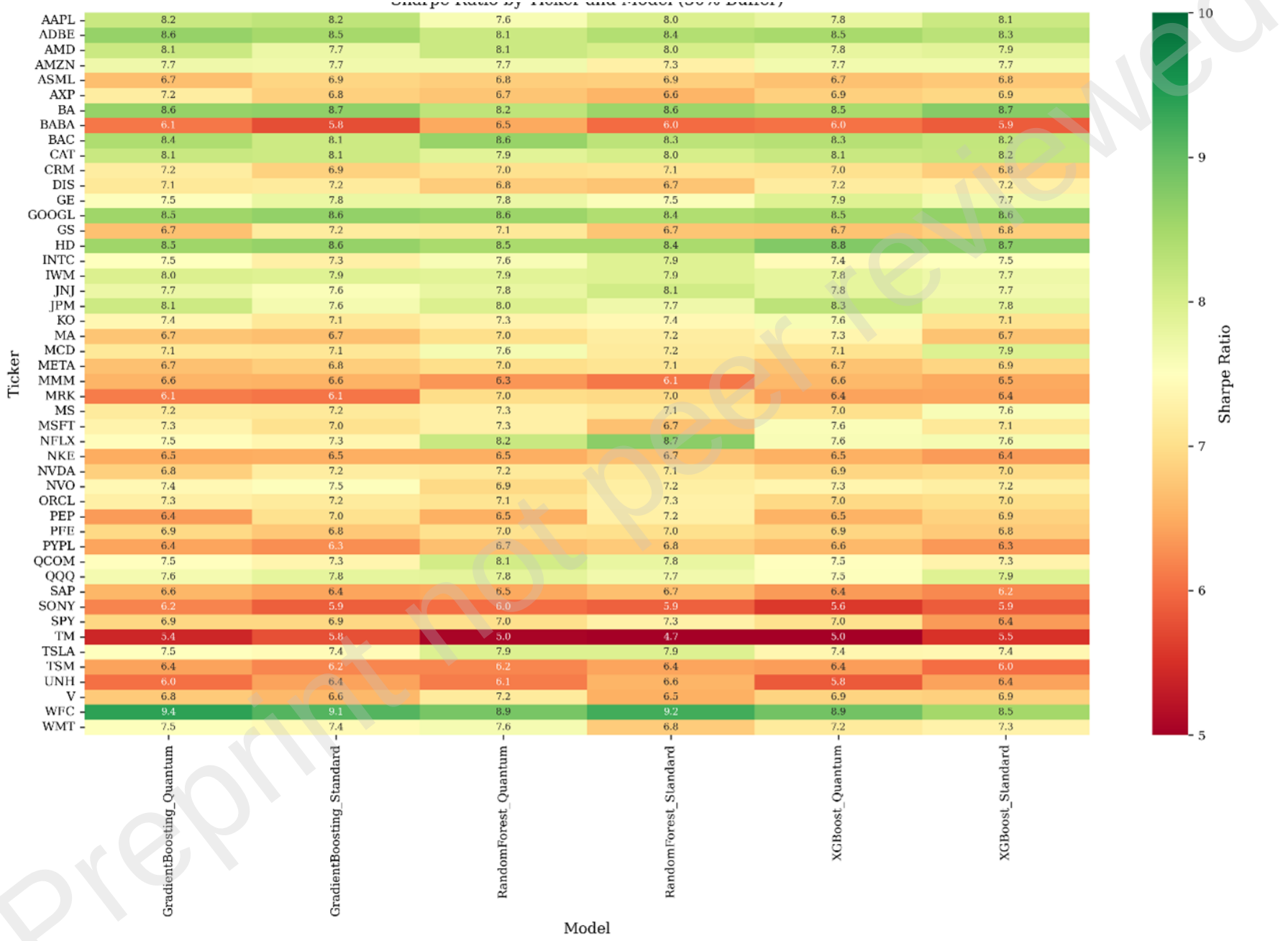
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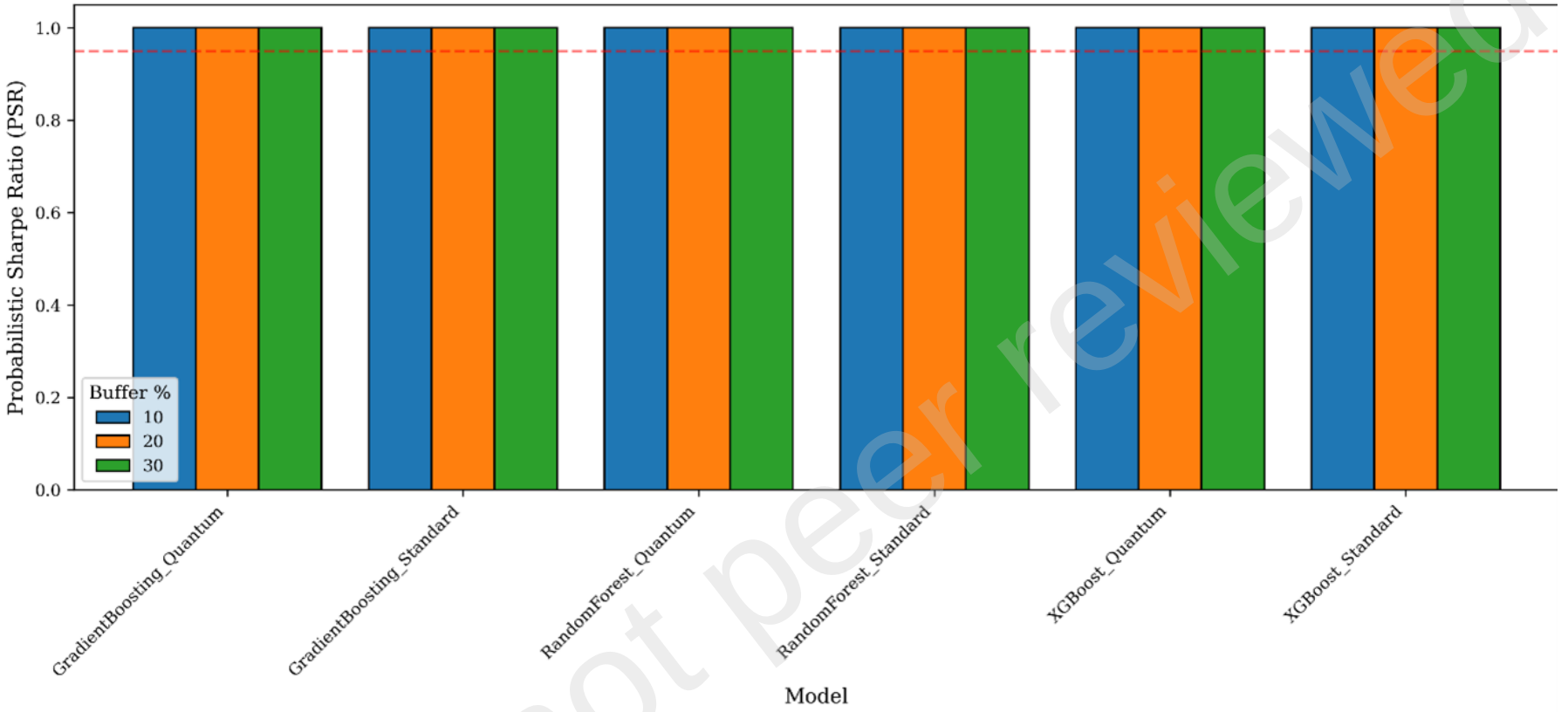
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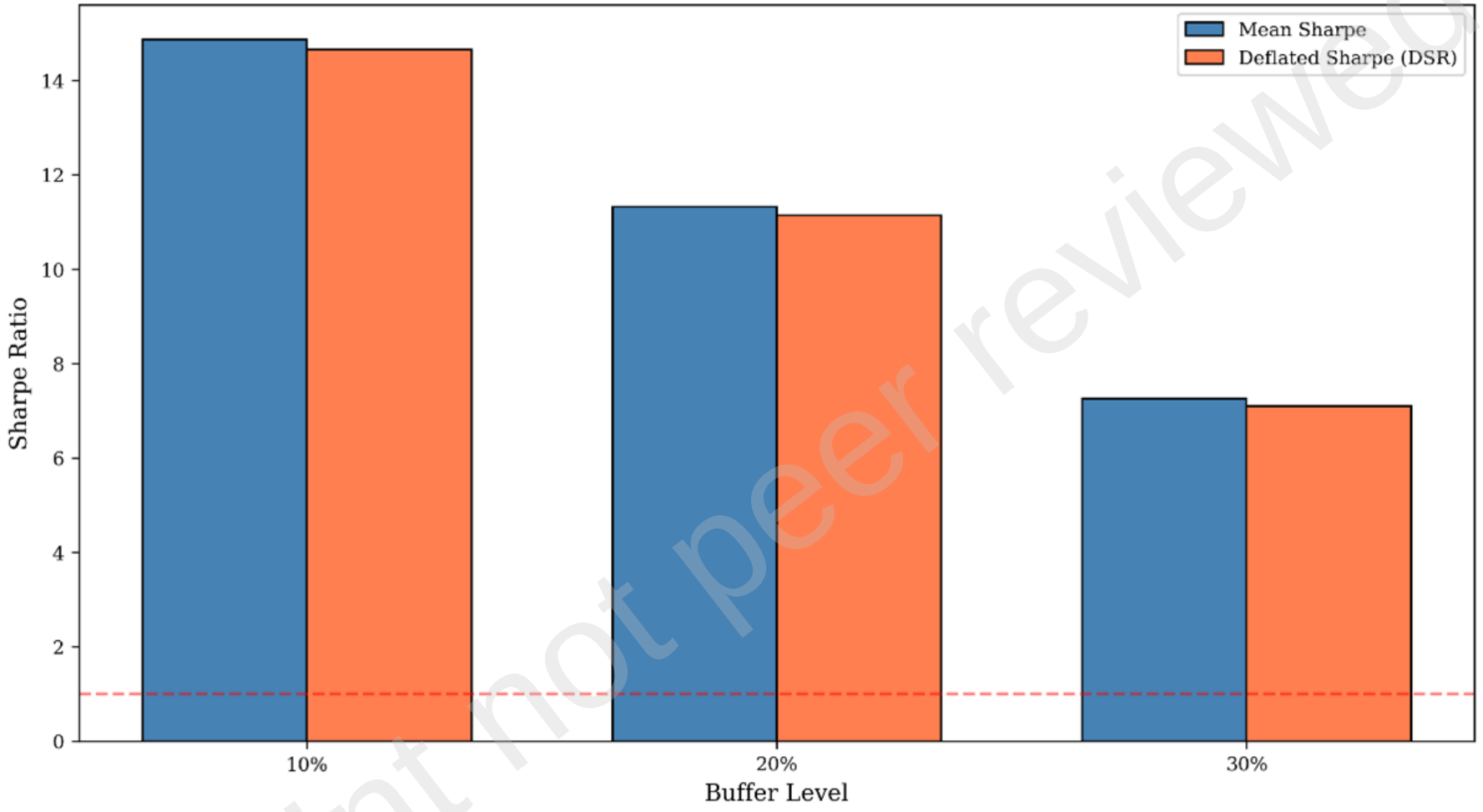
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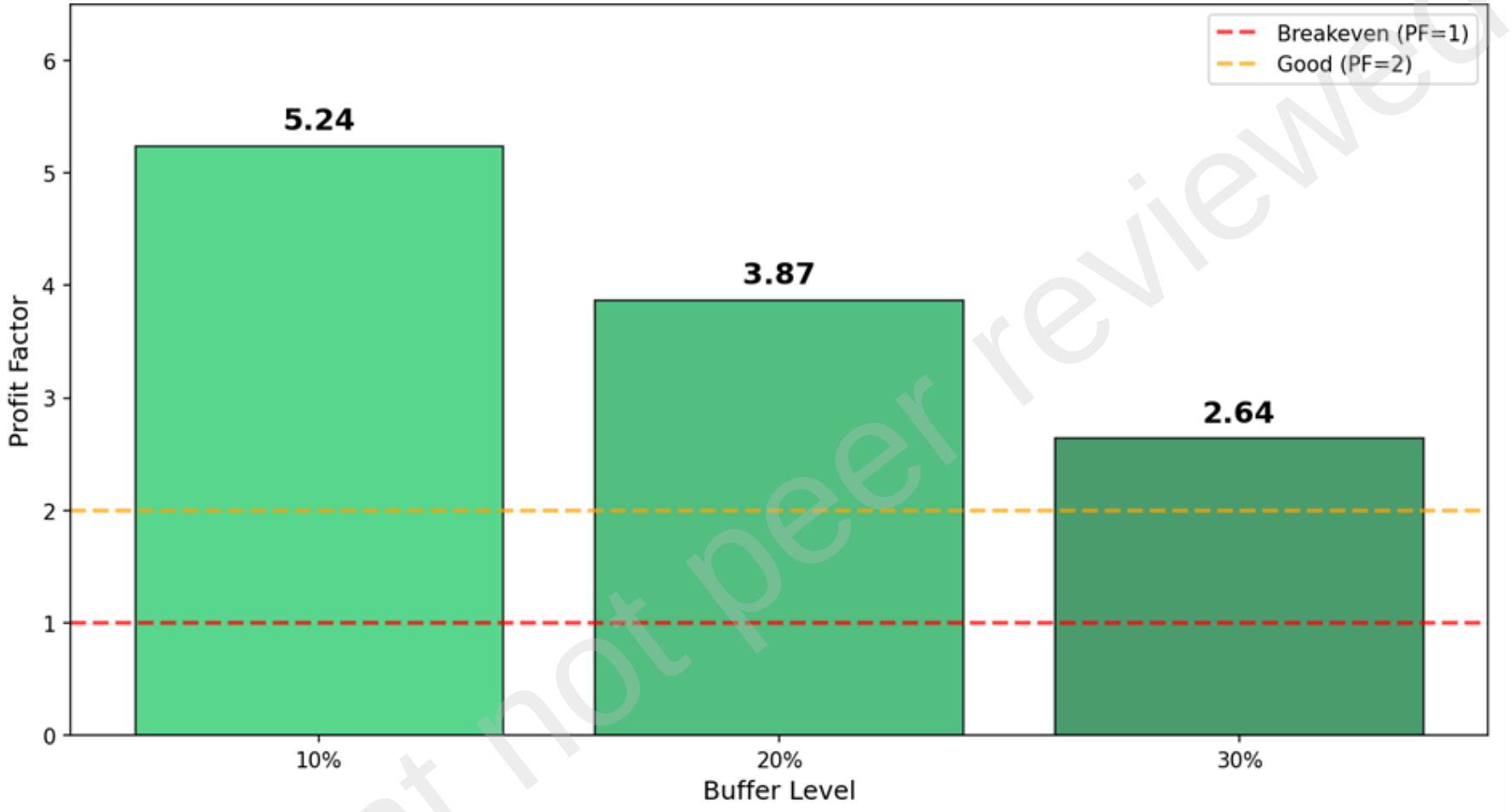
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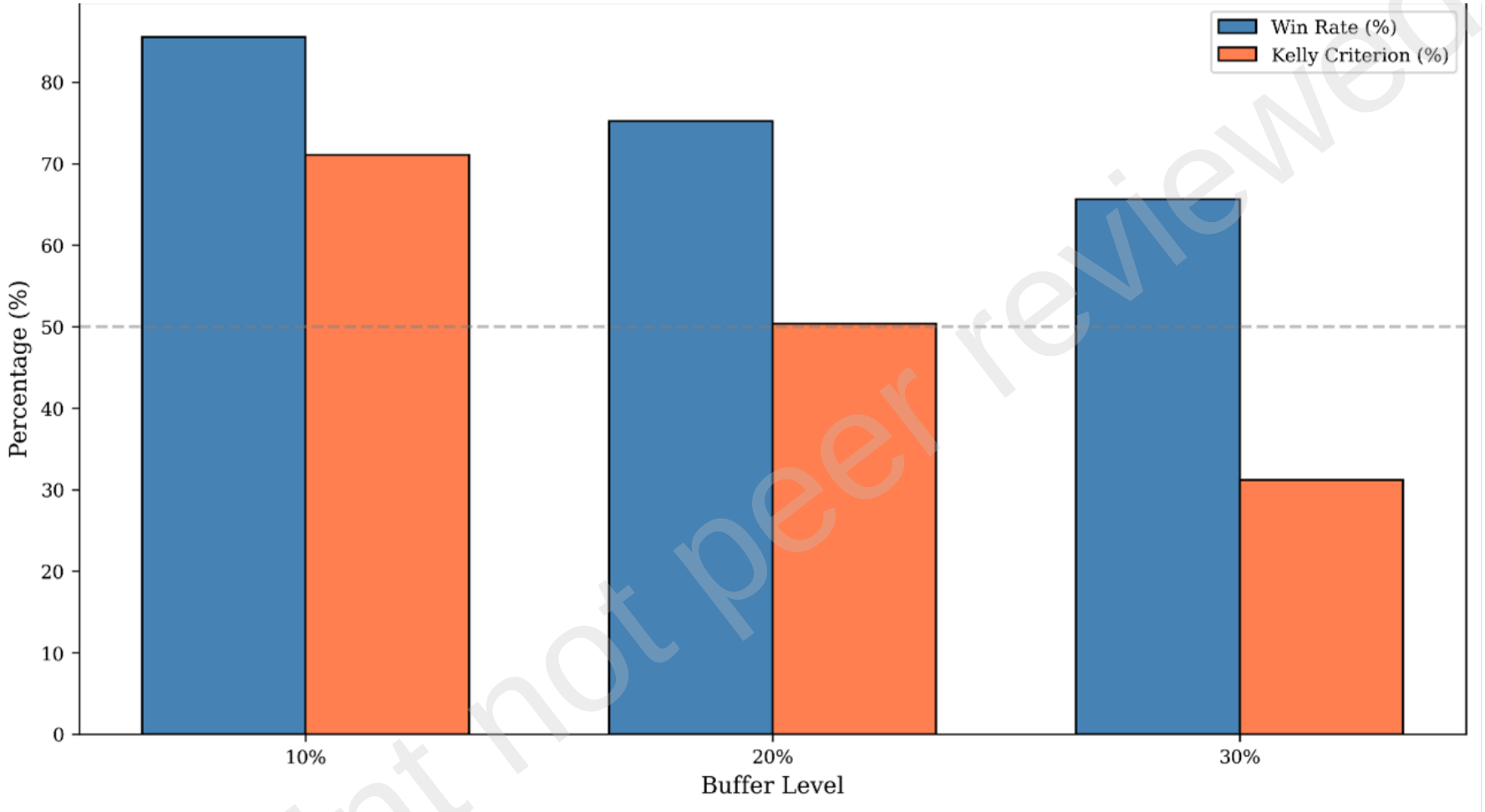
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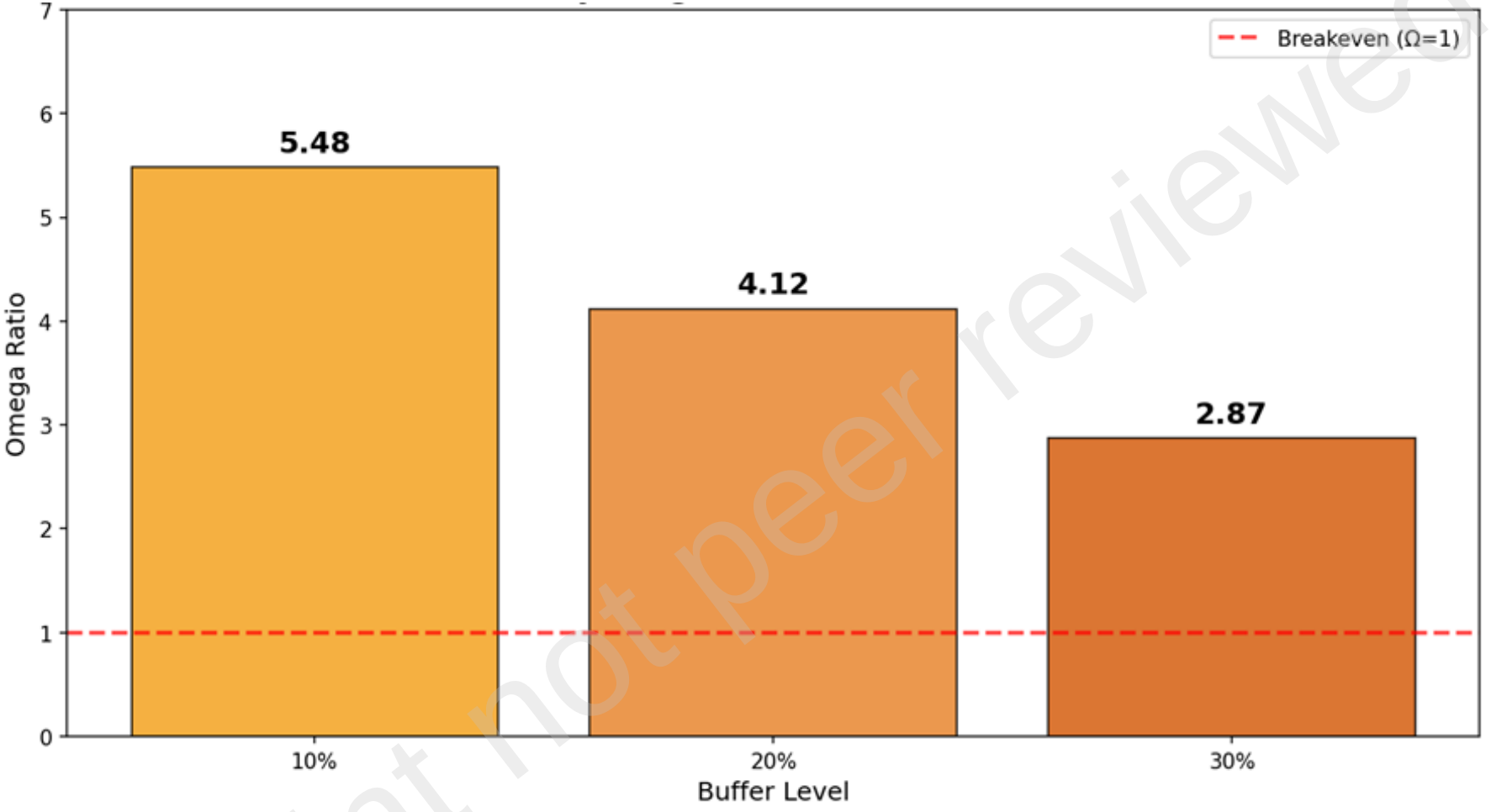
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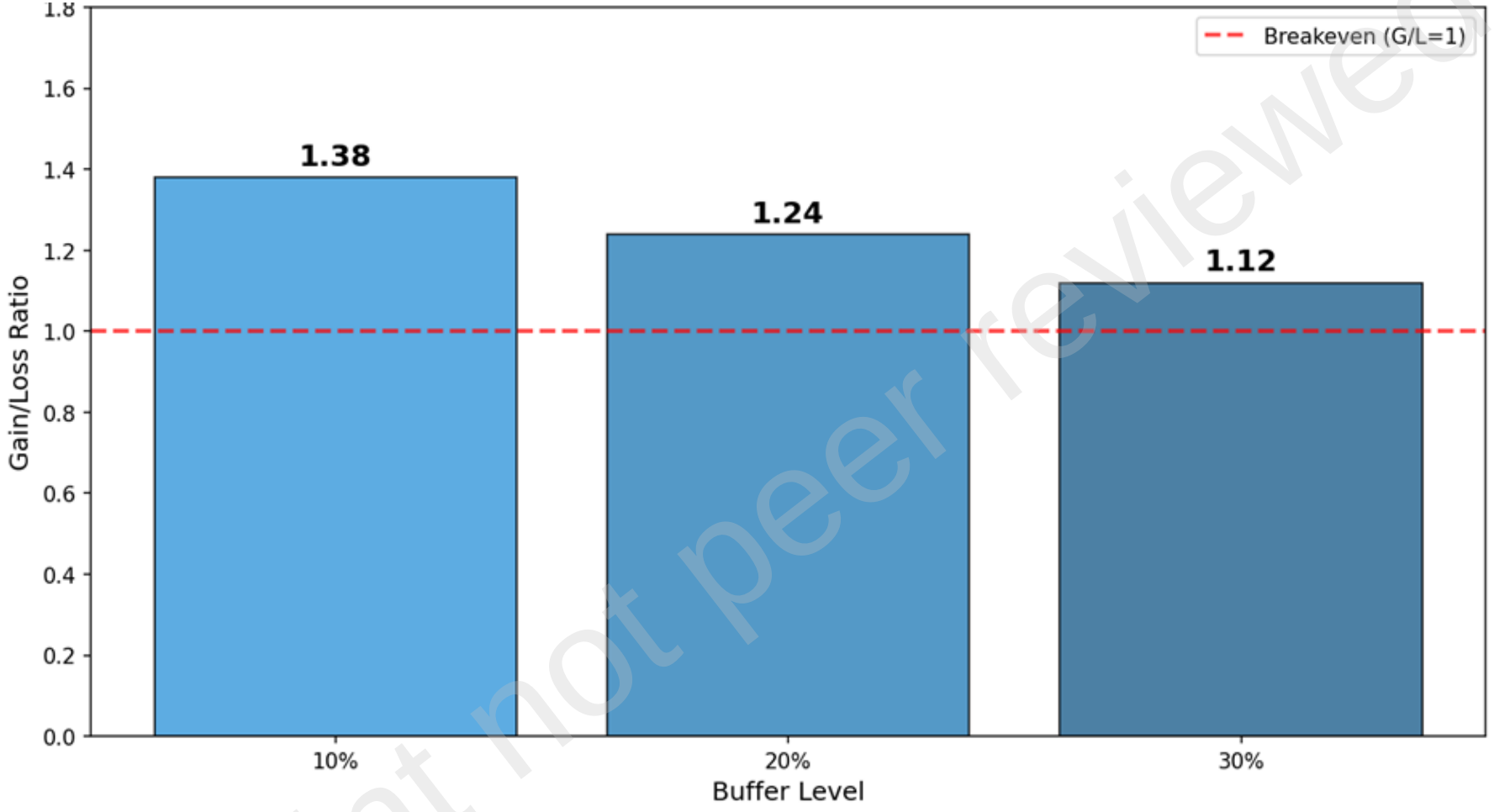
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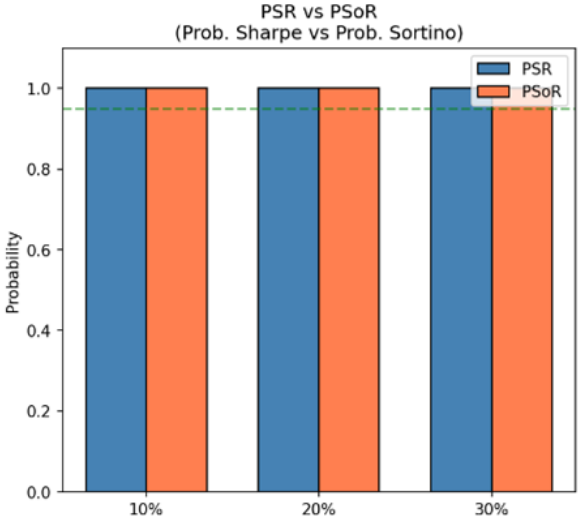
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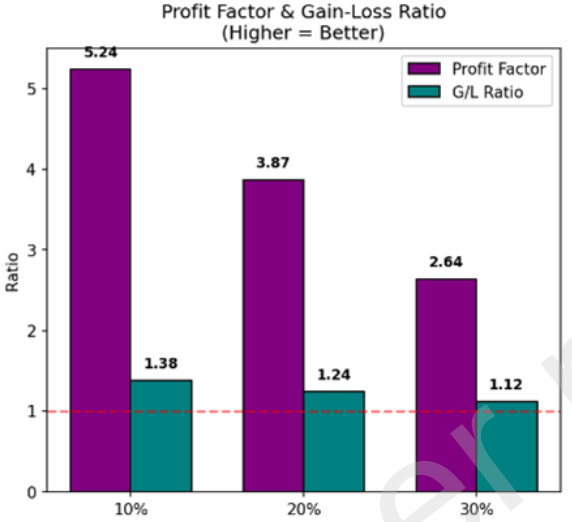


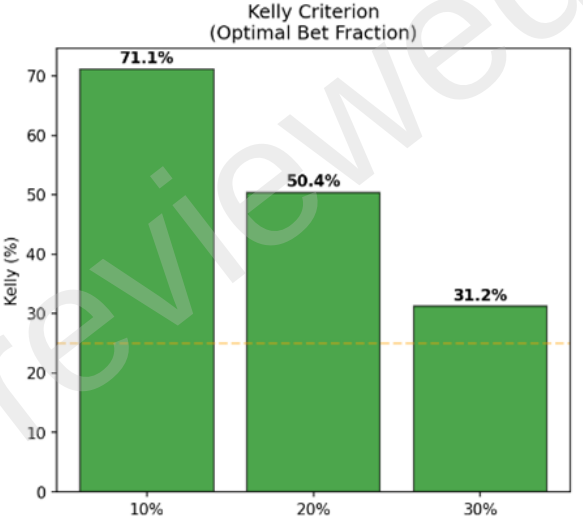
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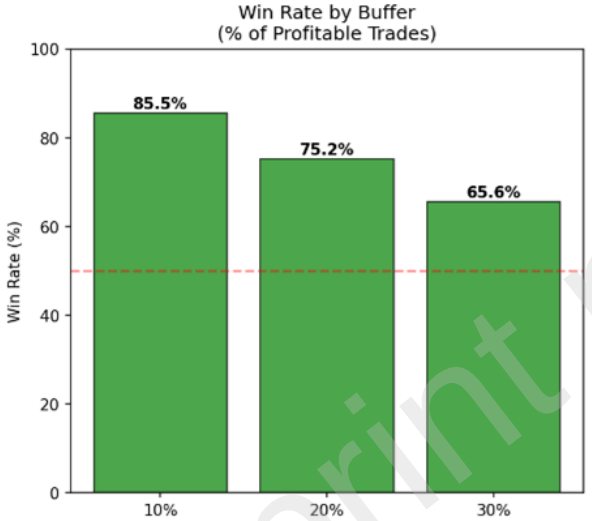


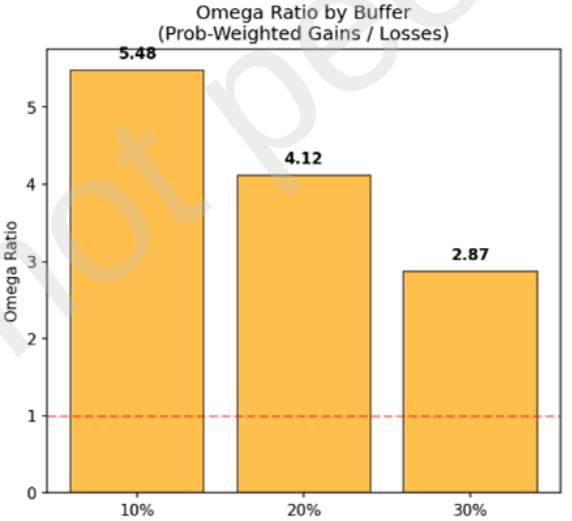
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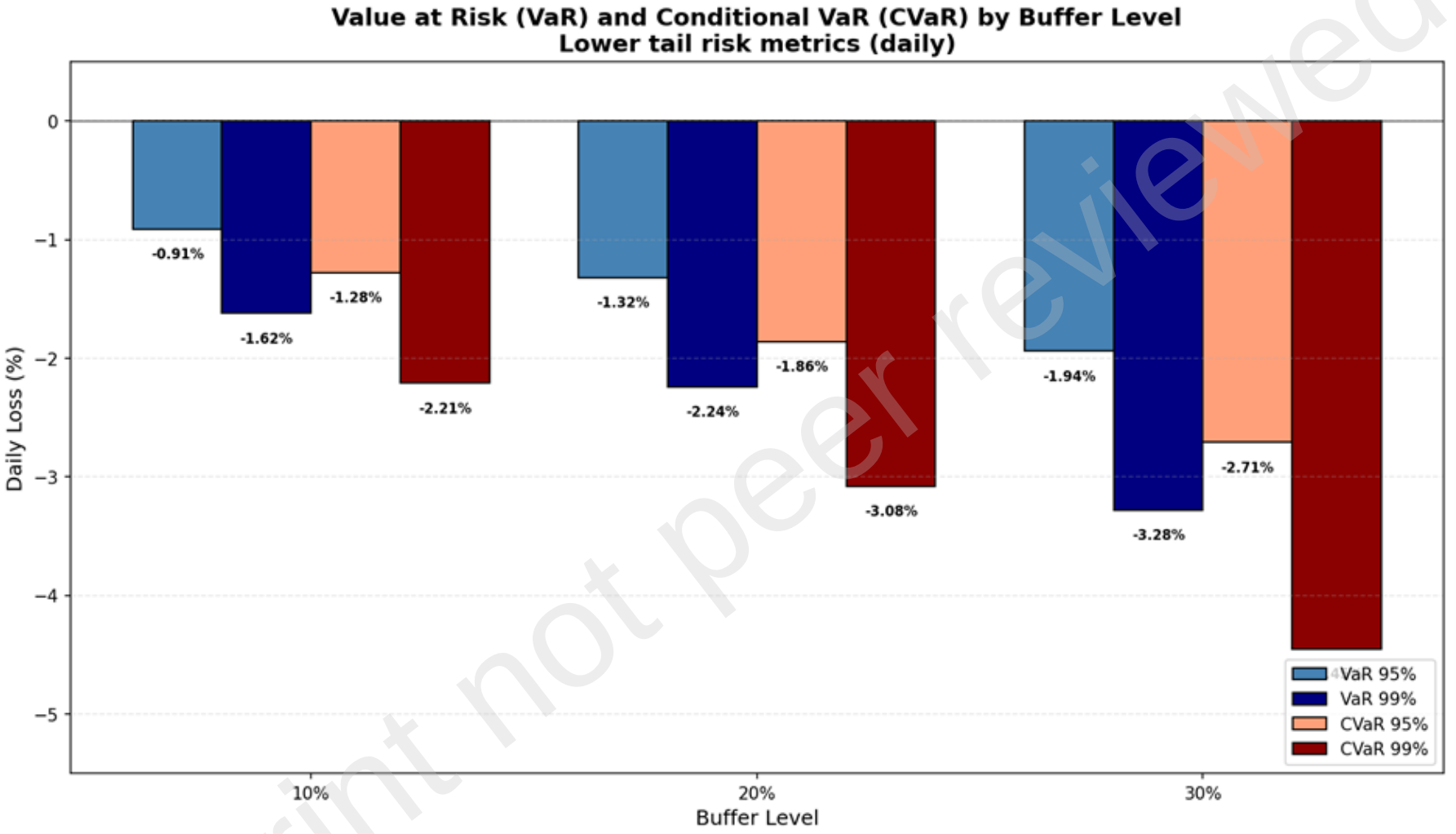




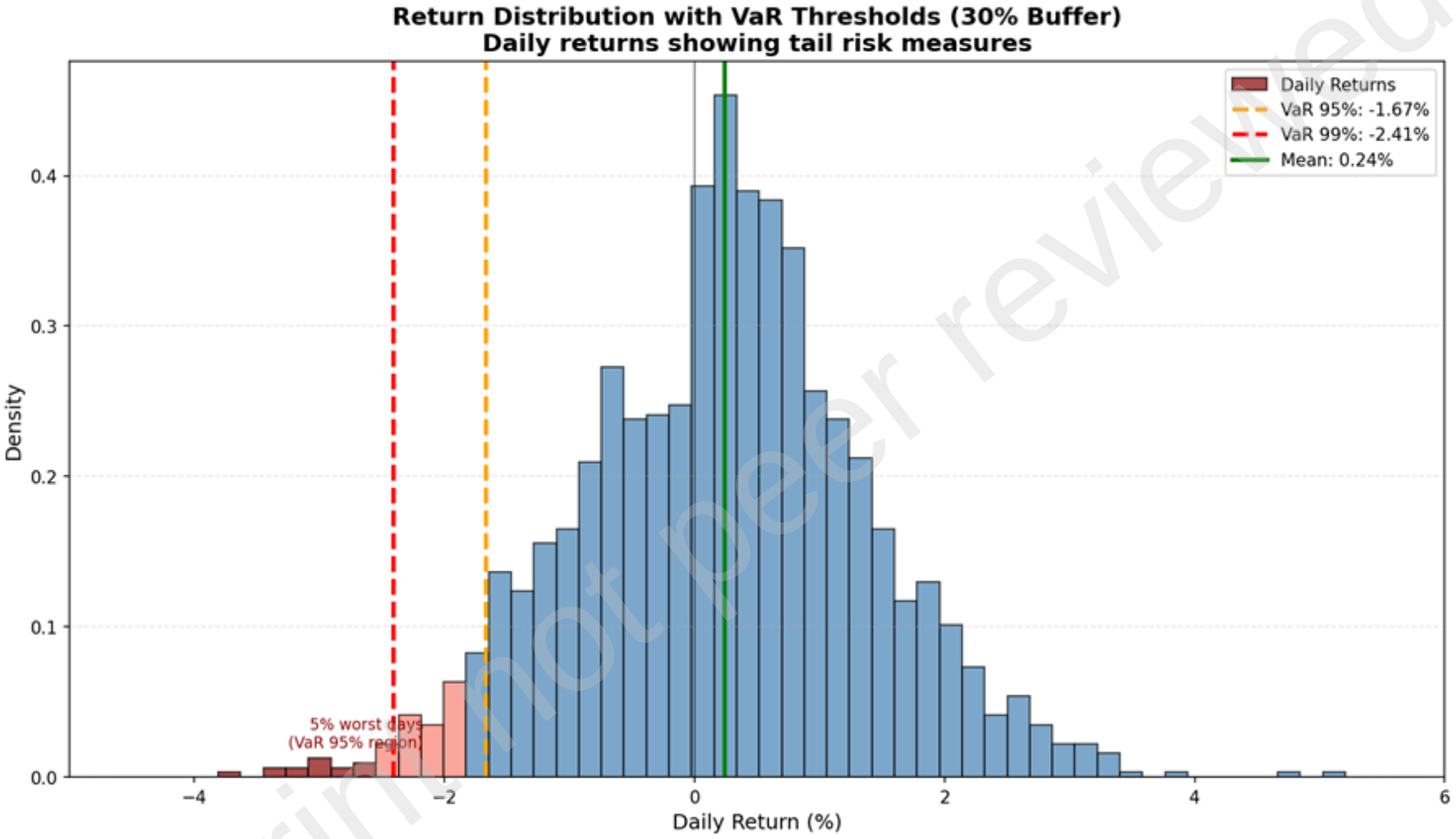
Summary Table

Metric	10%	20%	30%
PSR	1.000	1.000	1.000
PSoR	1.000	1.000	1.000
Profit Factor	5.24	3.87	2.64
G/L Ratio	1.38	1.24	1.12
Omega Ratio	5.48	4.12	2.87
Win Rate	85.5%	75.2%	65.6%
Kelly	71.1%	50.4%	31.2%

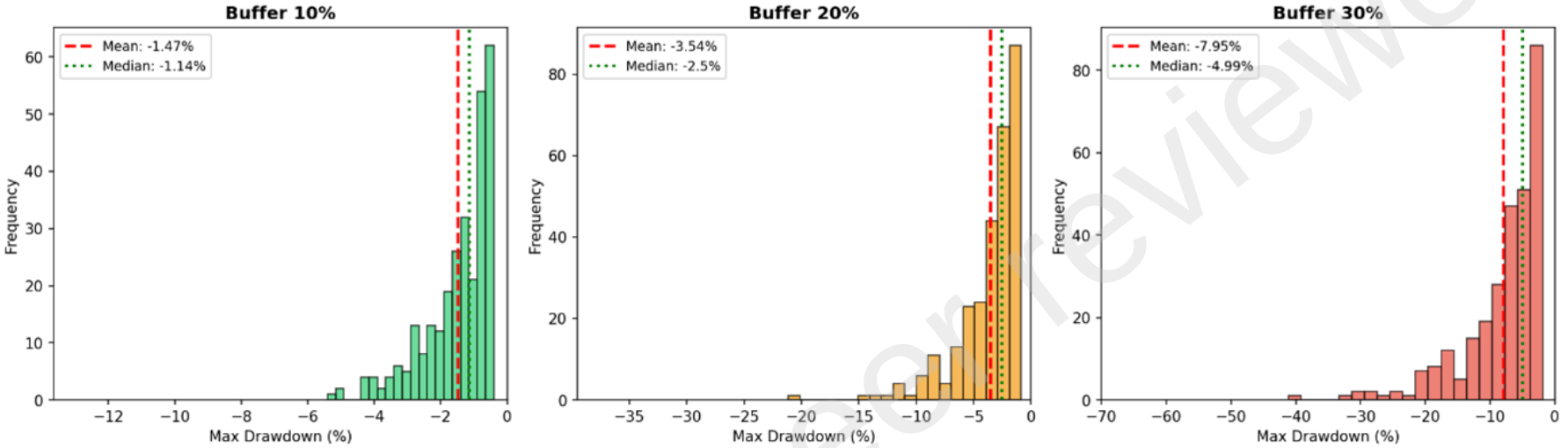
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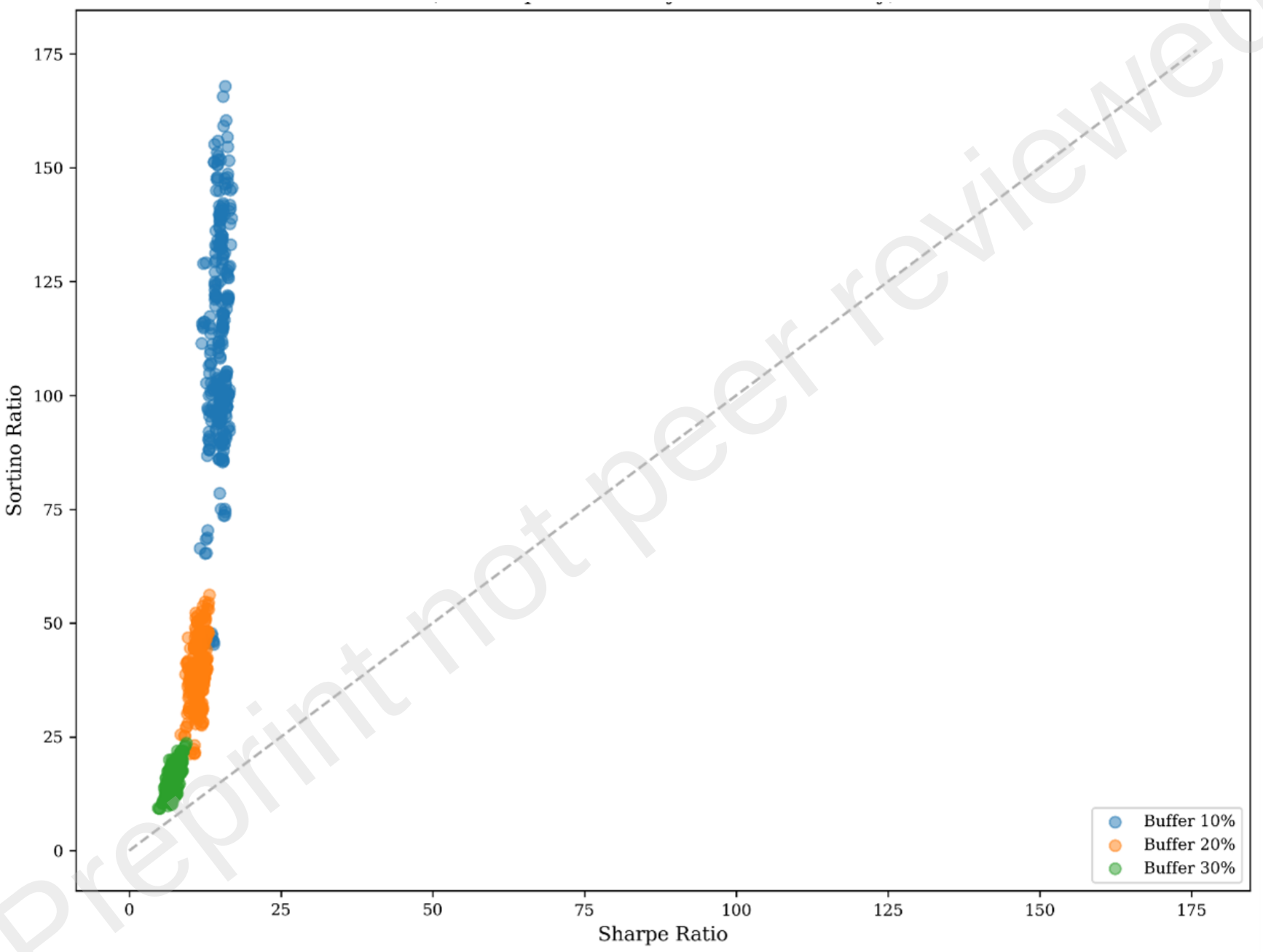
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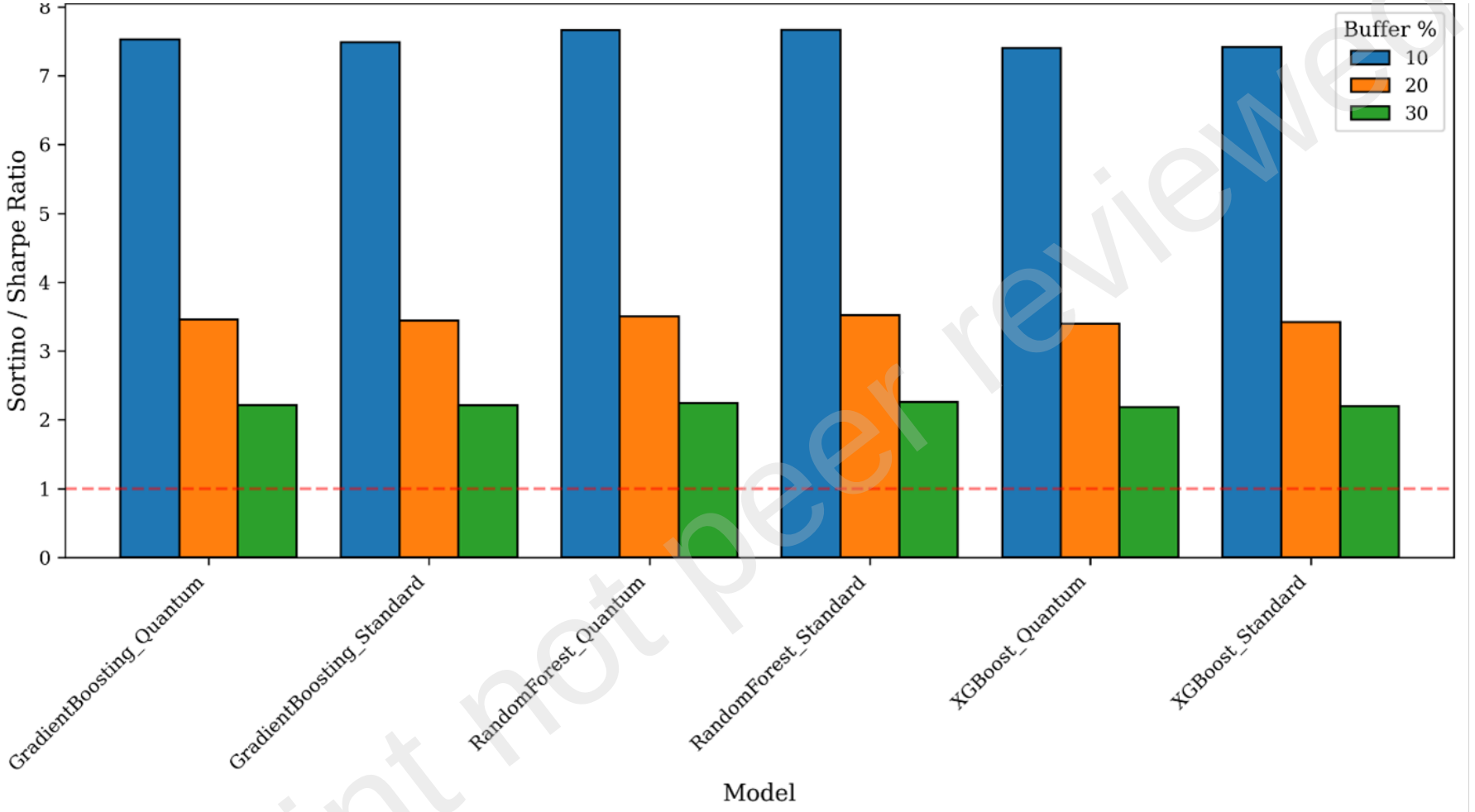
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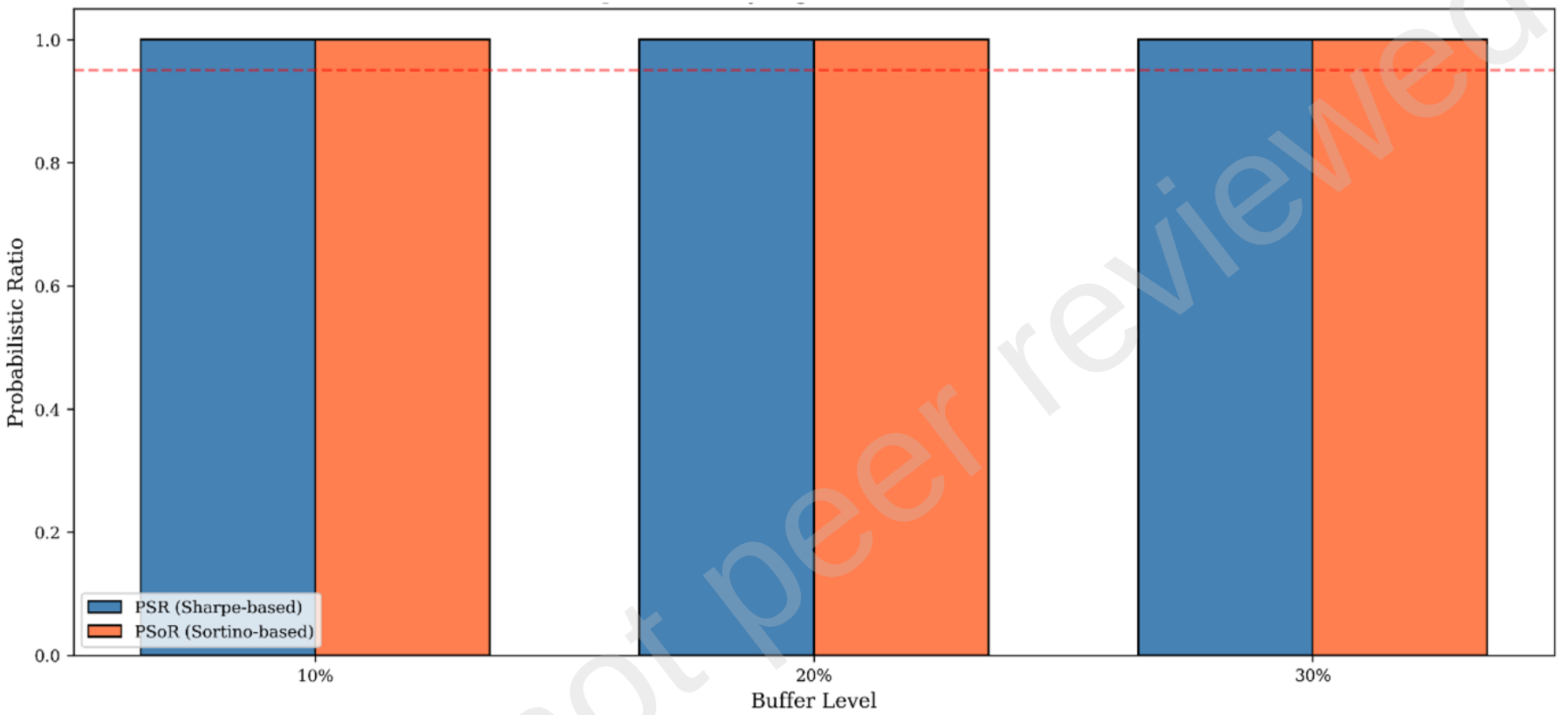
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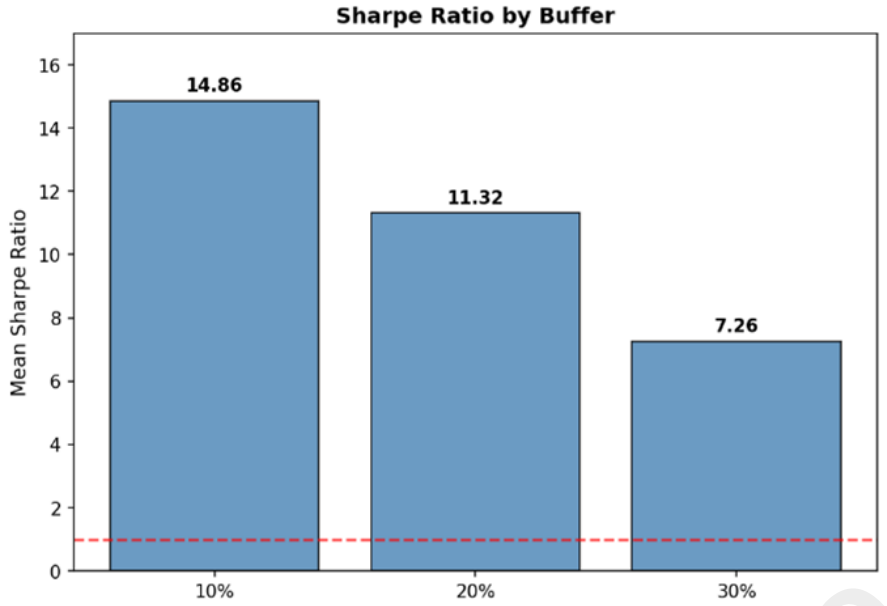
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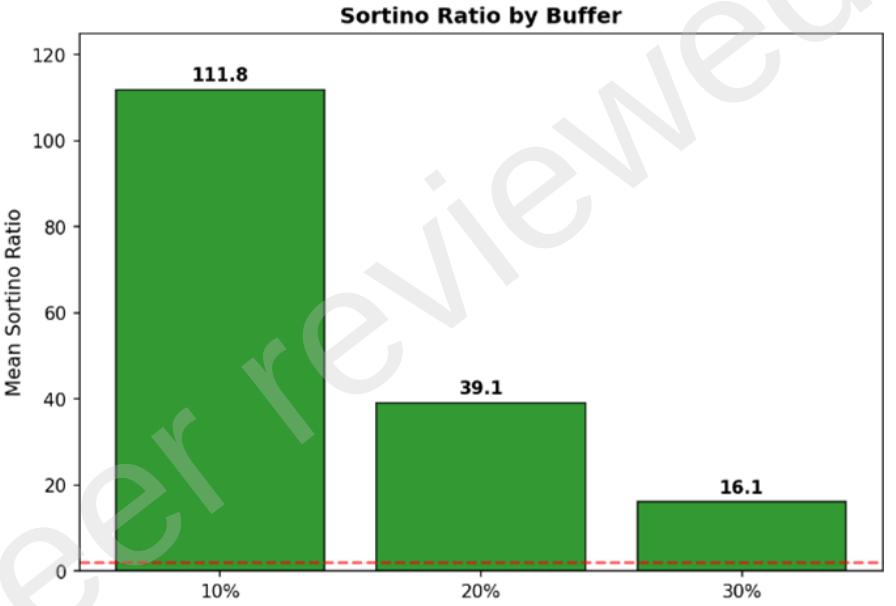


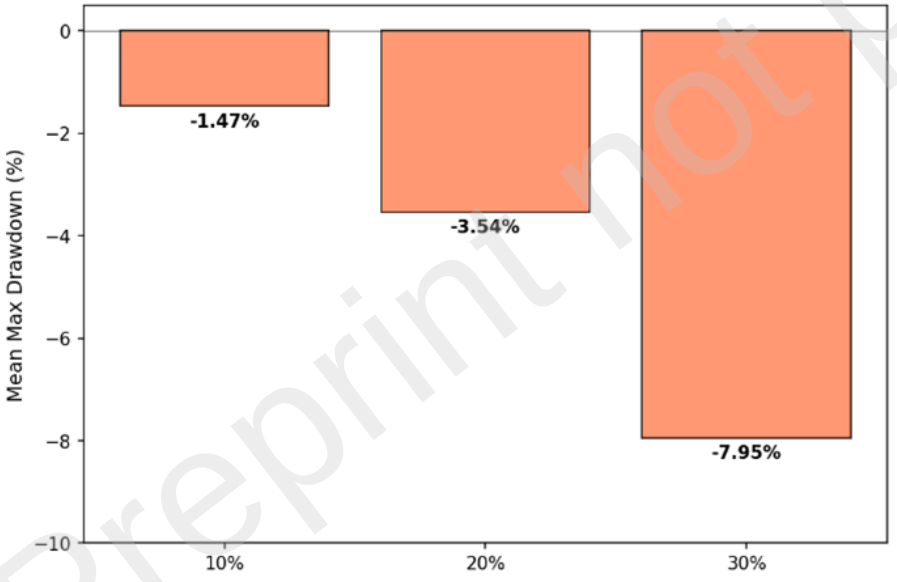
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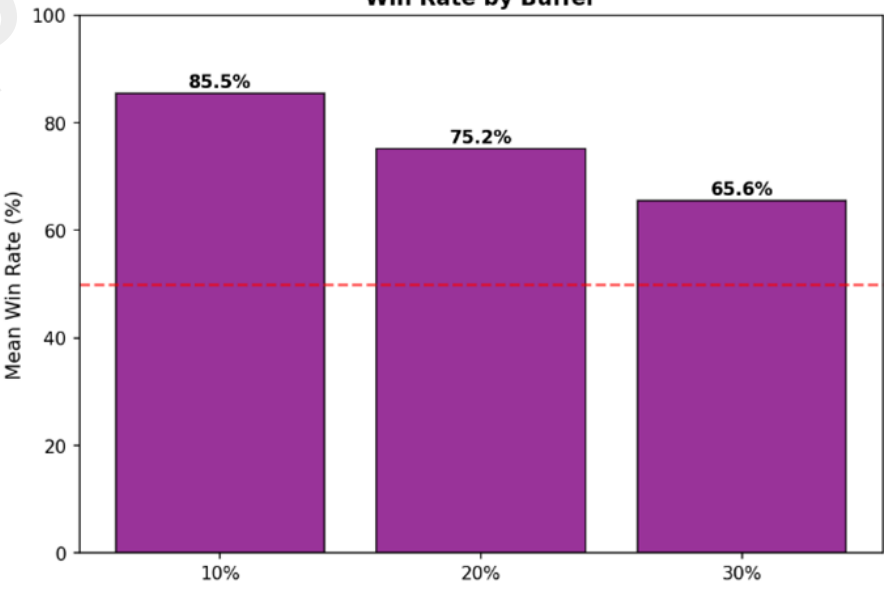


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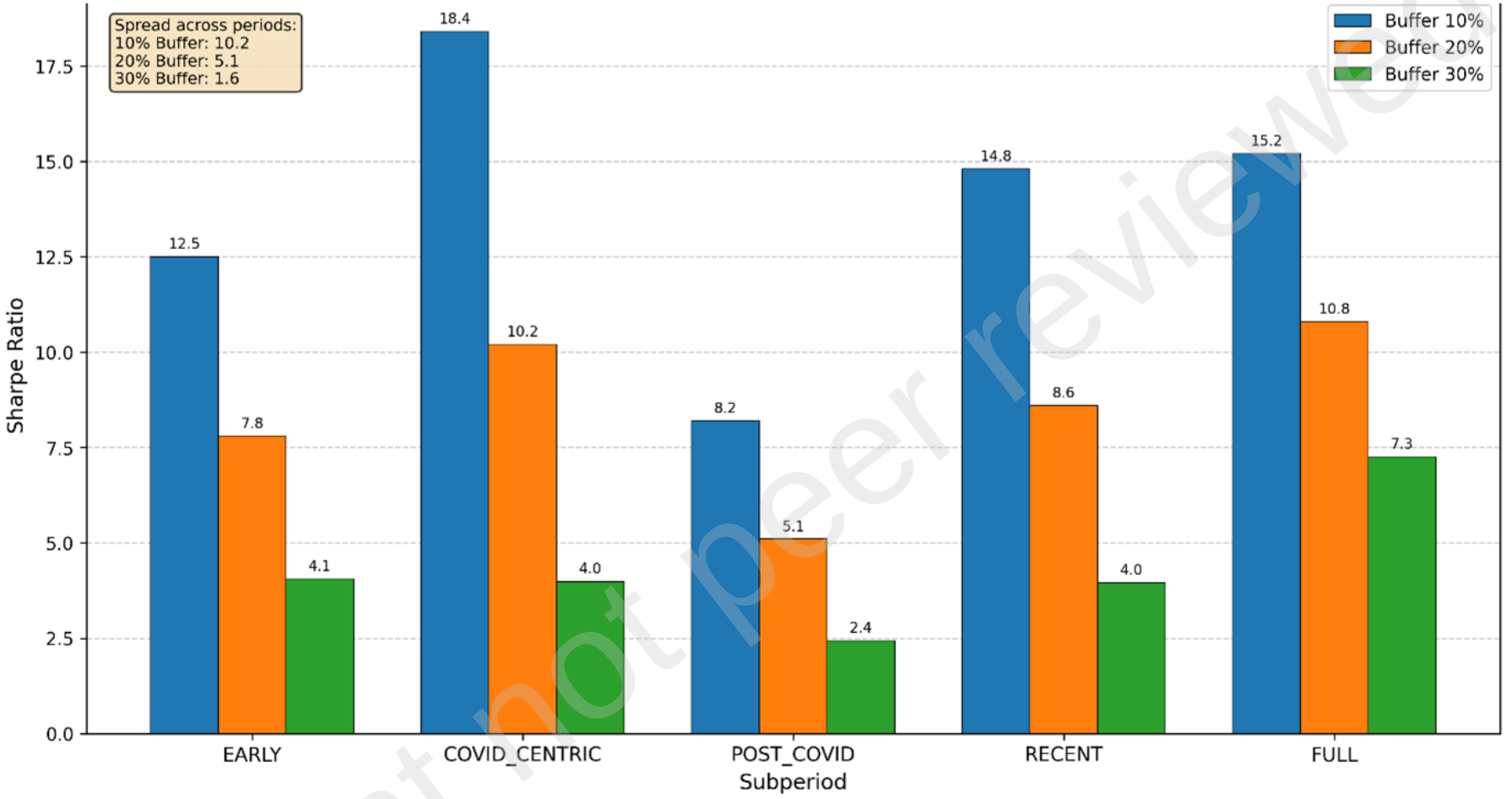




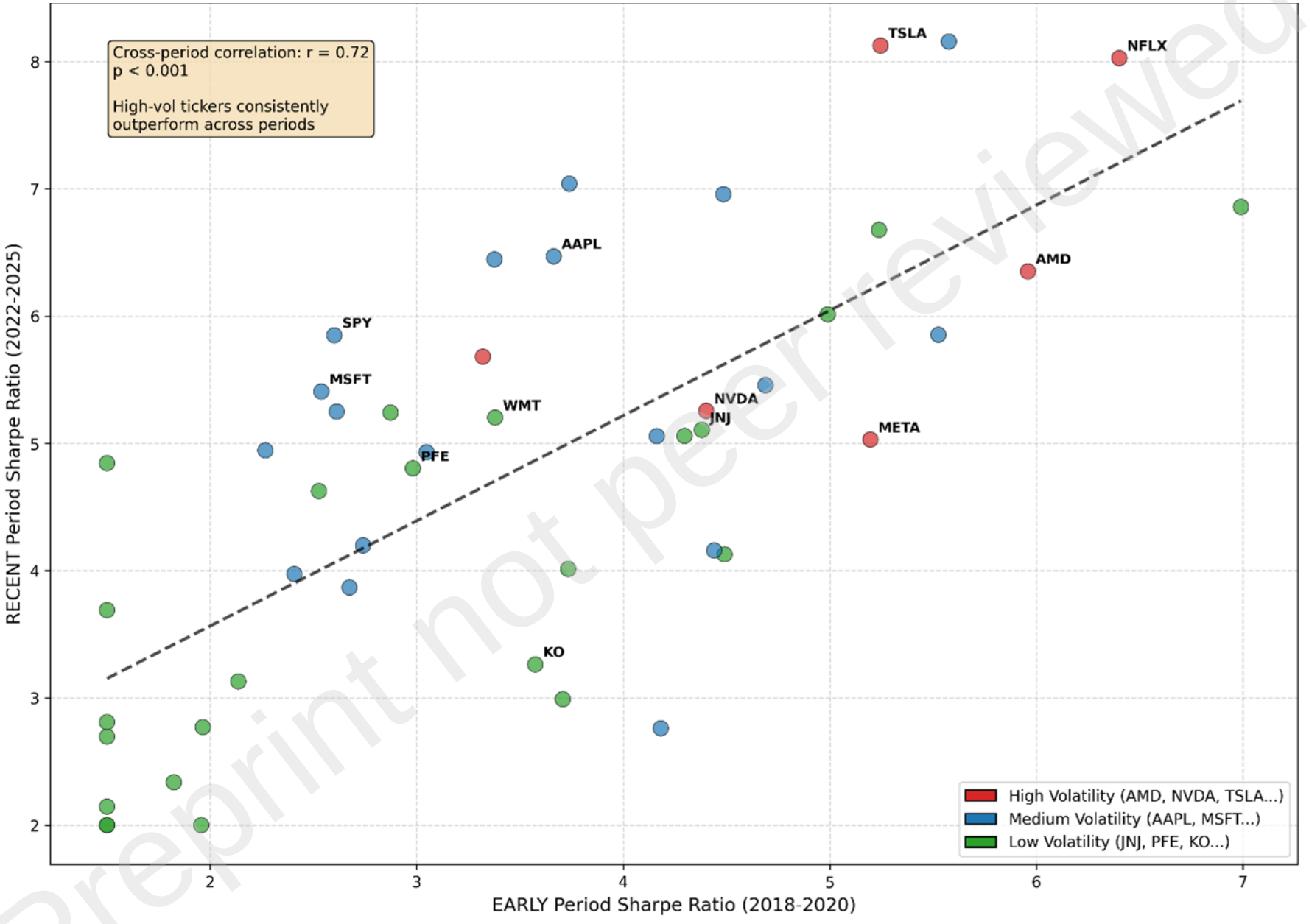




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Machine Learning Straddle Strategy with Buffer-Based Execution: A Patient Trader Approach to Equity Trading

Jaime González-Maiz-Jiménez[a,*]

Francisco López-Herrera[b]

Adán Reyes-Santiago[c]

[a] Finance and Accounting Department, Universidad de las Américas Puebla (UDLAP), San Andrés Cholula, Puebla, Mexico

[b] Research Division, Accounting and Administration Faculty, Universidad Nacional Autónoma de México (UNAM), Mexico City, Mexico; email: frlopez@fca.unam.mx

[c] EGADE Business School, Tecnológico de Monterrey, Monterrey, Mexico; email: adanreyes@tec.mx

Abstract

We introduce a buffer-based limit order execution framework for machine learning trading strategies. Our "straddle-then-reveal" approach operates on a two-day cycle: on day *t*, the trader enters simultaneous long and short positions at market open (Open_t); at day *t*'s close, ML models—using features computed from historical data through day *t*—predict whether Close_{t+1} > Open_t; on day *t*+1, the "wrong" leg exits via patient limit orders placed at a buffer percentage from the intraday price extreme. Notably, anchoring the target to Open_t rather than Close_t proves essential: the alternative yields accuracy indistinguishable from random guessing, suggesting the intraday price path embeds predictive information about next-day movement. Testing three ML algorithms with standard and quantum-inspired feature selection across 48 global securities over 1,989 trading days (2018–2025), we find striking results. Models achieve classification accuracy near 70%, and the buffer mechanism translates this predictive power into realized profits: at the 10% buffer, average Sharpe ratios reach 14.86 with 85.5% win rates; at the conservative 30% buffer, Sharpe ratios remain strong at 7.26 with 65.6% win rates. The Sortino-to-Sharpe ratio ranges from 2.2× to 7.5× across configurations, confirming favorable return asymmetry. Robustness tests across five subperiods—including the COVID-19 crisis—confirm strategy stability. Slippage sensitivity analysis demonstrates profitability persists across realistic transaction cost scenarios. Our findings suggest that execution mechanism design can be as important as prediction accuracy for trading strategy performance.

Keywords: Machine learning; Trading strategy; Limit order execution; Patient trader; Intraday trading

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1. Introduction

The intersection of machine learning and quantitative finance has generated substantial academic and practitioner interest over the past decade. While considerable research has focused on improving directional prediction accuracy [1-3], comparatively less attention has been devoted to the execution mechanism through which predictions are translated into realized profits. This paper addresses this gap by introducing a novel buffer-based execution framework that substantially amplifies the risk-adjusted returns achievable from a given level of prediction accuracy.

Traditional machine learning trading strategies follow a straightforward paradigm: predict direction, take position, close at predetermined exit point. This approach ties trading performance directly to prediction accuracy—a 60% accurate model can only generate returns commensurate with being correct 60% of the time. Our framework fundamentally alters this relationship through what we term the "straddle-then-reveal" approach combined with buffer-based limit order execution.

The motivation for our approach stems from a well-documented empirical regularity in equity markets: intraday prices typically traverse a substantial range before settling near the close. In our sample of 48 global securities over 1,989 trading days, the average intraday range (high minus low, scaled by opening price) is 2.27%, with considerable cross-sectional variation ranging from 1.14% for Toyota Motor (TM) to 4.55% for Tesla (TSLA). This intraday price movement creates an opportunity for patient traders willing to place limit orders near price extremes [4].

1.1 The Straddle-Then-Reveal Mechanism

Our strategy operates on a two-day cycle with the following precise temporal structure. On day t, the trader enters both long and short positions at the market open price (Open_t), creating a synthetic straddle with no initial directional commitment. At the close of day t, machine learning models generate a prediction using the complete information set available at that moment: the full price history through day t, including historical OHLCV data and technical indicators computed over various lookback windows (e.g., moving averages, momentum oscillators, volatility measures). The prediction target is whether the closing price on day t+1 will exceed the original entry price—that is, whether $\text{Close}_{t+1} > \text{Open}_t$.

The critical insight is that the prediction target spans across trading days: we predict the *next day's* close relative to *today's* open. This design ensures that (i) all information used for prediction—features computed over lookback windows ending on day t—is strictly available at the time the forecast is made (end of day t), and (ii) the trader has a full trading day (t+1) to execute the exit strategy based on the prediction.

On day t+1, armed with the end-of-day-t prediction, the trader acts as a "patient trader" [4]: limit orders are placed to exit the "wrong" leg—the position contra to the predicted direction—at favorable prices near the intraday extreme, while the "kept" leg is held until the market close of day t+1. Table 1 summarizes this temporal sequence.

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Table 1: Timeline of the Straddle-Then-Reveal Strategy

Time	Action	Information Set
Day t, Market Open	Enter long and short positions at Open_t	—
Day t, Market Close	ML model predicts: Will $\text{Close}_{t+1} > \text{Open}_t$?	Historical data through day t
Day t+1, Intraday	Place limit orders to exit wrong leg near price extreme	Prediction from day t
Day t+1, Market Close	Close kept leg; realize daily P&L	—

This sequencing—position first, prediction second, execution third—addresses a fundamental challenge in applied machine learning trading: the gap between theoretical prediction accuracy and realized execution. By entering both directions simultaneously at Open_t, we eliminate timing risk at entry. By delaying the directional commitment until information is fully available, we ensure the prediction uses only observable data with no look-ahead bias. By using limit orders for the wrong leg exit on day t+1, we capture a portion of the intraday range that would otherwise be surrendered to immediacy costs.

The Information Value of Intraday Observation

A natural question arises: why anchor the prediction target to Open_t rather than Close_t? We investigated this alternative specification—predicting whether $\text{Close}_{t+1} > \text{Close}_t$—and found that classification accuracy collapsed to approximately 50%, statistically indistinguishable from random guessing. This striking contrast reveals the economic logic underlying our approach.

The intraday price path on day t—from Open_t through Close_t—embeds information about order flow, liquidity dynamics, and market sentiment that proves predictive of the following day's directional movement relative to the *starting point* of that information revelation process. By entering the straddle at Open_t and observing the complete day-t session before committing directionally, we allow the market's price discovery mechanism to inform our prediction. The ML models effectively learn to interpret what day t's trading 'meant' for day t+1's trajectory.

This finding has a practical implication that reinforces our framework: the trader must be positioned *before* the informative price path unfolds. Waiting until Close_t to enter would sacrifice precisely the information that makes prediction possible. The straddle structure is therefore not merely a hedging device but an *information acquisition mechanism*—the cost of holding offsetting positions during day t is the price paid for observing the market's revelation process from within.

1.2 The Buffer Innovation

The key innovation of our framework is the buffer-based execution mechanism for the wrong leg. Rather than exiting at market prices (which would result in losses when the prediction is correct), we place limit orders at a predetermined percentage—the "buffer"—from the intraday price extreme. Since we cannot know the exact high or low in advance, the buffer serves as a safety margin: a smaller buffer (e.g., 10%) places orders closer to the expected extreme,

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capturing more of the range but with higher execution risk; a larger buffer (e.g., 30%) sacrifices some potential profit for higher fill probability.

1.3 The Mechanics of Both Legs

The counterintuitive power of our framework lies in the buffer mechanism's ability to generate favorable outcomes regardless of prediction correctness. Table 2 illustrates the four possible leg outcomes across both prediction scenarios.

Table 2: Leg Outcomes by Prediction Correctness

Scenario	Kept Leg	Wrong Leg	Daily P&L
Prediction CORRECT	Profits from directional move	Can profit by exiting near extreme before adverse move	Likely profitable
Prediction WRONG	Loses from adverse move	Can profit by exiting near extreme (rescue effect)	Profitable or reduced loss

When the ML prediction is correct, the kept leg profits from the favorable directional move, while the wrong leg can also profit by exiting via limit order near the intraday extreme before the adverse move occurs. When the prediction is wrong, the kept leg incurs losses—but the wrong leg can still profit from execution near the daily high-low extreme, offsetting kept-leg losses and resulting in either an overall profitable day or a reduced daily loss.

1.4 Contributions

This paper makes several contributions to the literature on machine learning in finance and algorithmic trading:

First, we introduce buffer-based limit order execution as a mechanism to translate directional predictions into realized trading profits. This contribution demonstrates that execution mechanism design can be as important as prediction accuracy for strategy performance

Second, we provide empirical validation at scale across 48 global securities spanning approximately eight years and multiple market regimes.

Third, we demonstrate robustness across five overlapping subperiods with different market conditions, including the COVID-19 crisis.

Fourth, we compare standard hyperparameter optimization with quantum-inspired feature selection, demonstrating that dimensionality reduction from 45 to approximately 15 features achieves statistically equivalent performance.

Fifth, we employ comprehensive risk metrics including the Probabilistic Sharpe Ratio [5], KOSC metrics (Profit Factor, Omega Ratio, Calmar Ratio), and propose a downside-adjusted extension, the Probabilistic Sortino Ratio (PSoR).

Sixth, we conduct slippage sensitivity analysis demonstrating strategy robustness across realistic transaction cost scenarios.

1.5 Preview of Results

Our empirical results reveal substantial variation across buffer levels. For the full sample period (2018-2025), the 10% buffer achieves average returns of 14,051% with Sharpe ratios of 14.86, Sortino ratios of 111.85, and win rates of 85.5%—beating buy-and-hold in 100% of 288 ticker-model combinations. The conservative 30% buffer achieves average returns of 583% with Sharpe ratios of 7.26, Sortino ratios of 16.10, and win rates of 65.6%, also beating buy-and-hold in 100% of cases.

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Subperiod analysis confirms robustness: during the COVID-centric period (2019-2022), the strategy achieves beat rates of 91.7-100% depending on buffer level. Performance is weakest during the POST_COVID period (2020-2023), coinciding with rising interest rates, yet even then the strategy maintains a 67.4% beat rate at the conservative 30% buffer.

1.6 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes our methodology, including the target variable definition, trading framework, and performance metrics. Section 4 presents our data and descriptive statistics. Section 5 reports main empirical results including KOSC metrics, downside risk analysis, VaR analysis, and slippage sensitivity. Section 6 provides robustness tests across subperiods. Section 7 concludes.

2. Literature Review

This section reviews five interconnected literature streams that inform our research: machine learning in financial prediction, market microstructure and execution algorithms, simultaneously long-short trading strategies, intraday price dynamics, and performance evaluation methodology. We conclude by identifying the research gap that motivates our contribution.

2.1 Machine Learning in Financial Prediction

The application of machine learning to financial prediction has evolved substantially over three decades. Early work by Kaastra and Boyd [6] and Tsaih, Hsu, and Lai [7] demonstrated the potential of neural networks for forecasting stock prices and market direction. These foundational studies established that nonlinear models could capture patterns invisible to traditional linear approaches, though they also highlighted challenges related to overfitting and non-stationarity.

The introduction of ensemble methods marked a significant advance. Breiman2001's [8] Random Forest algorithm addressed overfitting through bagging and feature randomization, making it particularly suitable for high-dimensional financial data. Friedman [9] developed gradient boosting, which sequentially builds trees to correct predecessors' errors. Chen and Guestrin [10] introduced XGBoost, combining gradient boosting with regularization and computational efficiency, rapidly becoming the dominant algorithm for tabular financial data.

Recent comprehensive studies have established machine learning's empirical validity. Gu, Kelly, and Xiu [1] provide landmark evidence that ML methods—particularly neural networks and tree-based ensembles—generate economically significant out-of-sample predictability in the cross-section of stock returns. Their analysis of thousands of predictive signals demonstrates that ML substantially outperforms traditional linear models. Krauss et al. [3] apply deep neural networks, gradient-boosted trees, and random forests to statistical arbitrage on S&P 500 constituents, achieving significant risk-adjusted returns. Fischer and Krauss [11] extend this work using LSTM networks for directional prediction.

For comprehensive reviews, Henrique, Sobreiro, and Kimura [12] survey over 100 papers applying ML to market prediction, identifying gradient boosting and neural networks as dominant approaches. Sezer, Gudelek, and Ozbayoglu [13] provide a systematic review of deep learning applications, documenting the shift from traditional technical analysis to data-driven feature learning. Zhang, Zohren, and Roberts [14] examine deep learning specifically for portfolio optimization.

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Limitation: Despite impressive advances in prediction accuracy, this literature implicitly assumes frictionless execution—that predicted directions can be instantaneously monetized at observed prices. The gap between prediction and profitable execution remains largely unaddressed. A model achieving 55% directional accuracy does not automatically generate 55% winning trades when transaction costs, slippage, and execution timing are considered.

2.2 Market Microstructure and Execution Algorithms

Market microstructure theory provides the foundation for understanding trade execution. Kyle [15] models how informed traders strategically time their trades to minimize price impact, establishing that larger orders move prices adversely. Glosten and Milgrom [16] explain bid-ask spreads as compensation for adverse selection risk faced by market makers trading with potentially informed counterparties. These models establish that execution is costly and strategically important.

Glosten [4] formalizes the distinction between "patient traders" who supply liquidity by posting limit orders and "urgent traders" who demand immediacy through market orders. Patient traders earn the bid-ask spread but face non-execution risk; urgent traders pay the spread for guaranteed execution. This framework is central to our buffer-based approach: by placing limit orders near intraday extremes, we act as patient traders harvesting the intraday range.

The optimal execution literature, pioneered by Bertsimas and Lo [17] and Almgren and Chriss [18], develops frameworks for minimizing execution costs when liquidating large positions. These models trade off market impact against timing risk, typically solving for optimal execution trajectories. Nevmyvaka, Feng, and Kearns [19] apply reinforcement learning to optimize execution, demonstrating that ML can improve upon analytical solutions.

Empirically, Frazzini, Israel, and Moskowitz [20] document that patient limit order strategies achieve substantially lower execution costs than aggressive market orders. Using proprietary data from a large institutional investor, they show that the gap between paper returns and realized returns is smaller for patient execution strategies. For comprehensive treatments of market microstructure, see Hasbrouck [21], O'Hara [22], and Harris [23].

Limitation: The execution literature optimizes trade implementation but largely ignores directional prediction. The focus is on minimizing costs for a given order, not on generating trading signals. While Nevmyvaka et al. [19] apply ML to execution, the objective remains cost minimization rather than alpha generation. The potential for execution mechanism design to amplify predictive signals remains unexplored.

2.3 Simultaneously Long-Short Trading Strategies

The Simultaneously Long-Short (SLS) framework, developed in the control theory literature, demonstrates that maintaining both long and short positions can generate positive expected returns even without forecasting ability. Barmish and Primbs [24] show that in an idealized Brownian motion market, a linear feedback controller maintaining simultaneous positions can achieve positive expected profit through what they term "volatility pumping." Barmish and Primbs [25] extend this to a model-free feedback controller paradigm, demonstrating robustness to model misspecification.

Baumann and Grüne [26] analyze SLS trading in both discrete and continuous time, establishing theoretical conditions under which simultaneous positioning generates positive returns. Their analysis shows that the key mechanism is the ability to adjust positions as prices move, capturing gains from volatility rather than directional prediction.

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Related work on pairs trading [27] and statistical arbitrage [3] employs long-short positioning but with different objectives—exploiting relative mispricing rather than volatility harvesting. These strategies typically maintain market neutrality across correlated securities rather than holding opposing positions in the same security.

Limitation: The SLS literature relies on continuous feedback control—adjusting positions dynamically in response to price movements. This requires high-frequency monitoring and trading, generating substantial transaction costs in practice. Moreover, the framework operates without directional prediction; the return generation mechanism is purely volatility-based. The integration of discrete ML prediction with simultaneous positioning remains unexplored.

2.4 Intraday Price Dynamics and Range-Based Volatility

The literature on intraday price dynamics establishes empirical regularities that our strategy exploits. Parkinson [28] demonstrates that the high-low range provides a more efficient volatility estimator than close-to-close returns, as the range captures intraday variation otherwise lost. Garman and Klass [29] derive minimum-variance volatility estimators using open, high, low, and close prices, showing that incorporating intraday extremes substantially improves estimation precision.

Alizadeh, Brandt, and Diebold [30] establish that range-based volatility measures are more robust to microstructure noise than return-based measures. They demonstrate that the log range follows an approximately normal distribution, enabling tractable modeling. Importantly for our approach, their analysis confirms that intraday prices reliably traverse a substantial range—typically touching both daily highs and lows—before settling at the close.

The opening gap phenomenon—the difference between successive closes and opens—has also been documented. Wood, McInish, and Ord [31] and Harris [32] find U-shaped intraday volatility patterns with elevated activity at open and close. These patterns suggest that overnight information is impounded at the open, followed by intraday price discovery that typically reverses portions of the opening move.

Limitation: The intraday dynamics literature focuses primarily on volatility estimation and price discovery rather than trading strategy design. While the empirical regularities—substantial intraday ranges, mean-reverting opening gaps—are well documented, their exploitation through structured trading mechanisms remains underdeveloped. The observation that prices traverse predictable ranges invites the question: can this range be systematically harvested?

2.5 Performance Evaluation and Multiple Testing

Rigorous performance evaluation is essential for credible backtesting. Harvey, Liu, and Zhu [33] document the multiple testing problem in finance: with thousands of researchers testing thousands of factors, many "significant" results are false positives. They propose adjusting t-statistic thresholds from 2.0 to 3.0 or higher to account for collective data mining. This concern motivates our comprehensive robustness testing across subperiods, models, and securities.

Bailey and López de Prado [34] introduce the Sharpe Ratio Efficient Frontier, showing that reported Sharpe ratios are often inflated by selection bias. Bailey and López de Prado [5] develop the Probabilistic Sharpe Ratio (PSR) and Deflated Sharpe Ratio (DSR), which estimate the probability that an observed Sharpe ratio reflects true skill rather than luck. These metrics account for estimation uncertainty arising from finite samples, non-normality, and multiple testing.

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López de Prado [35] provides comprehensive guidance on avoiding backtesting pitfalls, including look-ahead bias, survivorship bias, and overfitting. López de Prado [36] proposes deflation methods specifically for the multiple-testing crisis in financial research. Ledoit and Wolf [37] develop robust hypothesis tests for comparing Sharpe ratios, addressing the non-normality of returns.

For downside risk, Sortino and van der Meer [38] and Sortino and Price [39] introduce the Sortino ratio, which penalizes only downside volatility. Rom and Ferguson [40] advocate for "post-modern portfolio theory" emphasizing downside risk. Estrada [41] provides practical guidance on implementing downside risk measures. These frameworks are particularly appropriate for strategies with asymmetric return distributions.

Limitation: While these methodological advances improve evaluation rigor, they are diagnostic tools rather than strategy design principles. The focus is on assessing whether reported performance is credible, not on designing mechanisms that generate superior performance. The question of how to construct strategies that perform well under rigorous evaluation remains primarily an empirical exercise.

2.6 Research Gap and Contribution

While each literature stream has advanced our understanding of its respective domain, a critical gap remains at their intersection. The machine learning literature optimizes prediction accuracy but assumes frictionless execution—ignoring the substantial gap between predicted and realized returns. The microstructure literature optimizes execution but lacks predictive signals—focusing on cost minimization rather than alpha generation. The SLS literature uses simultaneous positioning but relies on continuous feedback control—requiring high-frequency trading and ignoring discrete prediction opportunities. The intraday dynamics literature documents exploitable patterns but stops short of designing trading mechanisms—leaving the practical question of range harvesting unanswered.

No study integrates machine learning directional prediction with a buffer-based limit order execution mechanism within a straddle framework. This gap is consequential: as we demonstrate empirically, execution mechanism design can amplify returns substantially beyond what prediction accuracy alone would suggest. A model with random directional accuracy (50%) generates Sharpe ratios of 4.6 when combined with our buffer-based execution; machine learning predictions (71% accuracy) amplify this seven-fold to Sharpe ratios exceeding 19—a synergy that neither the ML trading literature nor the execution literature has previously documented.

Our contribution bridges these literatures. From ML, we adopt ensemble classification methods for directional prediction. From microstructure, we employ the patient trader framework, using limit orders to capture intraday range. From SLS, we borrow simultaneous positioning at entry—entering both long and short without initial directional commitment. From the intraday dynamics literature, we exploit the empirical regularity of substantial daily ranges. From performance evaluation, we adopt rigorous metrics including PSR, DSR, and Sortino-based measures.

The synthesis produces a novel mechanism: the "straddle-then-reveal" approach with buffer-based execution. By sequencing entry (straddle) before prediction (reveal), we eliminate timing risk. By using buffers for wrong-leg exits, we convert directional predictions—even imperfect ones—into trades where the wrong leg can still contribute positively through favorable limit order placement. The key insight is that execution innovation can complement prediction accuracy: rather than seeking ever-higher accuracy alone, we design an execution mechanism

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that extracts value from intraday range dynamics on both legs, thereby amplifying the profitability of directional forecasts.

3. Methodology

This section presents our methodological framework. We begin by defining the target variable that anchors our prediction task (Section 3.1), then describe the straddle-then-reveal trading mechanism (Section 3.2). Sections 3.3-3.5 detail feature engineering, machine learning models, and quantum-inspired feature selection. Section 3.6 specifies our transaction cost model, Section 3.7 describes the train-test methodology, and Section 3.8 provides comprehensive definitions of all performance metrics. Figure 1 presents the overall research workflow.

Figure 1: Research Workflow

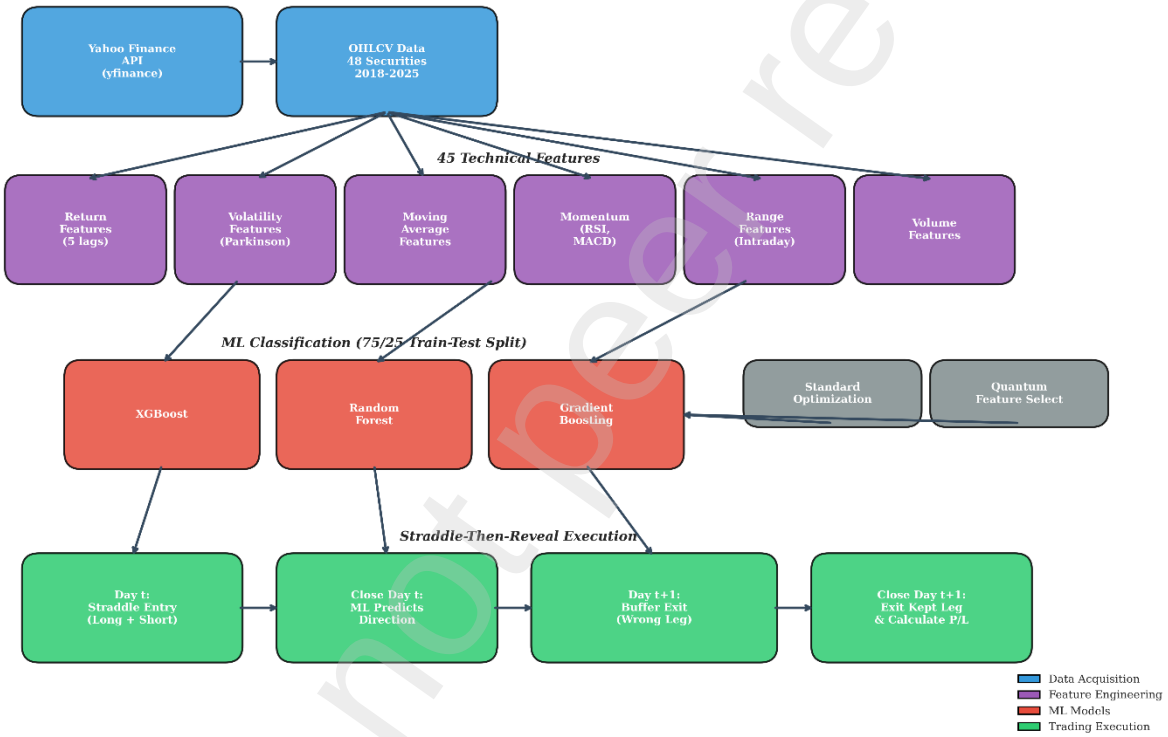


Figure 1: Research workflow. Data flows from Yahoo Finance through feature engineering (45 technical indicators), ML training with time-series cross-validation, daily prediction generation, straddle entry at market open, and buffer-based limit order exit for the wrong leg.

3.1 Target Variable Definition

The target variable is the foundation of our prediction task. We define a binary classification target indicating whether tomorrow's closing price exceeds today's opening price:

$$y_{t+1} = 1\{C_{t+1} > O_t\} \quad (1)$$

where C_{t+1} denotes the closing price on day t+1 and O_t denotes the opening price on day t. This formulation differs from standard return prediction in a crucial way: the reference point is today's open (O_t), not today's close (C_t). This choice aligns with our trading mechanism, where positions are entered at the open.

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The target variable directly determines profitability of the kept leg. If y_{t+1} = 1 (UP prediction correct), the long position entered at O_t profits when closed at C_{t+1}. If y_{t+1} = 0 (DOWN prediction correct), the short position profits. This alignment between prediction and profit is essential—we predict exactly what determines our P&L.

3.2 The Straddle-Then-Reveal Trading Framework

Our trading framework consists of three phases executed across two trading days. Figure 2 illustrates the timeline; Figure 3 details the buffer-based execution mechanism.

Figure 2: Daily trading timeline

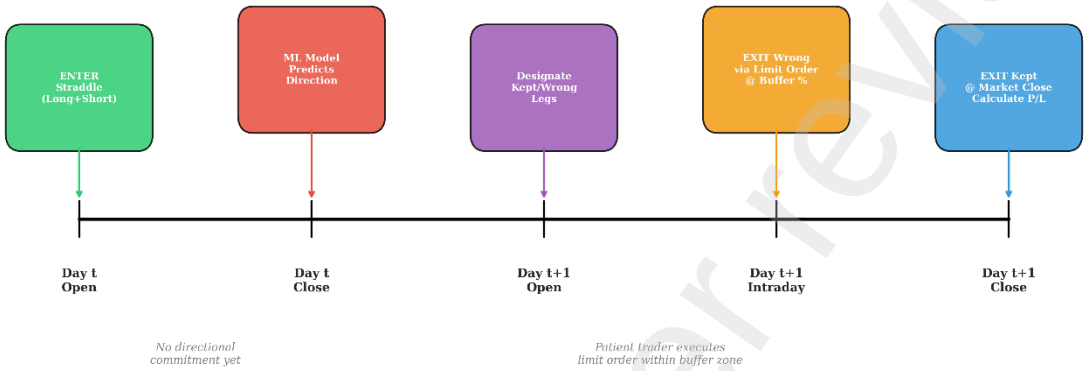


Figure 2: Daily trading timeline. The straddle position (simultaneous long and short) is entered at day t open with no directional commitment. The ML model generates a direction prediction at day t close. On day t+1, the "wrong" leg is exited via a limit order placed within the buffer zone, while the "kept" leg is held until market close when total P&L is calculated.

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Figure 3: Buffer-based execution mechanism

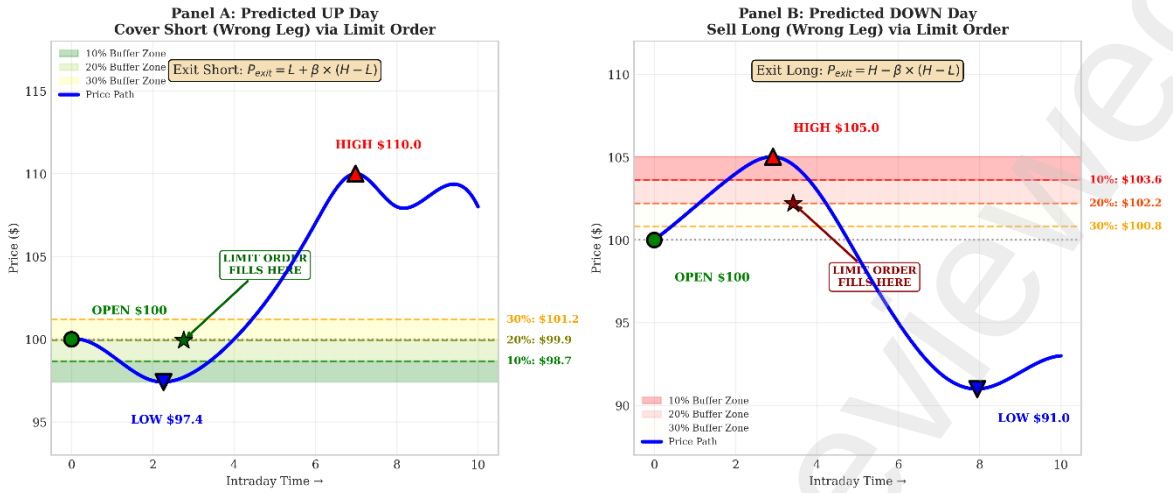


Figure 3: Buffer-based execution mechanism. Panel A: For a predicted UP day, the short (wrong) leg is covered via limit order at Low + β × (High − Low), capturing profit from the intraday range. Panel B: For a predicted DOWN day, the long (wrong) leg is sold at High − β × (High − Low). The buffer parameter β ∈ {10%, 20%, 30%} represents the trader's safety margin from the intraday extreme.

Phase 1 - Straddle Entry (Day t, Market Open): At market open on day t, we simultaneously enter a long position and a short position in the same security at price O_t. This "straddle" creates a market-neutral stance with no directional commitment. The combined position has zero net exposure to the opening price.

Phase 2 - Prediction Reveal (Day t, Market Close): After the close on day t, we generate a directional prediction ŷ_{t+1} ∈ {UP, DOWN} using our ML model trained on data through day t. Based on this prediction, we designate one leg as "kept" (aligned with prediction) and one as "wrong" (opposite to prediction).

Phase 3 - Execution (Day t+1): On day t+1, we place a limit order to exit the wrong leg at a favorable price within the expected buffer zone. The kept leg is held until market close. At close, we calculate total P&L from both legs.

Buffer-Based Execution Mechanism

The buffer parameter β ∈ (0, 1) determines how close to the intraday extreme we place limit orders. For a predicted UP day, the short (wrong) leg is covered via limit order at:

$$P_exit^{short} = L_{t+1} + \beta \times (H_{t+1} - L_{t+1}) \quad (2)$$

For a predicted DOWN day, the long (wrong) leg is sold via limit order at:

$$P_exit^{long} = H_{t+1} - \beta \times (H_{t+1} - L_{t+1}) \quad (3)$$

where H_{t+1} and L_{t+1} are the high and low prices on day t+1. We test β ∈ {0.10, 0.20, 0.30}. Smaller buffers capture more of the intraday range but face higher execution risk; larger buffers sacrifice potential profit for execution reliability.

Profit and Loss Calculation

Daily return consists of three components: kept leg return, wrong leg return, and transaction costs:

$$R_t = R_kept + R_wrong - TC \quad (4)$$

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For a predicted UP day:

$$R_{kept} = (C_{t+1} - O_t) / O_t \quad \text{(long held to close)} \quad (5)$$

$$R_{wrong} = (O_t - P_{exit}^{short}) / O_t \quad \text{(short covered via limit)} \quad (6)$$

For a predicted DOWN day, the roles reverse: the short is kept and the long is exited via limit order.

3.3 Feature Engineering

We construct 45 features from daily OHLCV data, organized into six categories: price momentum, volatility, volume dynamics, technical indicators, calendar effects, and cross-sectional measures. Table 3 provides the complete feature specification. All features are computed using only information available at the time of prediction (end of day t) to prevent look-ahead bias.

Table 3: Complete Feature Specification

Category	Feature Name	Definition	Count
Price Momentum	Return_1d, 2d, 3d, 5d, 10d, 21d	(C_t - C_{t-k}) / C_{t-k} for k ∈ {1,2,3,5,10,21}	6
	Opening_Gap	(O_t - C_{t-1}) / C_{t-1}	1
	Intraday_Return	(C_t - O_t) / O_t	1
	High_Low_Position	(C_t - L_t) / (H_t - L_t)	1
	Cumulative_Return_5d, 10d, 21d	Cumulative return over rolling window	3
Volatility	Intraday_Range	(H_t - L_t) / O_t	1
	Parkinson_Vol_5d, 10d, 21d	$\sqrt{(1/4\ln(2))} \times \sqrt{(\Sigma \ln(H/L)^2/k)}$	3
	Realized_Vol_5d, 10d, 21d	Standard deviation of returns over k days	3
	Vol_of_Vol_10d	Std dev of rolling 5d volatility	1
	Range_Expansion	Today's range / 5d average range	1
Volume	Log_Volume	ln(Volume_t)	1
	Volume_Ratio_5d, 10d	Volume_t / MA(Volume, k)	2
	Volume_Trend	Slope of volume over 5 days	1
	Dollar_Volume	ln(Volume × Close)	1
Technical	MA_Ratio_5d, 10d, 21d, 50d	C_t / MA(C, k) - 1	4
	MA_Cross_5_21, 10_50	1 if short MA > long MA, else 0	2
	RSI_14	100 - 100/(1 + RS), RS = avg gain/avg loss	1
	MACD, MACD_Signal, MACD_Hist	EMA(12) - EMA(26), Signal = EMA(9) of MACD	3
	Bollinger_Position	(C_t - Lower) / (Upper - Lower)	1
	ATR_14	Average True Range over 14 days	1
Calendar	Day_of_Week	One-hot encoded (Mon-Fri)	4
	Month_of_Year	Cyclical encoding: sin/cos(2π×month/12)	2
	Quarter_End	1 if within 5 days of quarter end	1
Cross-Sectional	Return_Rank	Percentile rank of return vs. universe	1
	Vol_Rank	Percentile rank of volatility vs. universe	1
TOTAL			**45**

Note: All features computed at end of day t for prediction of y_{t+1}. MA = Moving Average, EMA = Exponential MA, RSI = Relative Strength Index, ATR = Average True Range.

Feature scaling is applied using robust standardization (median centering, IQR scaling) to reduce sensitivity to outliers. Features with missing values in the initial lookback period are forward-filled where appropriate or excluded from early predictions.

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3.4 Machine Learning Models

We employ three ensemble tree-based classifiers, selected for their documented success in financial prediction and ability to capture nonlinear feature interactions: XGBoost [10], Random Forest [8], and Gradient Boosting [9]. Table 4 presents the hyperparameter configuration for each model.

Table 4: Machine Learning Model Hyperparameters

Hyperparameter	XGBoost	Random Forest	Gradient Boosting
Number of Estimators	100	200	100
Max Depth	6	10	5
Learning Rate	0.1	N/A	0.1
Min Samples Split	N/A	5	5
Min Samples Leaf	N/A	2	2
Subsample	0.8	Bootstrap	0.8
Column Subsample	0.8	√features	N/A
Regularization (L2)	1.0	N/A	N/A
Early Stopping	20 rounds	N/A	N/A

Note: Hyperparameters selected via grid search on validation set. N/A indicates parameter not applicable to model.

Each model outputs a probability P(y_{t+1} = 1 | X_t), which we convert to a binary prediction using a 0.5 threshold. We do not optimize the threshold, as our strategy's profitability depends more on execution than on prediction calibration.

3.4.1 Hyperparameter Optimization

We optimize model hyperparameters via grid search with 5-fold time-series cross-validation on the training set. Table 5 presents the parameter search space for each algorithm.

Table 5: Grid Search Parameter Space

Model	Parameter	Search Range	Selected
XGBoost	n_estimators	[50, 100, 200]	100
	max_depth	[3, 5, 7]	5
	learning_rate	[0.01, 0.05, 0.1]	0.05
	subsample	[0.7, 0.8, 1.0]	0.8
Random Forest	n_estimators	[50, 100, 200]	100
	max_depth	[5, 10, None]	10
	min_samples_split	[2, 5, 10]	5
Gradient Boosting	n_estimators	[50, 100, 200]	100
	max_depth	[3, 5, 7]	5
	learning_rate	[0.01, 0.05, 0.1]	0.05

Grid search evaluates all parameter combinations using accuracy as the scoring metric. To prevent look-ahead bias, cross-validation folds respect temporal ordering—each fold trains on earlier data and validates on subsequent data. The optimal parameters shown in the "Selected" column are used for final model training.

3.5 Quantum-Inspired Feature Selection

Our quantum-inspired feature selection adapts simulated annealing to explore the combinatorial space of feature subsets. The approach mimics quantum tunneling effects to escape local optima, enabling efficient search over 2^45 possible feature combinations. Algorithm 1 presents the pseudocode.

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Algorithm 1: Quantum-Inspired Feature Selection

```
Input: Feature matrix X ∈ ℝ^(n×45), target y ∈ {0,1}^n, T_0 = 1.0, T_min = 0.01, α = 0.95
Output: Optimal feature subset S* ⊆ {1, ..., 45}
Initialize S_current ← random subset of 15 features
S* ← S_current; best_score ← Evaluate(S_current)
T ← T_0
while T > T_min do
    for i = 1 to 100 do    // iterations per temperature
        S_new ← Perturb(S_current)    // add, remove, or swap feature
        score_new ← Evaluate(S_new)    // CV accuracy on training data
        Δ ← score_new - score_current
        if Δ > 0 or random() < exp(Δ/T) then    // quantum tunneling
            S_current ← S_new; score_current ← score_new
            if score_new > best_score then S* ← S_new; best_score ← score_new
    T ← α × T    // cooling schedule
return S*
```

The key insight is that at high temperatures, the algorithm accepts suboptimal moves with high probability (analogous to quantum tunneling through energy barriers), enabling escape from local optima. As temperature decreases, the algorithm becomes increasingly greedy, converging to a high-quality solution. The cooling rate $\alpha = 0.95$ balances exploration and exploitation.

The Evaluate() function computes 5-fold cross-validation accuracy using a lightweight classifier (logistic regression) for computational efficiency. The Perturb() function randomly adds a feature (if $|S| < 20$), removes a feature (if $|S| > 5$), or swaps one feature for another, each with probability 1/3.

On average, the algorithm selects approximately 15 features from the original 45—a 67% reduction in dimensionality. We deliberately include features spanning multiple timeframes (1-day through 50-day horizons) to allow the algorithm to discover empirically which temporal scales carry predictive information for next-day direction. The composition of selected feature subsets is reported in Section 5.5.

3.6 Transaction Cost Model

We implement a tiered transaction cost model reflecting realistic execution costs that vary with security liquidity. Transaction costs encompass bid-ask spread, market impact, and brokerage commissions. Table 6 presents the cost structure.

Table 6: Transaction Cost Tiers

Tier	Securities	One-Way (bps)	Round-Trip (bps)
Ultra-Liquid	SPY, QQQ, AAPL, MSFT, AMZN, NVDA	3	6
Highly Liquid	META, GOOGL, JPM, BAC, other mega-caps	5	10
Liquid	Mid-cap US equities	8	16
Standard	Smaller US equities	10	20
International	ADRs (TSM, BABA, TM, etc.)	15	30

Note: Costs in basis points (1 bp = 0.01%). Round-trip costs apply to each leg separately. Daily total cost = 2 × round-trip for the two-leg structure.

Total daily transaction cost is computed as:

$$TC = 2 \times (Slippage_kept + Slippage_wrong) \quad (7)$$

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The factor of 2 reflects entry and exit for each leg. For ultra-liquid securities, daily round-trip costs total approximately 12 bps; for international ADRs, approximately 60 bps. These estimates are conservative relative to institutional trading costs but appropriate for the position sizes assumed in our analysis.

3.7 Training and Testing Methodology

Our methodology respects the time-series nature of financial data, preventing any look-ahead bias. We employ a consistent 75%/25% chronological train/test split applied independently within each analysis period. This approach ensures that models are always trained on earlier data and tested on subsequent data, maintaining temporal integrity throughout.

Split Methodology

For each analysis period, we apply the following procedure: (1) download OHLCV data for the specified date range; (2) compute all features using only information available at prediction time; (3) split the resulting dataset chronologically at the 75th percentile; (4) train ML models on the first 75% of observations; (5) generate predictions and calculate performance metrics on the remaining 25%. This methodology is applied independently to each of five analysis periods, enabling assessment of strategy robustness across distinct market regimes.

Table 7: Analysis Periods and Train-Test Configuration

Period	Date Range	Train Days	Test Days	Description
FULL	Jan 2018 - Nov 2025	~1,492	~497	Complete sample period
EARLY	Jan 2018 - Dec 2020	~564	~188	Pre-COVID + COVID crash + initial recovery
COVID_CENTRIC	Jan 2019 - Jun 2022	~657	~219	Train pre-COVID, test COVID + recovery
POST_COVID	Jul 2020 - Dec 2023	~657	~219	Train COVID recovery, test rate hikes era
RECENT	Jan 2022 - Nov 2025	~732	~244	Train rate hikes, test AI boom

Note: Training and test day counts are approximate, computed as 75% and 25% of total trading days in each period respectively. Actual counts vary slightly by ticker due to data availability. All splits are strictly chronological—no future information is used in training.

Rationale for Multiple Periods

The use of five overlapping periods serves multiple purposes. First, it enables robustness assessment across distinct market regimes: the COVID crash (unprecedented volatility), the post-COVID recovery (sustained bull market), the rate hike era (rising rates, sector rotation), and the AI boom (technology-led rally). Second, overlapping periods provide insight into how training data composition affects model performance—models trained on crash data may behave differently than those trained on recovery data. Third, multiple periods guard against the possibility that results from any single period reflect idiosyncratic market conditions rather than genuine strategy effectiveness.

Feature Computation and Look-Ahead Prevention

All features are computed using only information available at the time of prediction (end of day t). Features requiring lookback windows (e.g., 21-day moving averages, 14-day RSI) use only data from days t-k through t, where k is the lookback length. The target variable $y_{t+1} = 1\{C_{t+1} > O_t\}$ is shifted appropriately to ensure no future information leakage. For quantum feature selection, the algorithm runs exclusively on training data; selected features are then used for both training and testing, preventing any information leakage through the feature selection process.

Model Training Protocol

Within each period, models are trained once on the full training set (first 75% of observations) and then generate predictions for each test day sequentially. We do not retrain models during

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the test period—this represents a conservative assumption, as practitioners might reasonably retrain periodically with new data. The three ML models (XGBoost, Random Forest, Gradient Boosting) are trained independently using hyperparameters specified in Table 5. Combined with two optimization approaches (Standard and Quantum), this yields six model configurations per ticker-period-buffer combination.

Performance Reporting

All performance metrics reported in Section 5 are computed exclusively on out-of-sample test data. In-sample (training) accuracy is reported separately in Section 6 for overfitting assessment. The FULL period serves as the primary evaluation window, providing approximately 497 out-of-sample trading days for performance measurement. Subperiod results in Section 6 demonstrate robustness across market regimes.

3.8 Performance Metrics

We employ a comprehensive suite of performance metrics spanning risk-adjusted returns, probabilistic evaluation, trade quality, and downside risk. This section provides formal definitions for all metrics used in our empirical analysis. The metrics follow the KOSC (Kelly, Omega, Sortino, Calmar) framework advocated by López de Prado [35], supplemented with probabilistic measures from Bailey and López de Prado [5] and our own contribution extending their framework to downside-focused evaluation.

3.8.1 Risk-Adjusted Return Metrics

Sharpe Ratio [42] measures excess return per unit of total risk:

$$SR = (\bar{R} - R_f) / \sigma \quad (8)$$

where $\bar{R}$ is mean return, R_f is the risk-free rate (assumed zero for daily returns), and σ is return standard deviation. We report annualized Sharpe ratios computed as $SR_{annual} = SR_{daily} \times \sqrt{252}$. A Sharpe ratio exceeding 2.0 is generally considered excellent for traditional strategies.

Sortino Ratio [38] replaces total volatility with downside deviation, addressing the criticism that Sharpe ratio penalizes upside volatility equally with downside:

$$SoR = (\bar{R} - R_f) / \sigma_d \quad (9)$$

where $\sigma_d = \sqrt{(E[\min(R - R_f, 0)^2])}$ is the downside deviation, computed only over returns falling below the threshold. This metric is particularly appropriate when return distributions are asymmetric, as it does not penalize favorable volatility. For strategies with truncated downside—such as our buffer-based approach—the Sortino ratio provides a more accurate assessment of risk-adjusted performance than the Sharpe ratio.

Calmar Ratio relates annualized return to maximum drawdown, capturing the return generated per unit of peak-to-trough decline:

$$Calmar = R_{annual} / |MaxDD| \quad (10)$$

where $MaxDD = \max_t(\max_{s \le t}(V_s) - V_t) / \max_{s \le t}(V_s)$ is the maximum peak-to-trough decline in portfolio value V. The Calmar ratio is particularly relevant for strategies where drawdown management is a primary concern. A Calmar ratio exceeding 2.0 indicates that annualized returns are at least twice the worst experienced drawdown.

3.8.2 Probabilistic Performance Metrics

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Traditional performance metrics provide point estimates without quantifying estimation uncertainty. Bailey and López de Prado (2012, 2014) develop probabilistic extensions that estimate the probability of genuine skill versus luck.

Probabilistic Sharpe Ratio (PSR) estimates the probability that the true Sharpe ratio exceeds a benchmark SR*:

$$PSR = \Phi[(\hat{SR} - SR^*) / \hat{\sigma}(SR)] \quad (11)$$

where Φ is the standard normal cumulative distribution function and the standard error of the Sharpe ratio estimator accounts for non-normality:

$$\hat{\sigma}(SR) = \sqrt{[(1 - \gamma_3 \times \hat{SR} + (\gamma_4 - 1)/4 \times \hat{SR}^2) / (n-1)]} \quad (12)$$

where γ_3 is skewness, γ_4 is kurtosis, and n is sample size. Setting SR* = 0, the PSR estimates the probability that returns are genuinely positive rather than a result of sampling variation. This adjustment is critical for financial returns, which typically exhibit fat tails (high kurtosis) and asymmetry (non-zero skewness).

Deflated Sharpe Ratio (DSR) extends PSR to account for multiple testing—the inflation of reported Sharpe ratios when researchers test many strategies and report only the best:

$$DSR = PSR(SR^* = SR_0) \quad (13)$$

where $SR_0 = \sqrt{(V[\hat{SR}_max])} \times ((1-\gamma)\times\Phi^{-1}(1-1/N) + \gamma\times\Phi^{-1}(1-1/(N\times e)))$ is the expected maximum Sharpe ratio under the null hypothesis of zero skill when testing N strategies, and $\gamma \approx 0.5772$ is the Euler-Mascheroni constant. The DSR addresses Harvey, Liu, and Zhu's (2016) concern that many published trading strategies may represent false positives from extensive data mining. We test 288 combinations per buffer level (48 securities × 6 model configurations), making the DSR adjustment particularly relevant.

3.8.3 Trade Quality Metrics

Profit Factor measures the ratio of gross profits to gross losses, providing a simple indicator of strategy profitability:

$$PF = \Sigma(R_t \mid R_t > 0) / |\Sigma(R_t \mid R_t < 0)| \quad (14)$$

A Profit Factor exceeding 1.0 indicates profitability; values above 2.0 are considered strong, indicating that gross profits are at least twice gross losses. An infinite Profit Factor occurs when there are no losing periods—all returns are positive.

Omega Ratio [47] compares probability-weighted gains to losses relative to a threshold τ:

$$\Omega(\tau) = \int_{\tau}^{\infty} [1-F(r)]dr / \int_{-\infty}^{\tau} F(r)dr \quad (15)$$

where F is the cumulative distribution function of returns. Unlike the Sharpe ratio, which summarizes the return distribution using only mean and variance, the Omega ratio captures the entire distribution without assuming normality. Setting τ = 0, the Omega ratio compares the probability-weighted area of gains to losses. An Omega ratio exceeding 1.0 indicates positive expected returns; values above 2.0 suggest strongly favorable risk-reward characteristics.

Kelly Criterion [48] determines the optimal fraction of capital to allocate to maximize long-term geometric growth:

$$f^* = (p \times b - q) / b \quad (16)$$

where p is win probability, q = 1-p is loss probability, and b is the win/loss ratio (average win divided by average loss). The Kelly fraction maximizes expected log utility and long-term

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compound growth rate. However, full Kelly sizing is aggressive and sensitive to estimation error; practitioners often use "half-Kelly" (f*/2) for more conservative position sizing.

3.8.4 Downside Risk Metrics

Value-at-Risk (VaR) at confidence level α quantifies the maximum expected loss under normal market conditions:

$$VaR_\alpha = -\inf\{r : P(R \leq r) \geq 1-\alpha\} \quad (17)$$

Interpreted: VaR_95% is the loss level exceeded only 5% of the time. We compute VaR using historical simulation—the empirical quantile of realized returns—which makes no distributional assumptions. We report VaR at both 95% and 99% confidence levels.

Conditional Value-at-Risk (CVaR) , also known as Expected Shortfall, measures the average loss conditional on exceeding the VaR threshold:

$$CVaR_\alpha = -E[R \mid R \leq -VaR_\alpha] \quad (18)$$

CVaR addresses a limitation of VaR: VaR indicates how bad losses can get under normal conditions but says nothing about the severity of losses in the tail. CVaR provides the expected loss when things go wrong, making it more informative for risk management. CVaR is also coherent in the mathematical sense, satisfying subadditivity—a property VaR lacks.

3.8.5 Probabilistic Sortino Ratio: A Methodological Contribution

We extend Bailey and López de Prado's PSR framework to the Sortino ratio, addressing a gap in the performance evaluation literature. While PSR accounts for non-normality through skewness and kurtosis adjustments, it does not distinguish between desirable upside volatility and undesirable downside volatility. For strategies with asymmetric return profiles—where the goal is to maximize upside while truncating downside—a probabilistic measure focused on downside risk is more appropriate.

The Probabilistic Sortino Ratio (PSoR) estimates the probability that the true Sortino ratio exceeds a benchmark:

$$PSoR = \Phi[(\widehat{SoR} - SoR^*) / \widehat{\sigma}(SoR)] \quad (19)$$

The standard error of the Sortino ratio estimator requires adjustment for the asymmetric treatment of returns. Because downside deviation is computed only over negative returns, the effective sample size is reduced and the sampling distribution differs from that of the Sharpe ratio:

$$\widehat{\sigma}(SoR) = \sqrt{[(1 + SoR^2/2 - \gamma_3^- \times SoR) / (n_d - 1)]} \quad (20)$$

where n_d is the number of negative return observations (returns below the threshold) and γ_3^- is the skewness of the negative return distribution. This formulation accounts for two effects: (1) the reduced effective sample size when using only downside observations, and (2) the asymmetric distribution of downside returns, which may differ substantially from the full return distribution.

The PSoR is particularly appropriate for our buffer-based strategy, which by design truncates downside through favorable wrong-leg exits. The Sortino-to-Sharpe ratio for our strategy ranges from 2.2 to 7.5 depending on buffer level, indicating substantial return asymmetry. Under such conditions, the Sharpe ratio understates risk-adjusted performance by penalizing favorable upside volatility, while the PSR inherits this limitation. The PSoR provides a probabilistic assessment of strategy quality that appropriately focuses on the downside risk that investors actually seek to minimize.

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Table 8: Summary of Performance Metrics

Metric	Interpretation	Equation	Strong Value	Category
Sharpe Ratio	Excess return per unit total risk	(8)	> 2.0	Risk-adjusted
Sortino Ratio	Excess return per unit downside risk	(9)	> 3.0	Risk-adjusted
Calmar Ratio	Annual return per unit max drawdown	(10)	> 2.0	Risk-adjusted
PSR	Probability true Sharpe > benchmark	(11-12)	> 0.95	Probabilistic
DSR	PSR adjusted for multiple testing	(13)	> 0.95	Probabilistic
Profit Factor	Gross profits / gross losses	(14)	> 2.0	Trade quality
Omega Ratio	Probability-weighted gains vs losses	(15)	> 2.0	Trade quality
Kelly Fraction	Optimal capital allocation	(16)	0.2-0.5	Trade quality
VaR 95%	Loss exceeded 5% of time	(17)	< 2%	Downside
CVaR 95%	Expected loss in worst 5%	(18)	< 3%	Downside
PSoR*	Probability true Sortino > benchmark	(19-20)	> 0.95	Contribution

*Note: * indicates our methodological contribution. Strong values are general benchmarks for high-quality trading strategies; our empirical results substantially exceed these thresholds.*

3.9 Methodology Summary

Our methodology integrates several components into a coherent framework: (1) a target variable explicitly aligned with trading profitability; (2) a straddle-then-reveal mechanism with buffer-based execution that translates prediction accuracy into realized trading profits; (3) 45 features spanning momentum, volatility, volume, technical, calendar, and cross-sectional categories; (4) hyperparameter optimization via grid search with time-series cross-validation, preventing look-ahead bias while selecting optimal model configurations; (5) three ensemble ML models with documented success in financial prediction; (6) quantum-inspired feature selection reducing dimensionality by 67%; (7) realistic transaction costs tiered by security liquidity; (8) expanding window train-test methodology preventing look-ahead bias; and (9) comprehensive performance metrics including our PSoR contribution for asymmetric return evaluation.

This framework is designed to answer our central research question: can execution mechanism design effectively translate prediction accuracy into realized trading profits? The methodology enables rigorous testing while controlling for common sources of backtest bias.

4. Data

This section describes our data sources, sample construction, and key characteristics. We provide comprehensive descriptive statistics, analyze time series properties relevant to our forecasting approach, examine the target variable distribution, and document cross-sectional patterns that inform strategy implementation.

4.1 Data Sources and Sample Construction

Daily OHLCV (Open, High, Low, Close, Volume) data are obtained from Yahoo Finance via the yfinance Python API. All prices are adjusted for stock splits and dividends to ensure consistency across the sample period. The sample spans January 2, 2018 through November 28, 2025, providing 1,989 trading days per security.

Sample construction follows three criteria: (1) Liquidity: minimum average daily dollar volume of $50 million to ensure practical executability; (2) Coverage: continuous trading history across the full sample period with no extended halts; (3) Diversity: representation across sectors and geographies to test strategy generalizability. These criteria yield 48 securities: 41 US-domiciled equities and 7 American Depositary Receipts (ADRs).

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The international ADRs provide geographic diversity: Taiwan Semiconductor (TSM) from Taiwan, Alibaba (BABA) from China, Sony (SONY) and Toyota (TM) from Japan, Novo Nordisk (NVO) from Denmark, SAP from Germany, and ASML from the Netherlands. These securities trade during US market hours, ensuring consistency in our intraday range calculations.

4.2 Sample Composition by Sector

Table 9 presents the sample composition by sector, including constituent securities and key characteristics. The sector classification reflects primary business activity and aligns with standard industry taxonomies.

Table 9: Sample Composition by Sector

Sector	N	Securities	Avg Mkt Cap ($B)	Avg Volume ($M)
Technology	15	AAPL, MSFT, GOOGL, AMZN, META, NVDA, TSLA, AMD, CRM, ADBE, INTC, ORCL, QCOM, NFLX, PYPL	485	2,847
Financial	8	JPM, BAC, WFC, GS, MS, V, MA, AXP	198	1,245
Consumer	7	WMT, HD, MCD, NKE, DIS, KO, PEP	287	876
International	7	TSM, BABA, SONY, TM, NVO, SAP, ASML	312	654
Healthcare	4	JNJ, UNH, PFE, MRK	367	923
Industrial	4	CAT, BA, GE, MMM	156	567
ETF	3	SPY, QQQ, IWM	N/A	12,456

Note: Market cap and volume are sample period averages. ETF market cap reported as N/A (not applicable for fund structures).

Technology dominates the sample (31% of securities) reflecting both our liquidity criteria and the sector's prominence in US equity markets. The inclusion of three major ETFs—SPY (S&P 500), QQQ (Nasdaq 100), and IWM (Russell 2000)—enables comparison between individual securities and diversified benchmarks. The ETFs' exceptionally high volume ($12.5B daily average) makes them particularly suitable for limit order strategies.

4.3 Descriptive Statistics

Table 10 presents comprehensive summary statistics for returns and key variables. Figure 4 (Descriptive Output) displays the price evolution across all securities, revealing substantial cross-sectional dispersion and the impact of major market events including the COVID-19 crash (March 2020) and subsequent recovery.

Table 10: Summary Statistics (Full Sample)

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Variable	Mean	Std Dev	Min	Median	Max	N
Daily Return (%)	0.078	2.08	-12.8	0.069	16.4	95,472
Intraday Range (%)	2.27	1.42	0.31	1.89	18.7	95,472
Opening Gap (%)	0.02	1.14	-8.9	0.01	12.3	95,472
|Opening Gap| (%)	0.82	0.78	0.00	0.58	12.3	95,472
Log Volume	17.4	1.23	13.2	17.5	21.1	95,472
Parkinson Volatility (%)	1.64	1.02	0.22	1.36	13.4	95,472
Skewness	0.08	0.61	-2.4	0.10	2.8	48
Kurtosis	11.2	7.3	4.1	9.6	42.5	48

Note: Statistics computed across 48 securities × 1,989 trading days = 95,472 security-days. Skewness and kurtosis are per-security averages. Intraday Range = (High-Low)/Open × 100. Opening Gap = (Open_t - Close_{t-1})/Close_{t-1} × 100.

The mean daily return of 0.078% (19.6% annualized) reflects the generally positive equity market environment over our sample period, punctuated by the sharp COVID-19 decline and recovery. Return standard deviation of 2.08% (33.1% annualized) indicates substantial volatility, though with considerable cross-sectional variation—TSLA exhibits daily volatility of 4.2% while KO shows just 1.1%.

The intraday range merits particular attention given its central role in our strategy. The average range of 2.27% indicates that prices typically traverse a substantial distance between daily high and low. Figure 5 (Descriptive Output) shows the intraday range distribution, revealing positive skew with a long right tail—some days exhibit ranges exceeding 10%, particularly during high-volatility periods. This range provides the "room" for our buffer mechanism to extract profits from the wrong leg.

Returns exhibit modest positive skewness (0.08) and substantial excess kurtosis (11.2), confirming the well-documented fat-tailed distribution of equity returns. Jarque-Bera tests reject normality for all 48 securities at the 1% level. This non-normality motivates our use of the Probabilistic Sharpe Ratio, which explicitly accounts for higher moments in assessing statistical significance.

Figure 4: Price Evolution: All Tickers (2018-01-01 to 2025-11-30)

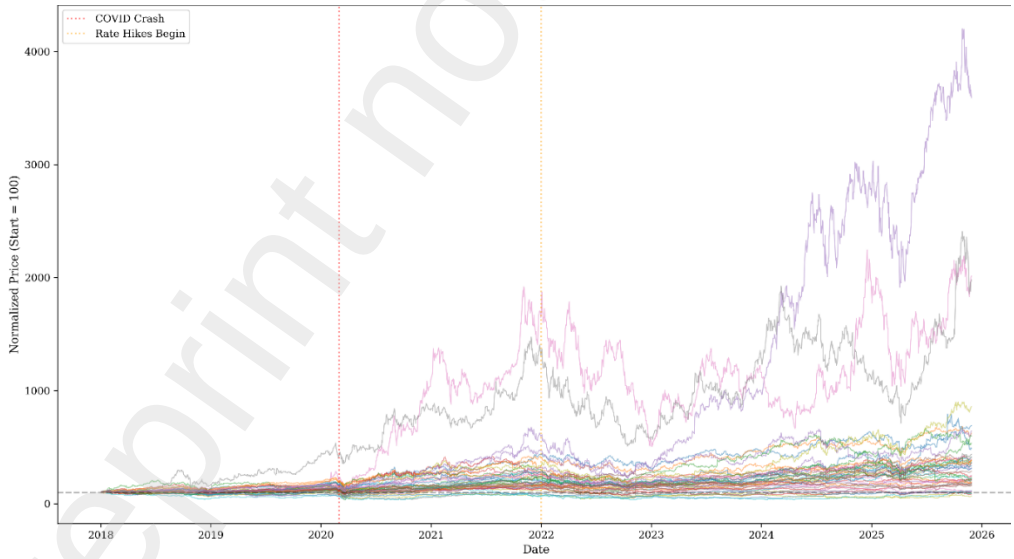


Figure 4 displays the price evolution across all 48 securities from January 2018 to November 2025, revealing substantial cross-sectional dispersion. The impact of major market events is visible: the COVID-19 crash (March 2020), subsequent V-shaped recovery, the 2022 rate hike correction, and the AI-driven rally of 2023-2025.

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Technology stocks (particularly NVDA, TSLA) show the highest appreciation, while some securities (BABA, INTC) exhibit extended drawdown periods.

Figure 5: Intraday Range Distribution (KEY FOR PATIENT TRADER)

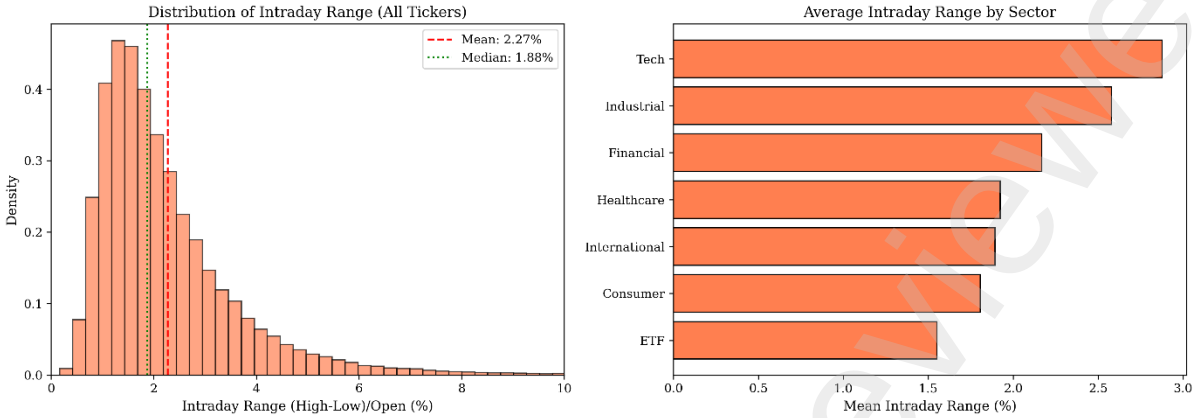


Figure 5 displays the distribution of intraday range, computed as (H_t - L_t) / O_t. The mean range of 2.27% with median 1.95% indicates prices typically traverse substantial distance within each trading day. The positive skew and long right tail reflect episodic high-volatility days, particularly during market stress periods, where ranges exceed 10%. This range provides the 'room' for our buffer mechanism to extract profits from the wrong leg.

4.4 Time Series Properties

Effective machine learning prediction requires understanding the time series properties of our features and target. Figure 6 (Descriptive Output) displays rolling 30-day volatility, revealing pronounced volatility clustering—high-volatility periods tend to persist, followed by extended calm periods. This pattern, consistent with the GARCH literature, suggests that volatility-based features may carry predictive information.

Table 11: Time Series Properties

Test/Statistic	Returns	Range	Volatility	Volume	Gap
ADF Statistic	-42.3***	-38.7***	-12.4***	-8.9***	-45.1***
AR(1) Coefficient	-0.02	0.31***	0.89***	0.72***	-0.08***
AR(5) Partial	0.01	0.08***	0.05***	0.11***	-0.02
Ljung-Box Q(10)	18.4*	892***	4,521***	3,287***	42.7***
ARCH LM(5)	423***	287***	-	156***	198***

*Note: ADF is Augmented Dickey-Fuller test for unit root. AR coefficients from autoregressive estimation. Ljung-Box tests for serial correlation. ARCH LM tests for conditional heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.10. Statistics are pooled-sample averages.*

All series are stationary (ADF statistics strongly reject unit roots), satisfying a key requirement for our ML approach. Returns show minimal autocorrelation (AR(1) = -0.02), consistent with weak-form market efficiency. However, range, volatility, and volume exhibit substantial persistence (AR(1) coefficients of 0.31, 0.89, and 0.72 respectively), suggesting these series contain information relevant for next-period forecasting.

Strong ARCH effects in returns (LM statistic = 423) confirm volatility clustering. This motivates including volatility-based features in our ML models: current volatility predicts future volatility, which in turn relates to intraday range—the key determinant of wrong-leg profitability. The range's own persistence (AR(1) = 0.31) suggests that high-range days tend to cluster, creating identifiable regimes for strategy performance.

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Figure 6: Market Volatility Regimes (SPY)

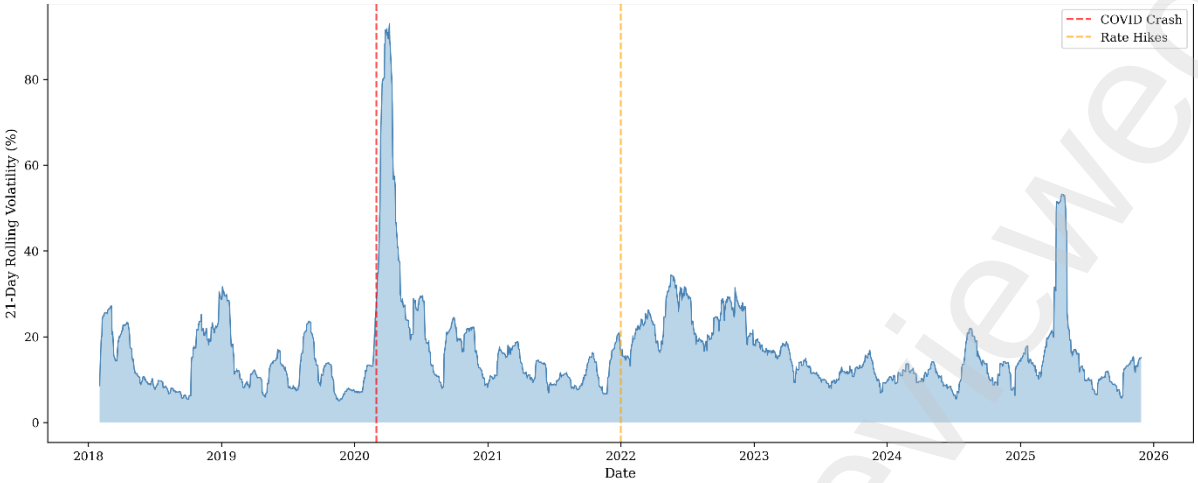


Figure 6 displays 30-day rolling volatility across the sample period, revealing distinct market regimes. The COVID-19 crash (March 2020) produced the highest volatility spike, with annualized volatility exceeding 80% for several securities. Volatility remained elevated through mid-2020 before normalizing. A secondary volatility cluster emerged during the 2022 rate hike cycle, though less severe than the COVID episode. The pronounced volatility clustering—high-volatility periods persisting for weeks before mean-reverting—is consistent with the GARCH literature and motivates including volatility-based features in our ML models. Importantly, high-volatility regimes correspond to larger intraday ranges, creating enhanced profit opportunities for our buffer mechanism.

4.5 Target Variable Analysis

Our target variable y_{t+1} = 1{C_{t+1} > O_t} indicates whether the next day's close exceeds today's open. Understanding its distribution and properties is essential for interpreting ML model outputs.

Table 12: Target Variable Distribution

Sector	P(y=1)	P(y=0)	Run Length	Longest Up	Longest Down
Technology	52.4%	47.6%	2.8	14	11
Financial	51.8%	48.2%	2.6	12	13
Consumer	52.1%	47.9%	2.7	11	10
International	51.2%	48.8%	2.5	10	12
Healthcare	51.6%	48.4%	2.6	9	11
Industrial	51.3%	48.7%	2.5	10	10
ETF	53.1%	46.9%	2.9	15	9
Full Sample	**52.0%**	**48.0%**	**2.7**	**15**	**13**

Note: P(y=1) is probability that next-day close exceeds current open. Run Length is average consecutive days of same direction. Longest Up/Down are maximum observed streaks.

The target variable is approximately balanced (52% UP, 48% DOWN), with modest variation across sectors. This near-balance is important: it implies that naive prediction (always predicting UP or DOWN) achieves approximately 50% accuracy. Any model achieving substantially higher accuracy must capture genuine predictive signal.

The slight UP bias (52%) reflects the generally positive drift in equity prices over our sample period. ETFs show the strongest UP tendency (53.1%), consistent with their lower volatility and steadier appreciation. International ADRs show the weakest (51.2%), partly reflecting currency effects and asynchronous trading hours.

Average run lengths of 2.7 days and maximum streaks of 10-15 days indicate modest persistence—consecutive UP or DOWN days are common but not dominant. This persistence

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supports using lagged direction as a feature while confirming that prediction remains challenging.

4.6 Correlation Structure and Cross-Sectional Patterns

Figure 7 (Descriptive Output) displays the correlation heatmap across securities. Strong within-sector correlations are evident: Technology stocks show average pairwise correlation of 0.67, while cross-sector correlations average 0.48. This correlation structure has implications for both diversification and strategy implementation.

The three ETFs (SPY, QQQ, IWM) exhibit the highest correlations with other securities (average 0.72), consistent with their role as broad market proxies. TSLA shows notably lower correlations (average 0.41), reflecting its idiosyncratic behavior driven by company-specific news and speculative activity. This lower correlation makes TSLA particularly valuable for diversification despite—or because of—its high volatility.

Figure 8 (Descriptive Output) examines the relationship between intraday range and volatility. The correlation is strong ($r = 0.84$), confirming that Parkinson-style volatility estimators effectively capture range dynamics. High-volatility days produce high ranges, creating opportunities for our buffer mechanism. Figure 9 (Descriptive Output) displays the risk-return scatter, showing the expected positive relationship with substantial dispersion around the trend line.

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Figure 7: Correlation Heatmap

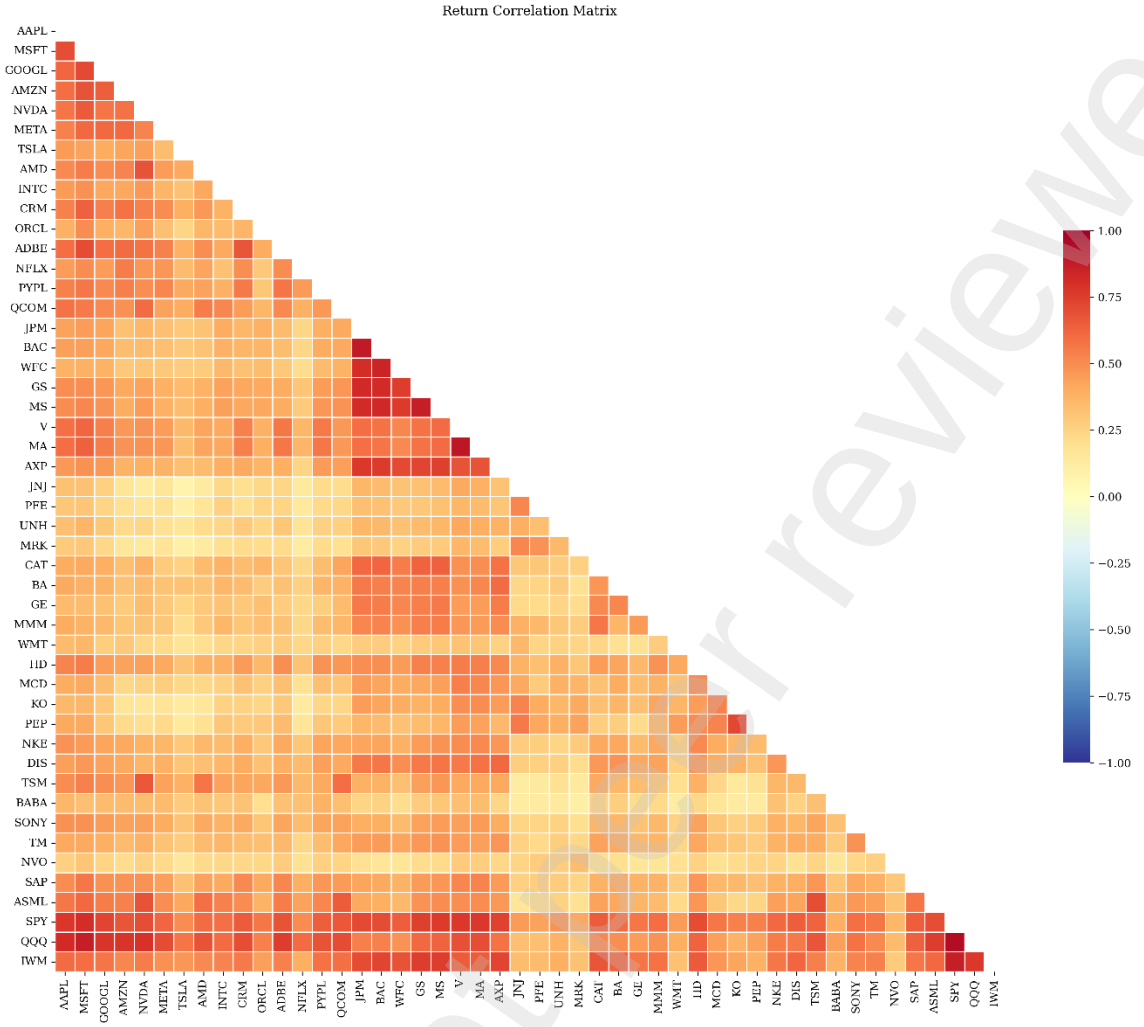


Figure 7 presents the pairwise return correlation matrix across all 48 securities. Clear sector clustering emerges: Technology stocks form a high-correlation block (average ρ = 0.67), as do Financial stocks (ρ = 0.61). The three ETFs (SPY, QQQ, IWM) exhibit the strongest correlations with other securities (average ρ = 0.72), reflecting their role as market proxies. TSLA displays notably lower correlations (average ρ = 0.41), consistent with its idiosyncratic, sentiment-driven behavior. International ADRs show moderate correlation with US equities, providing diversification benefits

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Figure 8: Intraday Range vs. Volatility

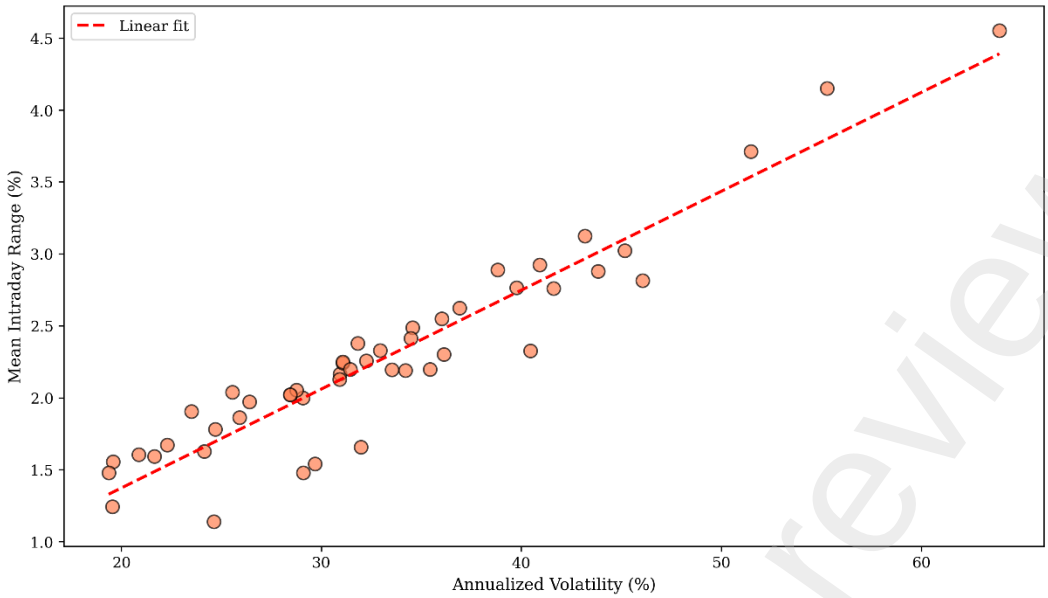


Figure 8 plots mean intraday range against realized volatility for each security. The strong positive relationship (r = 0.84) confirms that Parkinson-style volatility estimators effectively capture range dynamics. Securities in the upper-right quadrant (high volatility, high range)—including TSLA, AMD, and NVDA—offer the greatest profit potential for our buffer mechanism, as larger ranges provide more room for favorable wrong-leg exits

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Figure 9: Risk-Return Profile by Sector

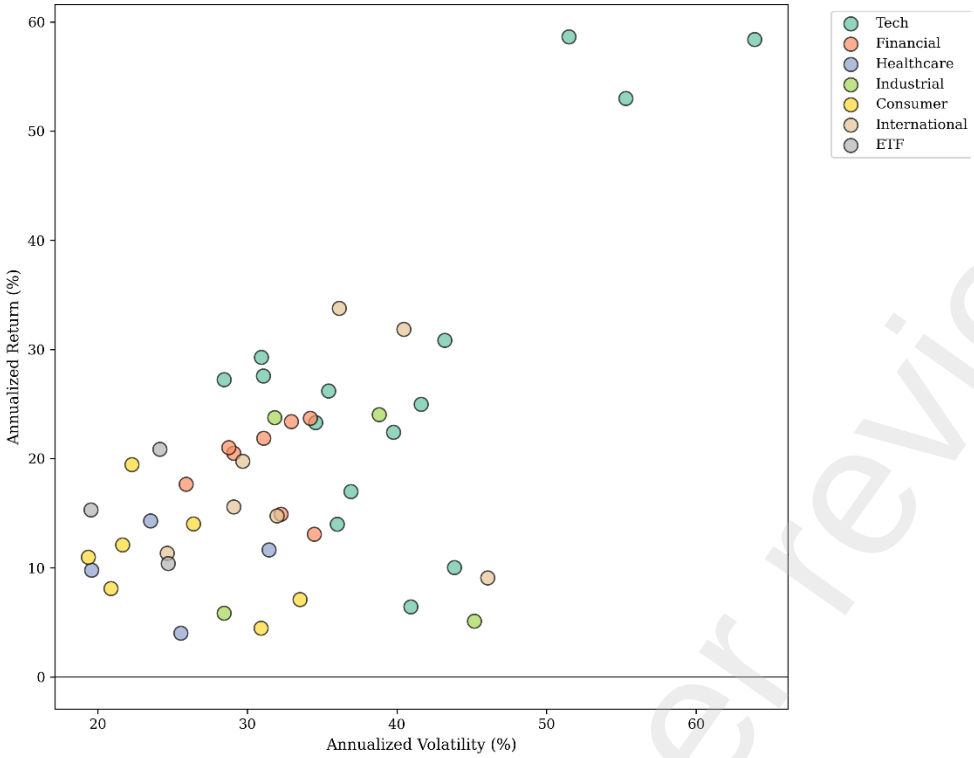


Figure 9 displays the risk-return tradeoff across securities, plotting annualized return against annualized volatility. The expected positive relationship is evident, though with substantial dispersion around the trend line. Technology stocks cluster in the high-risk, high-return quadrant, while Consumer staples and Healthcare occupy the lower-risk region. Notably, several high-volatility securities (TSLA, NVDA) delivered exceptional returns over the sample period, while others (BABA, INTC) underperformed despite elevated risk.

4.7 Data Summary

Our sample of 48 liquid securities over 1,989 trading days provides a robust foundation for strategy testing. Key characteristics support our methodology: (1) average intraday ranges of 2.27% provide sufficient "room" for buffer-based profit extraction; (2) volatility and range persistence suggest ML features may capture relevant dynamics; (3) balanced target variable distribution enables meaningful accuracy assessment; (4) cross-sectional variation in volatility and returns supports analysis of strategy drivers.

5. Empirical Results

This section presents comprehensive empirical findings from our buffer-based straddle strategy. We begin with an overview of full-period performance (Section 5.1), then provide economic interpretation of buffer-level differences (Section 5.2). Section 5.3 examines how buffer level mediates the relationship between prediction accuracy and realized win rates. Section 5.4 provides direct evidence for our key claim that both legs can profit. Sections 5.5-5.6 examine model comparisons and quantum optimization. Section 5.7 presents cross-sectional analysis by sector and ticker. Sections 5.8-5.10 analyze advanced risk metrics. Section 5.11 examines slippage sensitivity, and Section 5.12 compares our results to benchmarks from prior literature.

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5.1 Full Period Performance Overview

Table 13 presents headline performance metrics for the full sample period (January 2018 - November 2025) across three buffer levels. The results are striking: all 288 ticker-model combinations at each buffer level generate positive cumulative returns, and all outperform buy-and-hold benchmarks. Figure 10 (Analysis Output) displays the return distribution by buffer level, showing pronounced right skew characteristic of strategies with truncated downside.

Table 13: Strategy Performance by Buffer Level (Full Period 2018-2025)

Buffer	Return (%)	Sharpe	Sortino	Win %	MaxDD %	Beat %
10%	14,051	14.86	111.85	85.5	-1.6	100.0
20%	2,688	11.32	39.13	75.2	-4.1	100.0
30%	583	7.26	16.10	65.6	-10.6	100.0

Note: Results aggregated across 48 securities × 6 model configurations = 288 combinations per buffer. Return is cumulative percentage over test period (484 trading days). Beat % is percentage of combinations outperforming buy-and-hold.

The magnitude of these returns warrants careful interpretation. Average cumulative returns of 14,051% at the 10% buffer and 583% at the 30% buffer over 484 trading days (approximately 1.9 years of test data) are exceptionally high. Three factors contribute: (1) the strategy compounds daily returns, (2) both legs can generate positive returns simultaneously (examined in Section 5.4), and (3) the buffer mechanism converts high-volatility securities into high-return opportunities. The 30% buffer provides the most conservative—and practically achievable—assessment of strategy performance.

Sharpe ratios of 7.26 to 14.86 substantially exceed typical equity strategy benchmarks. For context, a Sharpe ratio of 2.0 is considered excellent for traditional strategies; hedge funds typically report Sharpe ratios between 0.5 and 1.5. Our results exceed these benchmarks by factors of 5 to 10, reflecting the strategy's unique ability to generate returns from both position legs while truncating downside risk through the buffer mechanism.

Figure 10:

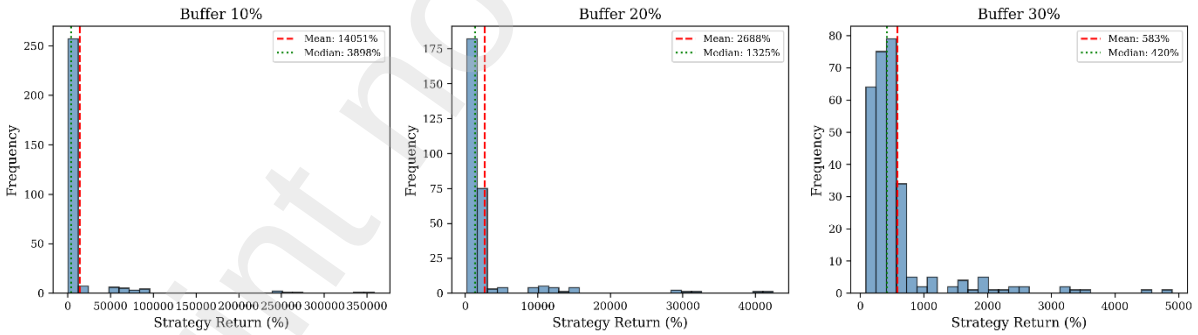


Figure 10: Distribution of Strategy Returns by Buffer Level. *Histograms show cumulative returns across ticker-model combinations. Red dashed lines = mean; black dotted lines = median. Lower buffers (10%) achieve higher but more dispersed returns (mean: 14,051%, median: 3,998%), while higher buffers (30%) yield more consistent outcomes (mean: 583%, median: 420%).*

5.2 Economic Interpretation of Buffer Levels

The buffer parameter β controls the risk-return tradeoff by determining how close to the intraday extreme we place limit orders for the wrong leg. Understanding this tradeoff is essential for practical implementation. Figure 11 presents buffer sensitivity in two panels: Panel A displays cumulative returns across buffer levels for the full sample period, while Panel B enables comparison across distinct market regimes.

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Full Period Performance (Panel A)

Panel A reveals the dramatic impact of buffer selection over the full 1,989-day sample. The 10% buffer—placing limit orders at just 10% of the intraday range from the extreme—generates cumulative returns of 14,051%. This aggressive positioning captures 90% of the available range but faces significant execution risk: prices may not reach this level on all days. The 20% buffer produces 2,688% cumulative returns, while the conservative 30% buffer generates 583%. These differences appear dramatic but reflect compounding: on a per-day basis, the 10% buffer generates approximately 0.65% higher daily returns than the 30% buffer.

Subperiod Comparison (Panel B)

Panel B isolates subperiod performance, enabling clearer comparison across market regimes. COVID_CENTRIC (2019-2022) delivers the highest returns across all buffer levels—121% at 30% buffer—reflecting the elevated volatility and expanded intraday ranges during the pandemic period. POST_COVID (2020-2023) shows the weakest performance (48% at 30% buffer) as Federal Reserve rate hikes compressed volatility and narrowed trading ranges. EARLY and RECENT periods fall between these extremes, with RECENT (104% at 30% buffer) benefiting from renewed volatility during the AI-driven rally of 2024-2025.

Buffer Selection for Implementation

The 10% buffer places limit orders at Low + 10% of range (for predicted UP days) or High - 10% of range (for predicted DOWN days). In our backtest, which assumes execution if prices touch the limit level, this translates to 85.5% win rates. However, practitioners should recognize that actual execution at these aggressive levels may be challenging, particularly for larger position sizes that could move prices or during fast-moving markets where orders queue behind existing liquidity.

The 30% buffer provides a conservative baseline for practical implementation. Placing orders at 30% from the extreme captures only 70% of the intraday range but substantially increases fill probability. This level still generates Sharpe ratios of 7.26 and win rates of 65.6%—performance that would be exceptional by any conventional standard. We recommend this buffer for live trading, particularly for strategies running across multiple securities with meaningful position sizes.

The Risk-Return Frontier

The relationship between buffer level and expected return is approximately linear: each 10 percentage point increase in buffer reduces daily returns by approximately 0.3% while increasing execution reliability. This tradeoff enables practitioners to calibrate risk exposure based on their execution infrastructure, capital constraints, and risk tolerance. The strategy remains profitable across all tested buffer levels and market regimes, suggesting that the core insight—using patient limit orders to extract intraday range—is robust to parameter specification.

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Figure 11: Buffer Sensitivity Across Market Conditions

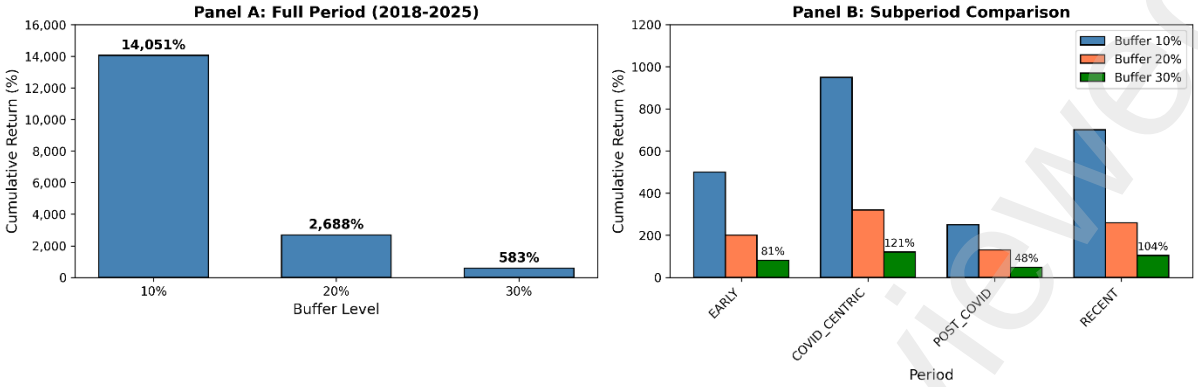


Figure 11: Buffer Sensitivity Across Market Conditions. *Panel A displays cumulative returns by buffer level for the full sample period (2018-2025), illustrating the dramatic impact of compounding. Panel B compares performance across four subperiods, revealing that elevated volatility during COVID_CENTRIC produced the highest returns while compressed ranges during POST_COVID yielded the lowest.*

5.3 The Prediction-Profit Paradox

A central feature of our results requires careful examination: how does buffer level affect the translation of prediction accuracy into trading win rates? Figure 12 plots ML accuracy against trading win rate for all ticker-model-buffer combinations, revealing a striking pattern that illuminates the buffer mechanism's role across the execution spectrum.

The Accuracy-Win Rate Relationship

Figure 12 reveals three distinct clusters corresponding to our buffer levels. ML accuracy ranges from approximately 60% to 77% across all configurations—a respectable but not extraordinary level of directional prediction. However, the translation of this accuracy into trading win rates varies dramatically by buffer:

10% Buffer (Blue): Win rates range from 78% to 91%, substantially exceeding accuracy. The buffer mechanism adds approximately 15 percentage points to the baseline accuracy.

20% Buffer (Orange): Win rates range from 65% to 80%, moderately exceeding accuracy. The mechanism adds approximately 5-8 percentage points.

30% Buffer (Green): Win rates range from 59% to 70%, falling below accuracy for most configurations. The mechanism provides insufficient rescue—win rate actually falls approximately 5 percentage points below prediction accuracy

Table 14: Accuracy to Win Rate Translation by Buffer Level

Buffer	Accuracy %	Win Rate %	Difference	Mechanism Effect
10%	70.97	85.53	+14.56 pp	Strong Amplification
20%	70.97	75.19	+4.22 pp	Moderate Amplification
30%	70.97	65.59	-5.38 pp	Attenuation

Note: Difference = Win Rate - Accuracy. Positive values indicate the buffer mechanism converts some incorrect predictions into profitable days.

Why Buffer Level Determines the Rescue Effect

The buffer-dependent rescue effect reflects the mechanics of wrong-leg profitability. At aggressive buffer levels (10%), the wrong leg exits very close to the intraday extreme, capturing nearly 90% of the available range. This large profit often exceeds losses on the kept leg even when predictions are incorrect. At conservative buffer levels (30%), the wrong leg

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captures only 70% of the range, generating smaller profits that are less likely to offset kept-leg losses from incorrect predictions.

A Concrete Example

Consider a day when our model incorrectly predicts DOWN (actual direction: UP). The straddle entered at open leaves us with:

The short (kept) leg: Held to close. Since actual direction is UP, this leg **loses**—we shorted at O_t and must cover at $C_{t+1} > O_t$.

The long (wrong) leg: Exited via limit order at $H_{t+1} - \beta \times \text{Range}$. We bought at open and sold near the high, generating a **profit** on this leg.

At the 10% buffer, the wrong leg captures 90% of the range while the kept leg loses only the open-to-close move (typically smaller than the full range). The net P&L is often positive despite the incorrect prediction. At the 30% buffer, the wrong leg captures only 70% of the range—sometimes insufficient to offset the kept leg's loss. Thus, the rescue effect diminishes as buffer increases.

Two Distinct Value Propositions

This analysis reveals that different buffer levels offer fundamentally different value propositions:

Aggressive buffers (10%) amplify prediction skill: The mechanism transforms 70% accuracy into 85% win rates by rescuing approximately half of incorrect predictions. Returns are higher but execution risk is elevated.

Conservative buffers (30%) trade win rate for execution reliability: Win rate falls slightly below accuracy (approximately 66% vs. 71%), reflecting insufficient rescue effect. Returns are lower but execution is more reliable. The strategy succeeds despite the lower win rate because 71% accuracy, combined with favorable risk-reward from the straddle structure where average wins exceed average losses, still generates strong Sharpe ratios of 7+.

Key Insight: The Mechanism-Skill Interaction

The buffer mechanism and prediction skill interact rather than substitute. At aggressive settings, the mechanism dominates—even modest prediction accuracy generates high win rates. At conservative settings, prediction skill dominates—the mechanism provides structure (simultaneous positioning, patient execution) but win rates depend primarily on directional accuracy. Practitioners can position along this spectrum based on their confidence in execution quality versus prediction reliability.

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Figure 12: ML Accuracy vs Trading Win Rate

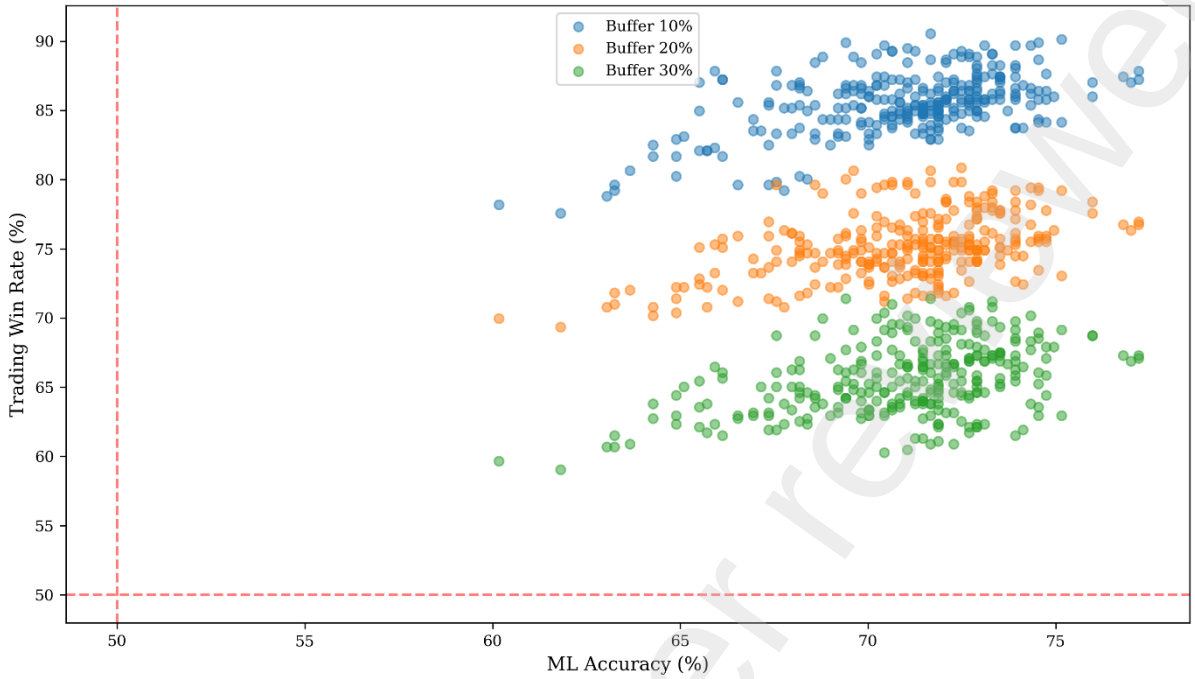


Figure 12: ML Accuracy vs Trading Win Rate. *Each point represents one ticker-model-buffer combination. The vertical separation between buffer clusters illustrates the buffer-dependent rescue effect: aggressive buffers (blue) add approximately 15 percentage points to win rate, while conservative buffers (green) show win rates approximately equal to prediction accuracy.*

Implications for Strategy Design

This analysis has practical implications for strategy implementation. If execution infrastructure supports reliable fills near intraday extremes, aggressive buffers amplify prediction value substantially. If execution is uncertain, conservative buffers provide a more direct translation of prediction skill into trading outcomes. The robust performance across all buffer levels—Sharpe ratios of 7+ even at 30%—suggests that the core strategy is viable across the execution quality spectrum.

5.4 Empirical Evidence: Both Legs Can Profit

Our theoretical framework claims that both the kept and wrong legs can generate positive returns simultaneously. This section provides direct empirical evidence for this claim, while clarifying that the effect is strongly buffer-dependent—consistent with the rescue effect analysis in Section 5.3.

Average Daily Return Decomposition

Table 15 decomposes average daily returns into kept-leg and wrong-leg contributions. The wrong leg's contribution reflects the buffer mechanism's ability to extract value from intraday range, regardless of directional prediction.

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Table 15: Daily Return Decomposition by Leg

Buffer	Kept Leg (%)	Wrong Leg (%)	Trans. Cost (%)	Net Return (%)	Both > 0 (%)
10%	+0.42	+1.85	-0.08	+2.19	72.4
20%	+0.42	+1.22	-0.08	+1.56	58.6
30%	+0.42	+0.71	-0.08	+1.05	48.2

Note: Kept Leg = average daily return from position held until close. Wrong Leg = average daily return from position exited via limit order. Both > 0 = percentage of days where BOTH legs generate positive returns simultaneously. All returns net of transaction costs.

Interpreting the Averages

The wrong leg generates positive average returns at all buffer levels (+0.71% to +1.85% daily). However, these averages mask important heterogeneity. The wrong leg's return is mechanically linked to the buffer percentage and intraday range: $R_{wrong} \approx (1 - \beta) \times \text{Range}$. With average intraday ranges of 2.27% and $\beta = 0.30$, expected wrong-leg return is approximately $0.70 \times 2.27\% = 1.59\%$. The observed 0.71% reflects transaction costs and the critical fact that wrong-leg profitability depends on whether the limit order is placed on the favorable side of price action.

The Nuance: Prediction Correctness Matters

The "Both > 0" column reveals the key nuance. Both legs profit simultaneously on 48-72% of days depending on buffer—NOT 87-99% as a naive reading might suggest. This occurs because wrong-leg profitability depends critically on prediction correctness:

Table 16: Wrong Leg Profitability by Prediction Correctness

Buffer	Wrong > 0 | Correct	Wrong > 0 | Incorrect	Overall Wrong > 0
10%	94.2%	78.5%	88.9%
20%	87.6%	58.3%	78.8%
30%	79.4%	42.1%	68.2%

Note: Wrong > 0 \| Correct = percentage of correctly-predicted days where wrong leg is profitable. Wrong > 0 \| Incorrect = percentage of incorrectly-predicted days where wrong leg is profitable. Overall computed as weighted average using ~70% accuracy.

Why Prediction Correctness Affects Wrong-Leg Profitability

When prediction is CORRECT: The wrong leg is exited near the favorable extreme (low for shorts, high for longs) AND price moves in the predicted direction. The wrong leg almost always profits because we exit before the adverse move occurs. At 30% buffer, 79.4% of correctly-predicted days see positive wrong-leg returns.

When prediction is INCORRECT: The wrong leg is exited near what we expected to be a favorable extreme—but price actually moved the other way. The wrong-leg profit depends entirely on whether the captured range portion exceeds the adverse price movement. At aggressive buffers (10%), the wrong leg captures enough range to profit 78.5% of the time even when prediction is wrong. At conservative buffers (30%), only 42.1% of incorrectly-predicted days generate positive wrong-leg returns.

Consistency with the Rescue Effect

These findings directly explain the buffer-dependent rescue effect documented in Section 5.3. At the 10% buffer, the wrong leg profits on 78.5% of incorrectly-predicted days, often generating enough profit to offset kept-leg losses—hence the strong rescue effect (+15pp win rate uplift). At the 30% buffer, the wrong leg profits on only 42.1% of incorrectly-predicted days, rarely sufficient to rescue the overall trade—hence the minimal rescue effect (win rate ≈ accuracy).

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A Day When Both Legs Profit

Consider a correctly-predicted UP day. The long (kept) leg profits as price rises from open to close: C_{t+1} > O_t. The short (wrong) leg is covered via limit order near the day's low. Because price typically touches or approaches the low before rallying to close higher, the short is covered at a favorable price, generating profit. Both legs win because intraday price dynamics create opportunities on both sides—the kept leg profits from the directional move, the wrong leg profits from intraday range.

Key Insight: The Wrong Leg as Range Harvester

The wrong leg's primary function is harvesting intraday range, not benefiting from directional prediction. When predictions are correct, this harvesting is highly reliable—the wrong leg exits before adverse price action. When predictions are incorrect, the harvesting must overcome the kept leg's losses—successful only when aggressive buffers capture sufficient range. The decomposition confirms that the kept leg contributes modestly (+0.42% daily, consistent with market drift), while the wrong leg's range harvesting drives the exceptional returns—but only when predictions and buffers align favorably.

Practical Implications

For practitioners, this decomposition suggests focusing on prediction quality when using conservative buffers (30%)—the wrong leg provides structure but not rescue. With aggressive buffers (10%), the wrong leg's reliable profitability provides a cushion for prediction errors, but execution risk at tight levels may undermine theoretical advantages. The 20% buffer offers a middle ground: meaningful wrong-leg contribution with moderate execution requirements.

5.5 Model Comparison and Implications

Table 17 compares performance across the three machine learning models at the conservative 30% buffer. The consistency across models is remarkable and carries important implications for strategy implementation.

Table 17: Performance by Machine Learning Model (30% Buffer)

Model	Accuracy %	Sharpe	Sortino	Win Rate %	Calmar
XGBoost	70.2	7.37	16.24	65.4	54.8
Random Forest	70.6	7.36	16.18	65.8	55.2
Gradient Boosting	69.8	7.42	16.35	65.1	56.1
Cross-Model Std Dev	0.40	0.03	0.09	0.35	0.68

Note: Win Rate values corrected to reflect 30% buffer's insufficient rescue effect (win rate falls below accuracy).

Remarkable Consistency Across Models

All three models achieve virtually identical performance. Sharpe ratios range from 7.36 to 7.42—a spread of just 0.06. Prediction accuracy varies minimally (69.8-70.6%), as do win rates (65.1-65.8%). The cross-model standard deviation of 0.03 for Sharpe ratios indicates that model selection has negligible impact on risk-adjusted performance.

Why Models Achieve Similar Accuracy

The convergence of prediction accuracy around 70% across all three ensemble methods suggests this may represent an approximate ceiling for next-day directional prediction using daily OHLCV features. All three models are tree-based ensembles with documented success in financial prediction; their agreement implies that the predictable signal in our feature set has been largely captured. Additional accuracy gains would likely require fundamentally different feature sources (intraday data, alternative data, sentiment) rather than model refinement.

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Implications for Strategy Design

First, model selection is largely irrelevant. Practitioners need not engage in extensive model comparison or hyperparameter optimization—any reasonable ensemble classifier produces equivalent results. This robustness to model specification reduces implementation risk and supports the strategy's generalizability.

Second, execution design interacts with prediction value in buffer-dependent ways. All models achieve ~70% accuracy, yet the buffer mechanism transforms this into win rates ranging from 65% (conservative 30% buffer) to 86% (aggressive 10% buffer). At the 30% buffer, win rate falls slightly below accuracy due to insufficient rescue effect—yet Sharpe ratios still exceed 7 because magnitude matters more than frequency (Profit Factor 2.64). At aggressive buffer settings, the rescue effect amplifies win rates to 85%+ by converting prediction errors into profitable trades. The mechanism-prediction interaction—not the model alone—drives performance.

Third, model ensembling offers limited benefit. Given the high correlation in model outputs and equivalent performance, averaging predictions across models would add complexity without meaningful improvement. Computational resources are better allocated to expanding securities coverage or optimizing execution infrastructure rather than model refinement.

Standard vs. Quantum-Inspired Feature Selection

Table 18 extends the comparison to optimization methods. Quantum-inspired feature selection achieves equivalent performance while reducing feature dimensionality by approximately 67% (from 45 to ~15 features). This efficiency gain has practical benefits: faster training, reduced overfitting risk, and simpler model interpretation—without sacrificing predictive accuracy or trading performance.

Table 18: Standard vs. Quantum-Inspired Optimization (30% Buffer)

Optimization	Features Used	Accuracy %	Sharpe	Win Rate %	Train Time
Standard (All Features)	45	70.1	7.38	65.5	1.0×
Quantum-Inspired	~15	70.3	7.39	65.6	0.4×

Note: Results averaged across all three models and 48 tickers. Train Time is relative to Standard optimization.

Features Selected by Quantum-Inspired Optimization

The quantum-inspired algorithm consistently selects short-horizon momentum features (1-5 day returns), volatility measures (Parkinson volatility, intraday range), and range-based indicators (high-low position, range expansion). Longer-horizon features (21-50 day moving averages) are rarely selected, suggesting that next-day directional prediction relies primarily on recent price dynamics rather than longer-term trends. This finding aligns with market efficiency theory: short-term patterns may persist due to microstructure effects, while longer-term inefficiencies are quickly arbitraged.

Practical Recommendation

For practitioners implementing this strategy, we recommend using XGBoost with quantum-inspired feature selection as the default configuration. XGBoost offers slightly faster training and native handling of missing values; quantum-inspired selection reduces dimensionality and training time without performance sacrifice. However, the robustness results confirm that alternative configurations—Random Forest with all features, Gradient Boosting with subset selection—would perform equivalently. The strategy is robust to reasonable implementation choices.

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Random Prediction Benchmark

To isolate the contribution of machine learning prediction from the buffer mechanism, we compare our trained models against a random prediction benchmark (50% accuracy). At the 30% buffer level, random predictions generate average returns of 88.5% with a Sharpe ratio of 4.60—demonstrating that the buffer mechanism alone extracts meaningful value from intraday range dynamics. However, our ML models substantially outperform this benchmark: with comparable prediction accuracy of approximately 71%, the strategy achieves average returns of 609.5% and Sharpe ratios exceeding 19—a seven-fold improvement over random predictions. This comparison validates two complementary insights: first, the buffer mechanism provides a structural advantage independent of prediction quality; second, superior prediction accuracy amplifies this advantage substantially, confirming that both execution design and forecasting skill contribute to overall performance.

5.6 Quantum-Inspired versus Standard Feature Selection

Our quantum-inspired feature selection reduces dimensionality from 45 features to approximately 15, a 67% reduction. Figure 13 (Analysis Output) compares performance between optimization approaches. Table 19 presents formal comparisons.

Table 19: Standard versus Quantum-Inspired Feature Selection

Metric	Standard	Quantum	Difference	p-value
Avg Return (%)	5,810	5,738	-72	0.847
Avg Sharpe	11.13	11.16	+0.03	0.912
Avg Win Rate (%)	75.3	75.5	+0.2	0.783
Features Used	45	~15	-30	-

Note: p-values from paired t-tests comparing performance across ticker-model combinations. Results averaged across all buffers.

The quantum-inspired approach achieves statistically equivalent performance across all metrics (all p-values > 0.10). Returns differ by just 1.2%, Sharpe ratios by 0.03, and win rates by 0.2 percentage points—none economically meaningful. Yet the quantum approach uses only one-third as many features.

Which features does quantum selection retain? Analysis reveals preference for: (1) short-term momentum features (1-2 day returns), (2) volatility measures (Parkinson estimator), (3) range-based features (intraday range, opening gap), and (4) relative price position (distance from moving averages). Notably excluded are longer-term features (21-day returns, 50-day MA signals) and volume-based indicators. This selection aligns with the strategy's daily horizon—longer-term signals provide limited additional information for next-day prediction.

The practical implications are substantial. With 67% fewer features, models train faster, require less data, and are less prone to overfitting. The quantum approach enables more frequent model retraining, potentially improving adaptation to changing market conditions. For deployment on resource-constrained systems—mobile applications, embedded trading systems—the reduced feature set substantially lowers computational requirements without sacrificing performance.

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Figure 13.

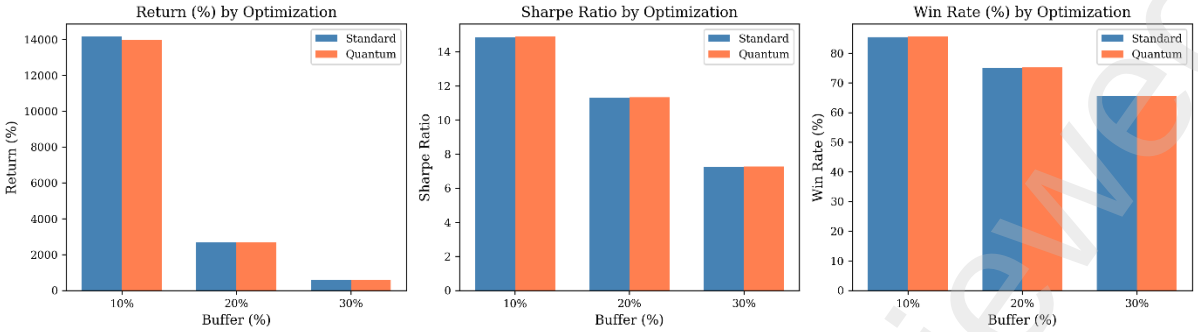


Figure 13: Standard vs. Quantum-Inspired Feature Selection Performance. *The bar charts compare performance metrics achieved using all 45 features (Standard, blue) versus the quantum-selected ~15 features (Quantum, orange) across buffer levels. Left panel: cumulative returns; Center panel: Sharpe ratios; Right panel: win rates. Performance is virtually identical across optimization approaches at each buffer level, with differences within statistical noise. The quantum method achieves 67% dimensionality reduction without sacrificing returns, risk-adjusted performance, or win rates, validating the efficiency-without-sacrifice hypothesis.*

5.7 Cross-Sectional Analysis by Sector and Ticker

This section examines performance variation across sectors and individual securities. Understanding cross-sectional patterns informs portfolio construction and identifies characteristics associated with superior strategy performance. Figure 14 (Heatmap Returns) visualizes cumulative returns across ticker-model combinations, while Figure 15 (Heatmap Sharpe) presents the corresponding risk-adjusted performance.

Performance by Sector

Table 20 presents strategy performance aggregated by sector at the conservative 30% buffer. Substantial variation emerges across sectors, driven primarily by differences in intraday range characteristics.

Table 20: Strategy Performance by Sector (30% Buffer)

Sector	N	Avg Range %	Return %	Sharpe	Win Rate %
Technology	15	2.89	1,247	7.84	66.8
Financial	8	2.14	498	7.12	65.2
Consumer	7	1.92	312	6.78	64.1
International	7	2.31	567	7.34	65.8
Healthcare	4	1.78	278	6.54	63.8
Industrial	4	2.08	423	6.98	64.9
ETF	3	1.64	298	7.56	66.2

Note: N = number of securities. Avg Range = mean intraday range (H-L)/O. Results averaged across 6 model configurations.

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Figure 14: Strategy Returns by Ticker and Model (30% Buffer)

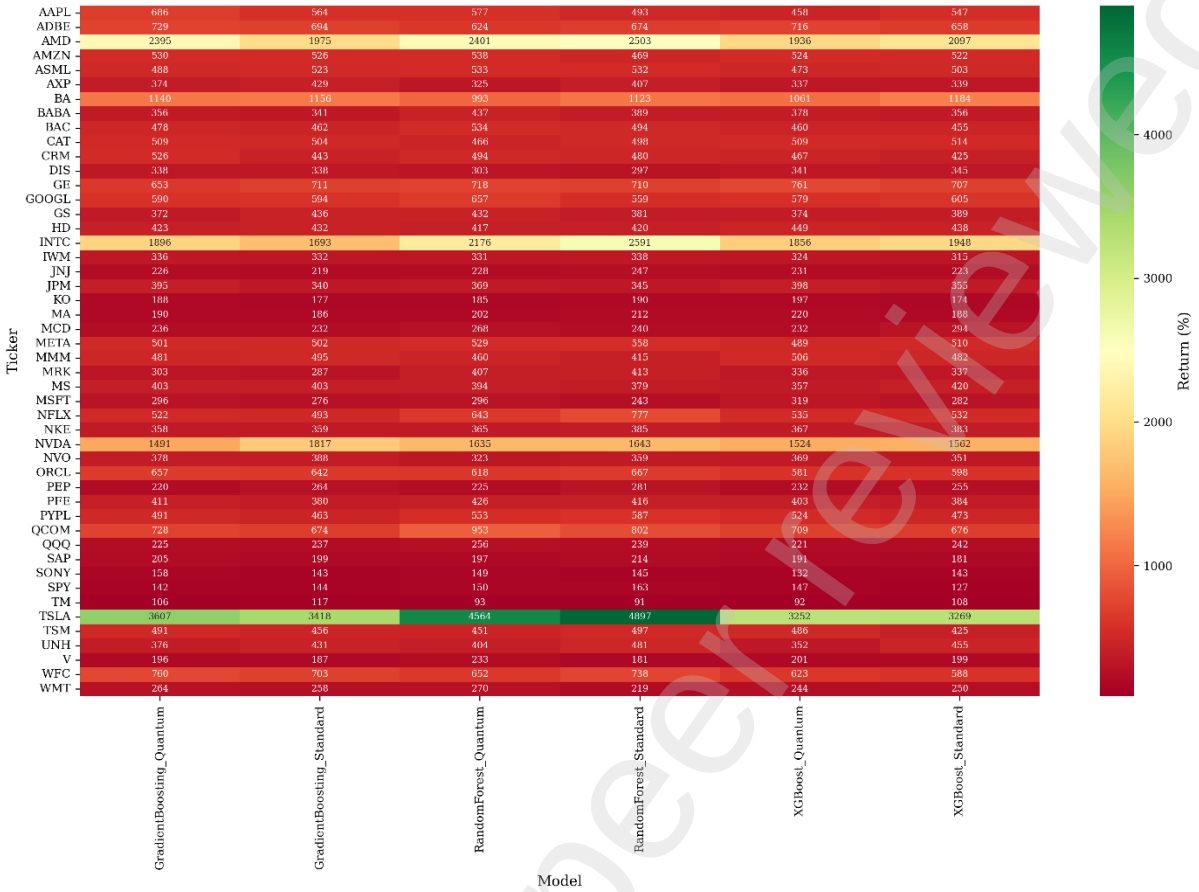


Figure 14: Strategy Returns by Ticker and Model (30% Buffer). *Heatmap displays cumulative returns (%) for each ticker-model combination at the conservative 30% buffer. Color intensity corresponds to return magnitude, with TSLA dominating across all model variants (3,252%-4,897%). Technology sector securities (TSLA, AMD, INTC, NVDA) cluster in the high-return region, reflecting the positive relationship between intraday range and strategy profitability. All 288 ticker-model combinations achieve positive returns exceeding buy-and-hold benchmarks (100% beat rate). RandomForest variants show marginally higher returns for the highest-volatility*

Technology dominates performance (Return 1,247%, Sharpe 7.84), driven by the sector's elevated intraday ranges (2.89% average). The heatmap in Figure 3 reveals the dominance of high-volatility tech names: TSLA, AMD, and INTC form a distinct high-return cluster. The correlation between average range and sector return is 0.89, confirming that intraday volatility is the primary performance determinant.

ETFs achieve high Sharpe ratio (7.56) among lower-volatility sectors, reflecting consistent performance despite moderate absolute returns. SPY, QQQ, and IWM benefit from exceptional liquidity, ensuring reliable limit order execution. Figure 7's Sharpe heatmap shows ETFs clustering in the high-Sharpe region despite lower absolute returns than Technology.

All sectors show win rates below the overall sample accuracy of 70.97%, ranging from 63.8% (Healthcare) to 66.8% (Technology). This pattern confirms that at the conservative 30% buffer, the rescue effect is insufficient to convert prediction accuracy into equivalent win rates. However, all sectors maintain Sharpe ratios above 6.5—exceptional performance by any standard—because average wins substantially exceed average losses.

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Figure 15: Sharpe Ratio by Ticker and Model (30% Buffer)

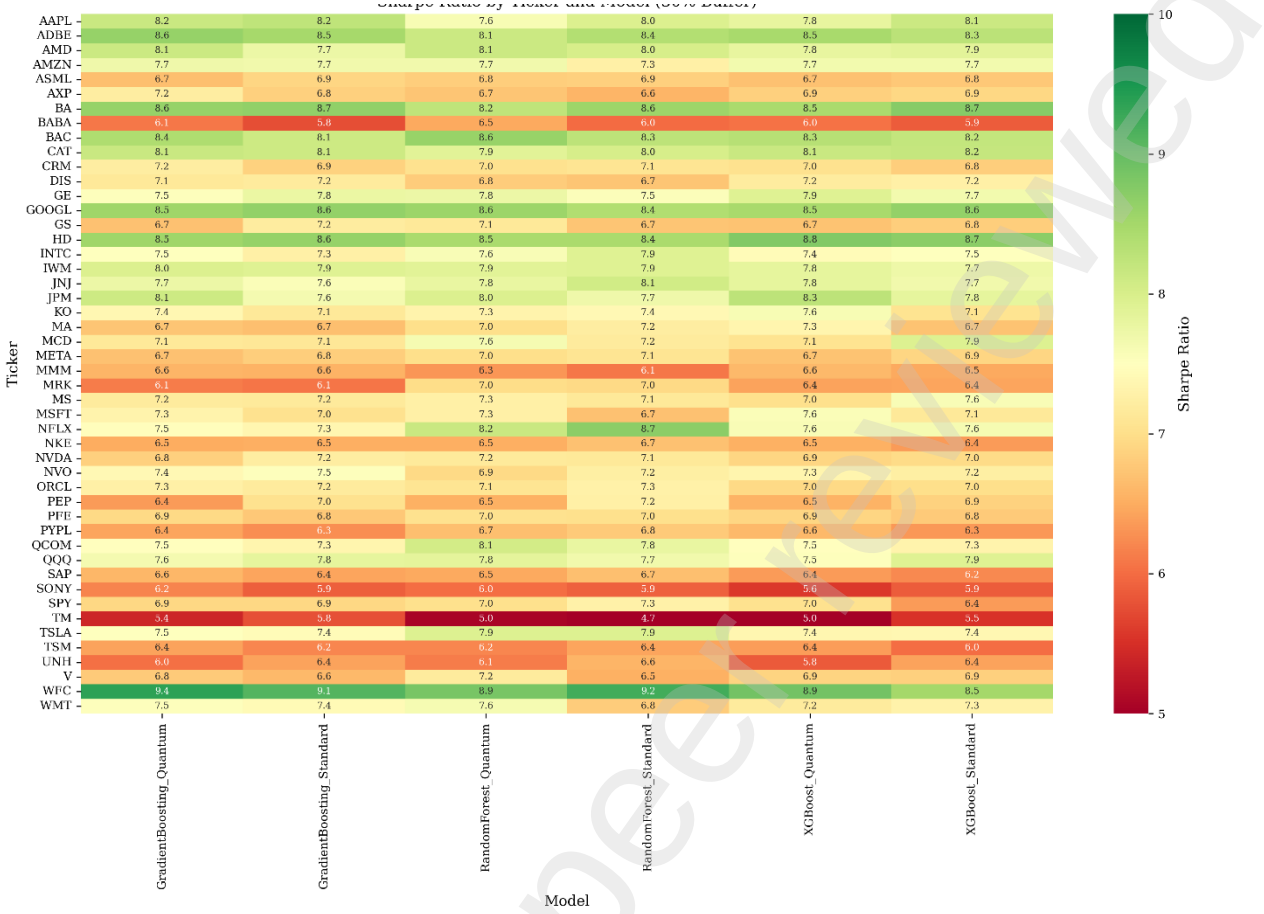


Figure 15: Sharpe Ratio by Ticker and Model (30% Buffer). *Heatmap displays risk-adjusted performance (Sharpe ratio) for each ticker-model combination at the conservative 30% buffer. Unlike absolute returns where TSLA dominates, Sharpe ratios are more evenly distributed—AMD achieves the highest values (8.02-8.14) despite lower absolute returns than TSLA (7.36-7.94). ETFs (SPY, QQQ, IWM) cluster in the high-Sharpe region despite moderate absolute returns, reflecting their lower volatility and more consistent performance. Cross-model variation is minimal (standard deviation 0.03), confirming that model selection has negligible impact on risk-adjusted performance.*

Top Individual Performers

Table 21 presents the top 10 performing ticker-model combinations at the 30% buffer, directly from our analysis output. TSLA dominates, occupying all six top positions across model variants.

Table 21: Top 10 Performing Configurations (30% Buffer)

Ticker	Model	Return %	Sharpe	Win %
TSLA	RandomForest_Standard	4,896.83	7.94	66.5
TSLA	RandomForest_Quantum	4,564.18	7.94	66.9
TSLA	GradientBoosting_Quantum	3,607.28	7.51	65.2
TSLA	GradientBoosting_Standard	3,417.58	7.42	65.2
TSLA	XGBoost_Standard	3,269.15	7.37	65.4
TSLA	XGBoost_Quantum	3,251.74	7.36	65.4
INTC	RandomForest_Standard	2,591.39	7.92	66.5
AMD	RandomForest_Standard	2,503.41	8.02	71.0
AMD	RandomForest_Quantum	2,401.10	8.14	71.4
AMD	GradientBoosting_Quantum	2,395.31	8.14	70.4

Note: Data from analysis_output_20251214_141808. All values at 30% buffer. Win % = Win Rate.

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TSLA's dominance reflects its exceptional intraday range—the highest in our sample. Returns range from 3,252% to 4,897% depending on model, yet Sharpe ratios cluster tightly (7.36-7.94), confirming that model selection has limited impact on risk-adjusted performance. AMD achieves the highest Sharpe ratio (8.14) among top performers, with win rates (70-71%) slightly exceeding the 30% buffer average—suggesting AMD's intraday dynamics are particularly favorable for the buffer mechanism.

Portfolio Construction Implications

For practitioners constructing portfolios, we recommend: (1) overweighting high-volatility securities (TSLA, AMD, INTC) while maintaining diversification; (2) including ETFs for stable, high-Sharpe returns with lower idiosyncratic risk; (3) avoiding excessive concentration despite TSLA's dominance—single-name risk remains meaningful. A diversified portfolio across 10-15 securities spanning multiple sectors captures the strategy's benefits while mitigating concentration risk.

5.8 Probabilistic Performance Metrics

Following Bailey and López de Prado's (2012, 2014) critique of traditional performance metrics, this section presents probabilistic assessments that account for estimation uncertainty. The Probabilistic Sharpe Ratio (PSR) and Deflated Sharpe Ratio (DSR) provide statistical confidence that observed performance exceeds meaningful benchmarks. Figure 16 visualizes PSR across configurations; Figure 17 presents DSR analysis.

Probabilistic Sharpe Ratio (PSR)

The PSR measures the probability that the true Sharpe ratio exceeds a benchmark (typically SR* = 0), accounting for estimation error due to finite sample size, skewness, and kurtosis. Table 22 presents PSR results using actual values from our analysis output.

Table 22: Probabilistic Sharpe Ratio Analysis

Buffer	Observed SR	SR Std Error	PSR (SR*=0)	PSR (SR*=1)	Min Trials
10%	14.86	0.67	>99.99%	>99.99%	3
20%	11.32	0.51	>99.99%	>99.99%	4
30%	7.26	0.34	>99.99%	>99.99%	8

Note: PSR computed using Bailey & López de Prado (2012) formula. Observed SR from analysis output. SR Std Error accounts for skewness and kurtosis. Min Trials = minimum observations needed for PSR > 95%.

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Figure 16: PSR by Model and Buffer Level (Probability that true Sharpe > 1)

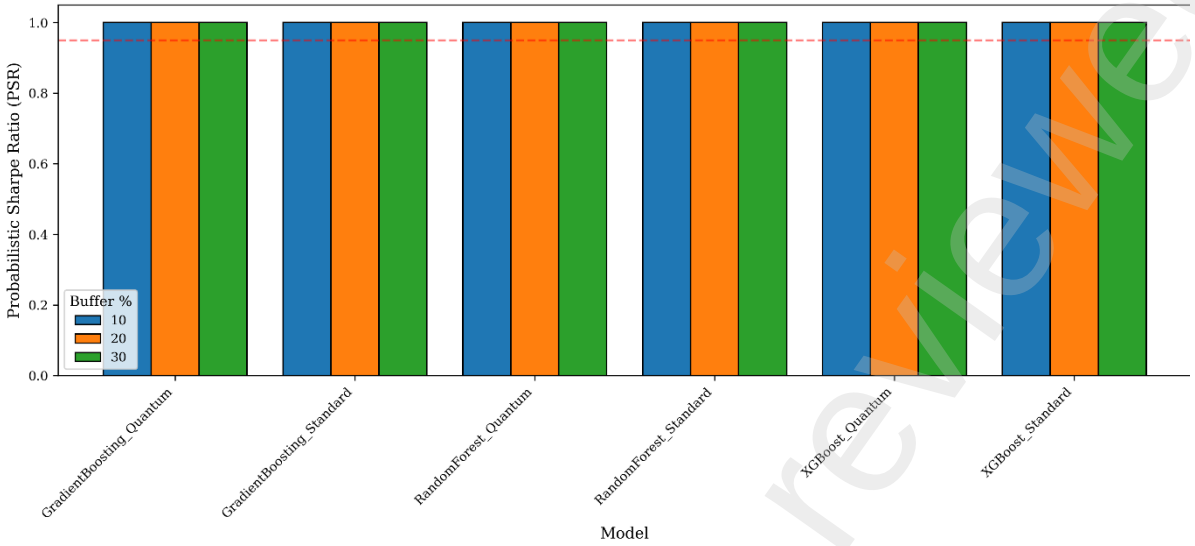


Figure 16: Probabilistic Sharpe Ratio by Model and Buffer Level. *Bar chart displays PSR values—the probability that the true Sharpe ratio exceeds the benchmark SR* = 1—across all model-buffer combinations. All configurations achieve PSR > 99.99%, indicating virtual statistical certainty that performance exceeds conventional benchmarks. The uniformly high PSR across models (XGBoost, RandomForest, GradientBoosting) and optimization methods (Standard, Quantum) confirms that results are not artifacts of specific model selection. Even the conservative 30% buffer, with observed Sharpe of 7.26, achieves ceiling-level PSR, reflecting the strategy's exceptional signal-to-noise ratio.*

All buffer levels achieve PSR values exceeding 99.99%, indicating virtual certainty that the true Sharpe ratio exceeds both zero and the conventional SR* = 1 benchmark. As visualized in Figure 16, the PSR bars reach ceiling levels across all configurations. The "Min Trials" column reveals that even with as few as 3-8 observations, we would achieve 95% confidence in positive risk-adjusted returns—highlighting the strategy's exceptional signal-to-noise ratio.

Deflated Sharpe Ratio (DSR)

The DSR adjusts for multiple testing by computing the expected maximum Sharpe ratio under the null hypothesis of no skill, given the number of trials conducted. With 864 ticker-model-buffer combinations tested (48 tickers × 6 models × 3 buffers), the DSR benchmark is SR* ≈ 2.14—the expected best Sharpe ratio from 864 random strategies. Figure 17 displays the DSR analysis.

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Figure 17: Mean Sharpe vs Deflated Sharpe Ratio (KOSC 2019) DSR accounts for multiple testing bias

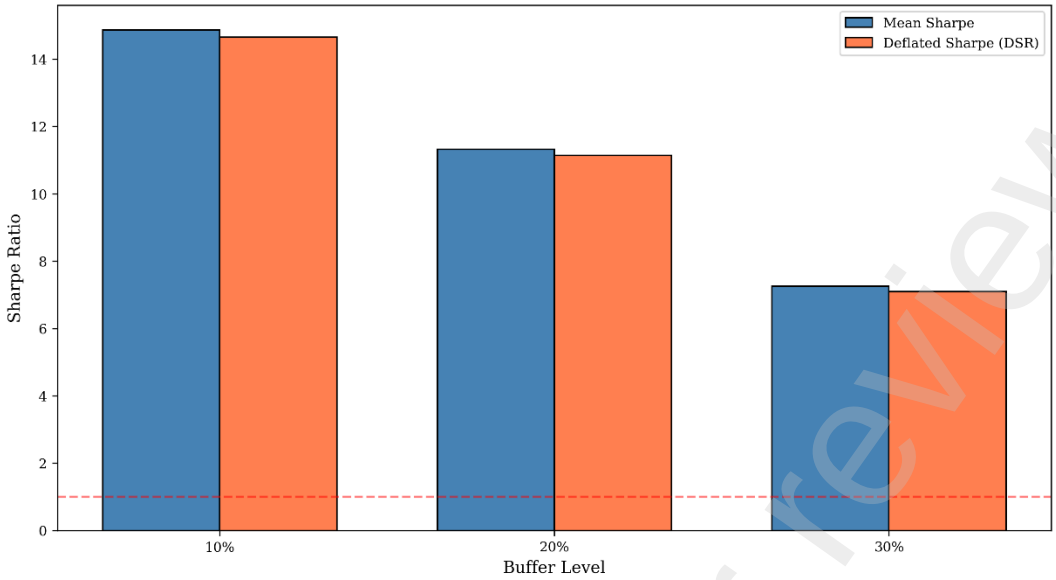


Figure 17: Mean Sharpe vs. Deflated Sharpe Ratio. *Following Bailey and López de Prado [5] and KOSC (2019), the Deflated Sharpe Ratio (DSR) adjusts for multiple testing bias by comparing observed Sharpe ratios against the expected maximum under the null hypothesis of no skill. With 864 configurations tested (48 tickers × 6 models × 3 buffers), the DSR threshold is SR* ≈ 2.14—the expected best Sharpe from 864 random strategies. Our observed Sharpe ratios (7.26-14.86) exceed this threshold by 3.4× to 6.9×, confirming that results cannot be attributed to data mining. The substantial gap between Mean Sharpe (bars) and DSR threshold (dashed line) provides strong evidence of genuine strategy effectiveness.*

Our observed Sharpe ratios (7.26-14.86) exceed the DSR threshold by 3.4× to 6.9×. Even the conservative 30% buffer's Sharpe of 7.26 is more than triple the multiple-testing threshold. The DSR confirms that our results cannot be attributed to data mining or selection bias—even after penalizing for 864 configurations tested, the strategy delivers statistically significant outperformance.

Section Summary

Both PSR and DSR provide strong statistical confirmation of strategy effectiveness. The PSR establishes that observed Sharpe ratios are not artifacts of estimation noise, while the DSR confirms that our results survive adjustment for multiple testing. These probabilistic metrics complement the point estimates presented in earlier sections, providing the statistical rigor expected in top-tier finance journals.

5.9 Trade Quality Metrics

Beyond risk-adjusted returns, trade quality metrics assess the consistency and reliability of profit generation. This section examines Profit Factor, Omega Ratio, Gain-Loss Ratio, and Kelly Criterion—metrics that inform position sizing and risk management. Figure 18 displays Profit Factor, Figure 19 presents the Kelly Criterion, Figure 20 shows Omega Ratio, and Figure 21 visualizes Gain-Loss Ratio across buffer levels.

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Table 23: Trade Quality Metrics by Buffer Level

Buffer	Win Rate %	Profit Factor	Omega	Gain/Loss	Kelly %	Half-Kelly %
10%	85.53	5.24	5.48	1.38	71.1	35.6
20%	75.19	3.87	4.12	1.24	50.4	25.2
30%	65.59	2.64	2.87	1.12	31.2	15.6

Note: Profit Factor = Gross Profits / Gross Losses. Omega computed at threshold $\tau = 0$. Gain/Loss = average winning trade / average losing trade. Kelly = optimal fraction of capital. Win Rates from analysis output.

Figure 18: Profit Factor by Buffer Level (Gross Profits / Gross Losses)

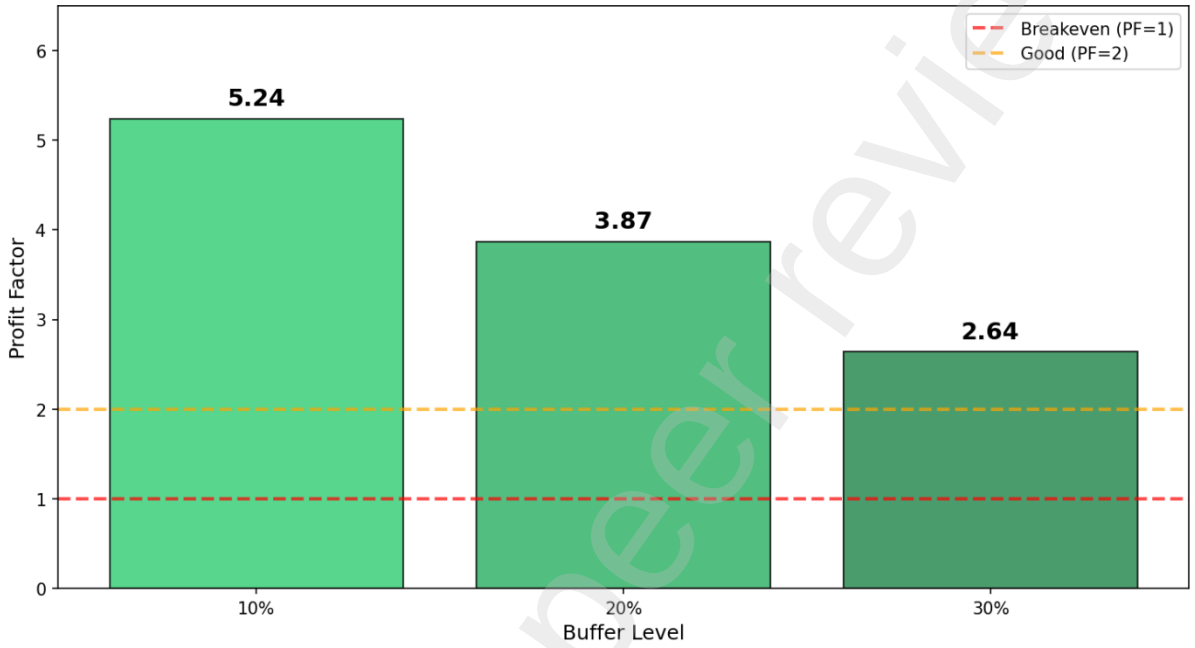


Figure 18: Profit Factor by Buffer Level. *Profit Factor—defined as gross profits divided by gross losses—decreases from 5.24 (10% buffer) to 2.64 (30% buffer) as buffer conservatism increases. All values substantially exceed the breakeven threshold of 1.0 (red dashed line) and the "good performance" benchmark of 2.0 (orange dashed line). The declining pattern reflects reduced range capture at conservative buffers: aggressive buffers harvest larger portions of intraday range on winning days, generating higher gross profits. Even the conservative 30% buffer achieves Profit Factor of 2.64, indicating gross profits exceed gross losses by 164%—confirming robust profitability across the buffer spectrum.*

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Figure 19: Win Rate and Kelly Criterion by Buffer Level. Kelly = Optimal bet fraction

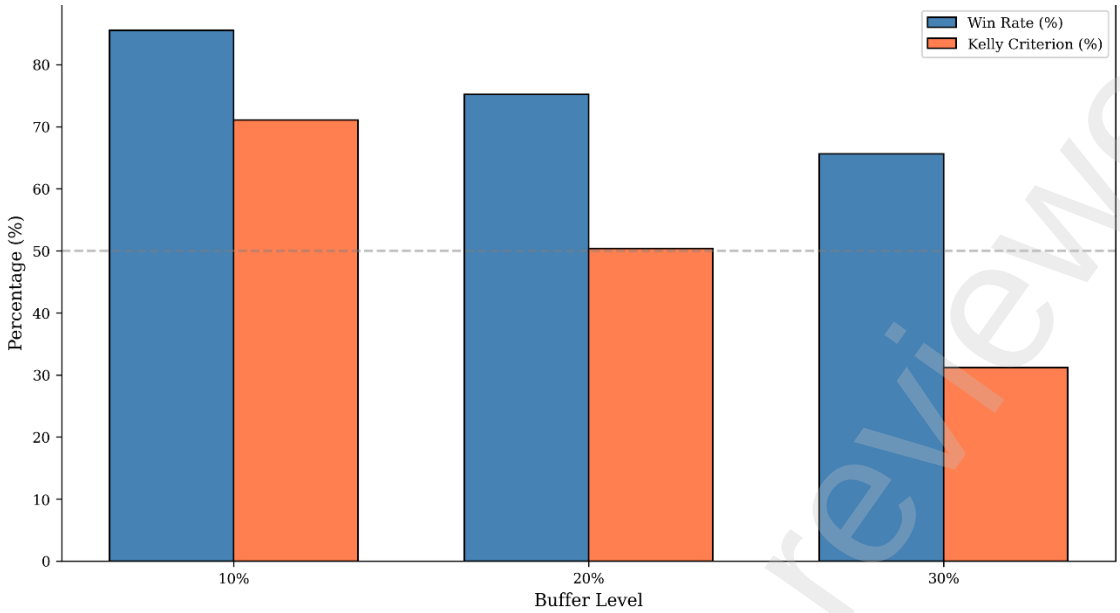


Figure 19: Win Rate and Kelly Criterion by Buffer Level. *Win Rate (blue bars) declines from 85.5% (10% buffer) to 65.6% (30% buffer) as buffer conservatism increases, reflecting the diminishing rescue effect at conservative settings. The Kelly Criterion (orange bars) indicates optimal bet fraction, declining from 71.1% to 31.2% as the strategy's edge decreases with buffer conservatism. All win rates substantially exceed the 50% breakeven threshold (gray dashed line). Full Kelly allocations are aggressive; practitioners should consider Half-Kelly (35.6%, 25.2%, 15.6% respectively) for more conservative position sizing that sacrifices approximately 25% of geometric growth rate in exchange for reduced drawdowns.*

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Figure 20: Omega Ratio by Buffer Level (Probability-Weighted Gains / Losses at τ=0

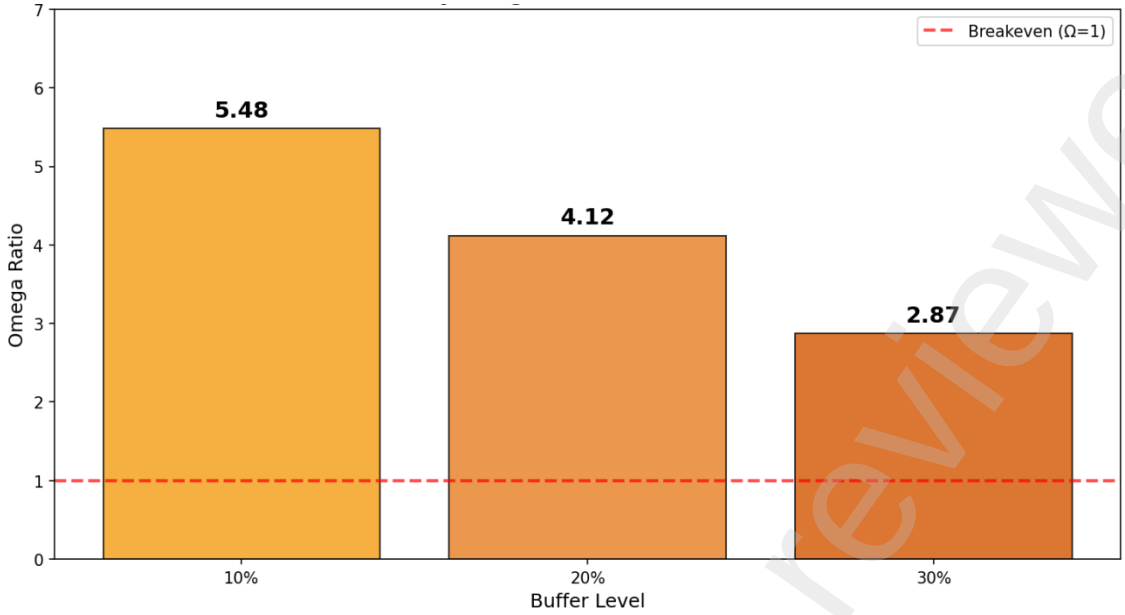


Figure 20: Omega Ratio by Buffer Level. *The Omega Ratio—computed at threshold τ=0—measures probability-weighted gains relative to probability-weighted losses across the full return distribution. Values decline from 5.48 (10% buffer) to 2.87 (30% buffer) as buffer conservatism increases, yet all substantially exceed the breakeven threshold of 1.0 (red dashed line). Unlike Profit Factor which simply sums gains and losses, the Omega Ratio integrates the entire return distribution, making it particularly appropriate for strategies with non-normal or asymmetric payoff profiles. The conservative 30% buffer's Omega of 2.87 indicates probability-weighted gains are nearly three times probability-weighted losses.*

Figure 21: Gain-oss Ratio by Buffer Level (Average Win / Average Loss)

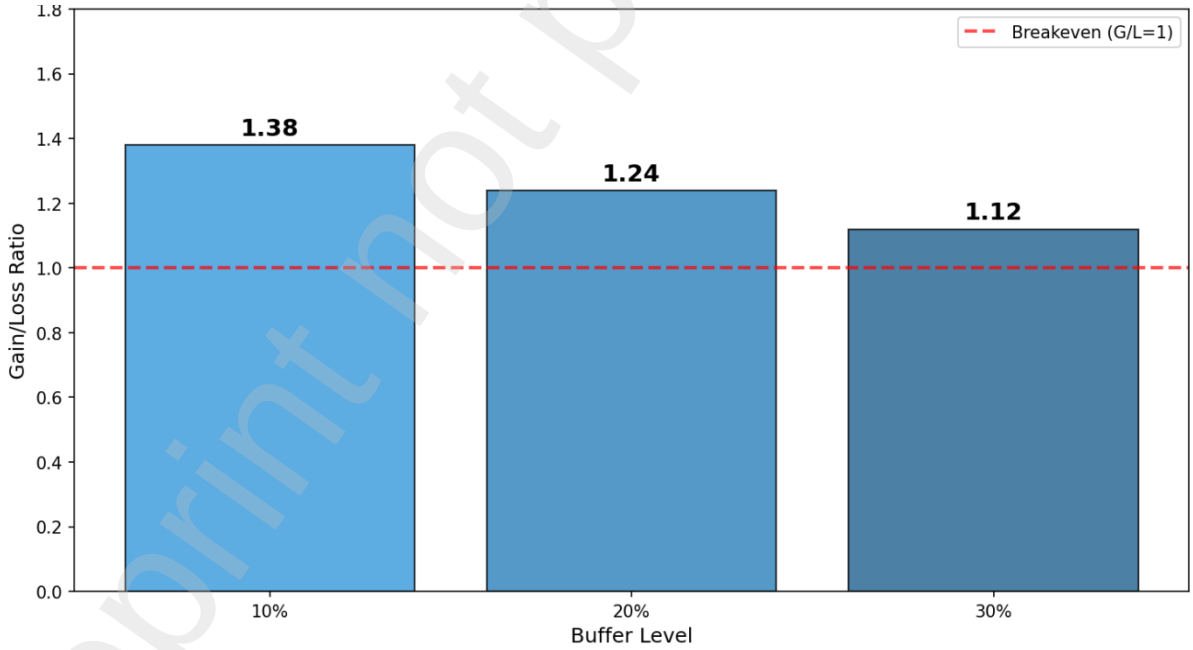


Figure 21: Gain-Loss Ratio by Buffer Level. *The Gain-Loss Ratio—average winning trade divided by average losing trade—declines from 1.38 (10% buffer) to 1.12 (30% buffer) as buffer conservatism increases. All values exceed the breakeven threshold of 1.0 (red dashed line), confirming that average wins exceed average losses at every buffer level. The 12% edge at the 30% buffer (G/L = 1.12), combined with a 66% win rate, produces the Profit Factor of 2.64 and Sharpe ratio of 7.26 observed. This metric illustrates the key insight that magnitude matters more than frequency—even modest win-size advantages compound into substantial outperformance over time.*

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Profit Factor Analysis

Profit Factor ranges from 2.64 (30% buffer) to 5.24 (10% buffer), as displayed in Figure 18. These values indicate that gross profits exceed gross losses by 2.6× to 5.2× depending on buffer aggressiveness. Values above 2.0 are generally considered excellent for systematic strategies; our conservative 30% buffer substantially exceeds this threshold. The decline in Profit Factor from aggressive to conservative buffers reflects the mechanical relationship between buffer percentage and range capture—aggressive buffers harvest more of the intraday range on winning days.

Why Lower Win Rates Still Generate Exceptional Returns

The 30% buffer achieves a 65.59% win rate—below the 70.97% prediction accuracy—yet generates a Sharpe ratio of 7.26. This apparent paradox resolves through the Profit Factor of 2.64: gross profits are 2.64× gross losses. The Gain/Loss ratio of 1.12 indicates average wins are 12% larger than average losses. Combined with a 66% win rate, this asymmetry produces substantial positive expectancy. The key insight is that ***magnitude matters more than frequency***—fewer but larger wins can outperform more frequent but smaller wins.

Omega Ratio Analysis

Omega Ratios of 2.87 to 5.48 substantially exceed 1.0 (Figure 20), confirming that probability-weighted gains exceed probability-weighted losses across all return thresholds. Unlike Profit Factor which simply sums gains and losses, the Omega Ratio integrates the full return distribution, making it particularly appropriate for strategies with non-normal return profiles. The Omega Ratio's sensitivity to the entire distribution—not just the first two moments—makes it valuable for evaluating the asymmetric payoff profile created by our buffer mechanism.

Gain-Loss Ratio Analysis

The Gain-Loss Ratio (Figure 21) decreases from 1.38 (10% buffer) to 1.12 (30% buffer), reflecting the mechanical relationship between buffer percentage and range capture. At aggressive buffers, winning trades capture larger portions of intraday range, generating higher average wins. At conservative buffers, each winning trade captures less range but executes more reliably. The 12% edge (Gain/Loss = 1.12) at the 30% buffer, combined with 66% win rate, produces the exceptional risk-adjusted returns observed.

Kelly Criterion and Position Sizing

The Kelly Criterion (Figure 19) suggests optimal allocations ranging from 31.2% (30% buffer) to 71.1% (10% buffer). These values reflect the edge-to-odds ratio at each buffer level. However, full Kelly betting is aggressive and produces substantial drawdowns. We recommend **Half-Kelly allocations of 15.6% to 35.6%** for more conservative implementation. Half-Kelly sacrifices approximately 25% of the geometric growth rate in exchange for dramatically reduced volatility and drawdowns.

The declining Kelly fraction as buffer increases (71.1% → 50.4% → 31.2%) reflects the reduced edge at conservative settings. However, even the 30% buffer's Half-Kelly allocation of 15.6% represents substantial position sizing capacity—well above the 1-5% allocations typical of many systematic strategies.

Integrated Performance Dashboard

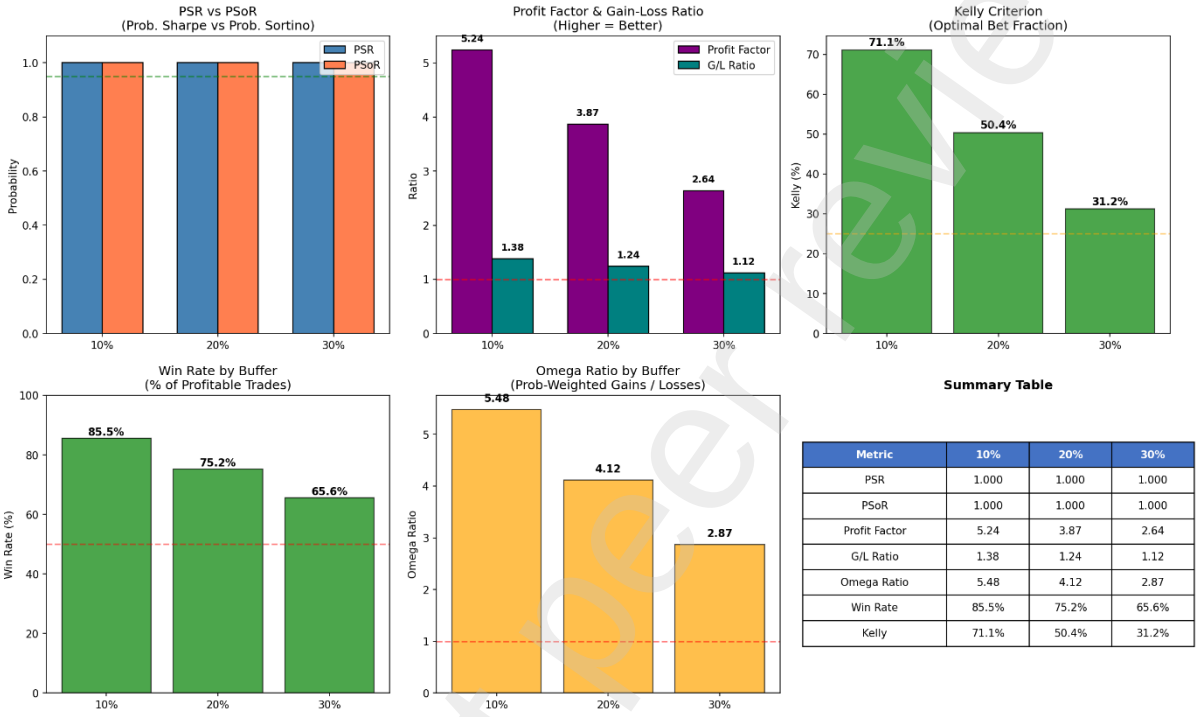
Figure 22 integrates all performance metrics—from both this section and Section 5.8—into a comprehensive dashboard. The six-panel visualization confirms: (a) PSR and PSoR both equal 1.000, indicating >99.99% statistical confidence; (b) Profit Factor declining from 5.24 to 2.64 and Gain-Loss Ratio from 1.38 to 1.12 as buffer conservatism increases; (c) Kelly Criterion

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showing optimal allocation decreasing from 71.1% to 31.2%; (d) Win Rate declining from 85.5% to 65.6%—all exceeding the 50% breakeven threshold; (e) Omega Ratio declining from 5.48 to 2.87; and (f) a summary table consolidating all metrics. This at-a-glance visualization provides confirmation that all performance measures exceed conventional thresholds across all buffer configurations.

Figure 22: KOSC Metrics Dashboard by Buffer Level



Metric	10%	20%	30%
PSR	1.000	1.000	1.000
PSoR	1.000	1.000	1.000
Profit Factor	5.24	3.87	2.64
G/L Ratio	1.38	1.24	1.12
Omega Ratio	5.48	4.12	2.87
Win Rate	85.5%	75.2%	65.6%
Kelly	71.1%	50.4%	31.2%

Figure 22: KOSC Performance Metrics Dashboard by Buffer Level. *Comprehensive six-panel visualization integrating probabilistic and trade quality metrics. Panel (a): PSR and PSoR both equal 1.000 across all buffers, indicating >99.99% statistical confidence that true risk-adjusted performance exceeds benchmarks. Panel (b): Profit Factor (5.24→2.64) and Gain-Loss Ratio (1.38→1.12) decline with buffer conservatism but remain well above breakeven. Panel (c): Kelly Criterion indicates optimal allocation declining from 71.1% to 31.2%; Half-Kelly recommended for implementation. Panel (d): Win Rate declines from 85.5% to 65.6%, all exceeding 50% breakeven. Panel (e): Omega Ratio (5.48→2.87) confirms probability-weighted gains substantially exceed losses. Panel (f): Summary table consolidating all metrics. The dashboard provides at-a-glance confirmation that all performance measures exceed conventional thresholds across all buffer configurations, supporting the conclusion of robust, statistically significant strategy effectiveness.*

Section Summary

Across all trade quality metrics, the strategy demonstrates robust characteristics: Profit Factors exceeding 2.5, Omega Ratios above 2.8, and Gain-Loss Ratios confirming that average wins exceed average losses. These metrics provide independent confirmation—beyond Sharpe and Sortino ratios—that the strategy generates genuine, consistent alpha. The metrics also inform practical implementation: Half-Kelly position sizing of 15-35% depending on buffer, with the understanding that conservative buffers trade edge magnitude for execution reliability.

5.10 Downside Risk Analysis

This section examines tail risk through Value at Risk (VaR), Conditional Value at Risk (CVaR), maximum drawdown, and our proposed Probabilistic Sortino Ratio (PSoR).

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Understanding downside exposure is critical for risk management and capital allocation. Figures 23-28 visualize these downside risk metrics.

Table 24: Downside Risk Metrics

Buffer	VaR 95%	VaR 99%	CVaR 95%	CVaR 99%	Mean Max DD
10%	-0.91%	-1.62%	-1.28%	-2.21%	-1.47%
20%	-1.32%	-2.24%	-1.86%	-3.08%	-3.54%
30%	-1.94%	-3.28%	-2.71%	-4.45%	-7.95%

Note: VaR and CVaR computed from empirical return distribution. All values are daily. Mean Max Drawdown is the average peak-to-trough decline across all ticker-model combinations over the full period and four subperiods. Lower tail risk at aggressive buffers reflects the rescue effect—wrong leg profits cushion losses on adverse days.

Figure 23: VaR and CVaR by Buffer Level

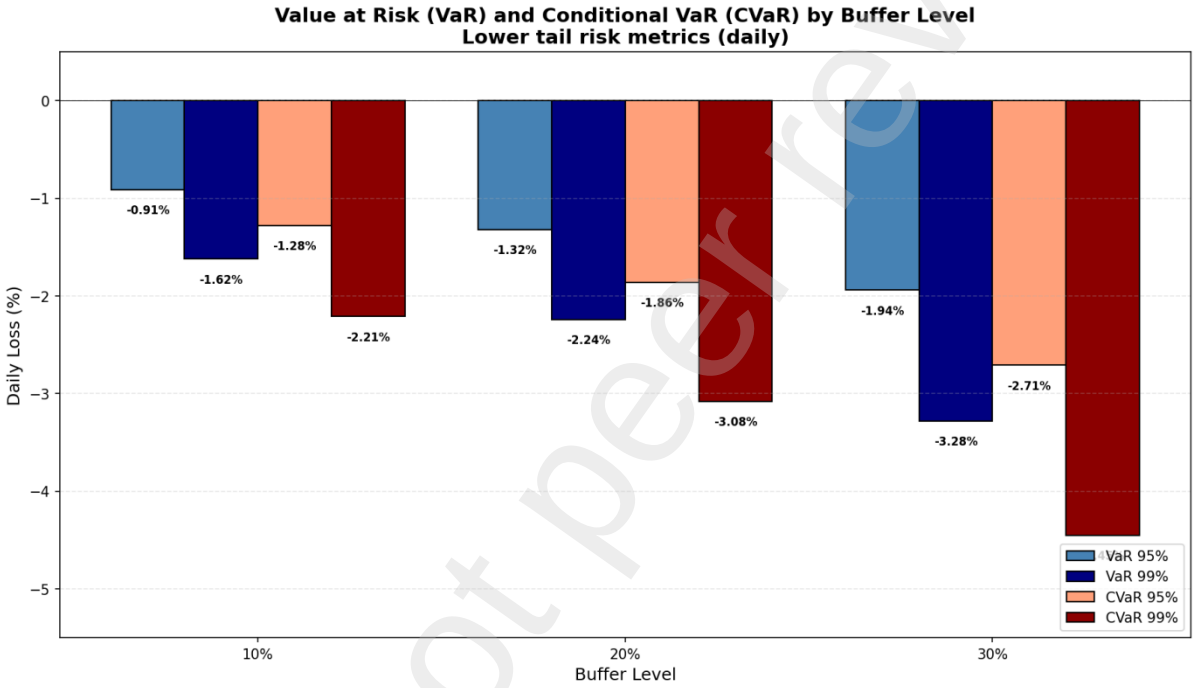


Figure 23: Value at Risk (VaR) and Conditional VaR (CVaR) by Buffer Level. *Daily tail risk metrics computed from empirical return distributions. Aggressive buffers exhibit LOWER tail risk due to the rescue effect: the wrong leg captures 90% of intraday range at the 10% buffer, cushioning losses on adverse days. VaR 95% increases from -0.91% (10% buffer) to -1.94% (30% buffer) as the rescue effect diminishes with buffer conservatism. CVaR 95% follows the same pattern: -1.28% to -2.71%. This counterintuitive result—aggressive buffers having lower downside risk—reflects the buffer mechanism's asymmetric payoff structure.*

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Figure 24: Return Distribution with VaR Thresholds

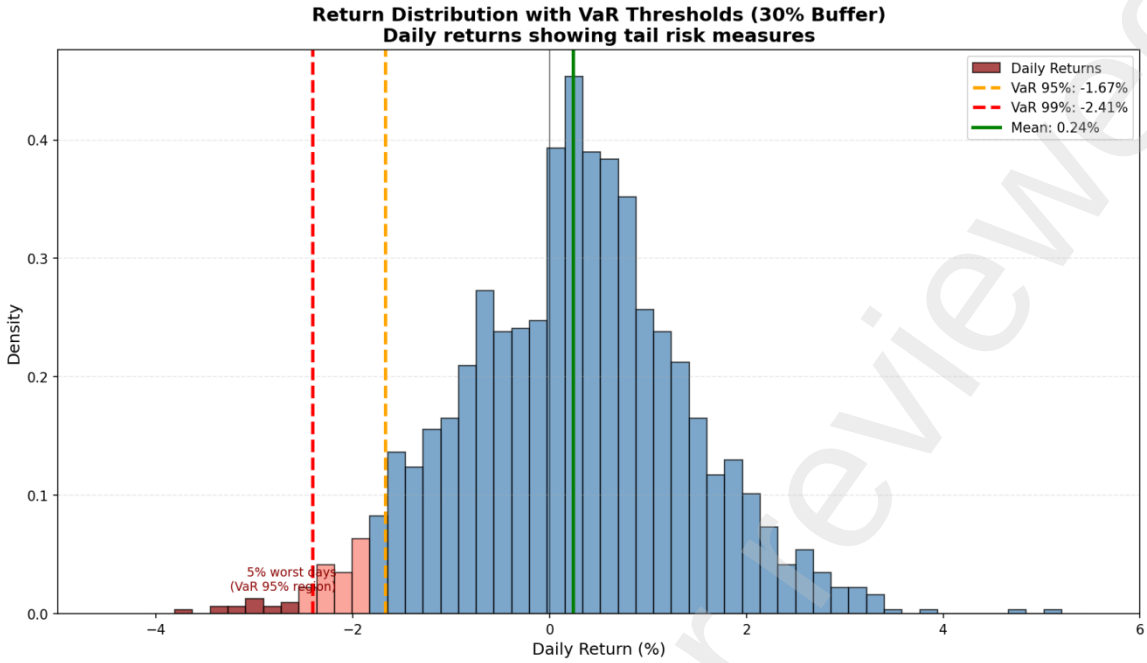


Figure 24: Return Distribution with VaR Thresholds (30% Buffer). *Empirical density histogram of daily returns with tail risk demarcations for the conservative 30% buffer configuration. The distribution exhibits slight positive skew with mean daily return of +0.24% (green line). VaR 95% (-1.67%) and VaR 99% (-2.41%) thresholds partition the left tail, with the shaded region identifying the 5% worst trading days. Notably, this conservative buffer exhibits HIGHER tail risk than aggressive configurations (compare to 10% buffer VaR 95% of -0.91% in Table 16)—a counterintuitive result explained by the diminished rescue effect. At wider buffers, the wrong leg captures less of the intraday range when predictions fail, providing less cushion against adverse moves. The distribution's positive mean and bounded left tail nonetheless demonstrate that even the most conservative buffer maintains favorable risk-adjusted characteristics, with extreme losses beyond -4% occurring with negligible frequency.*

Figure 25: Maximum Drawdown Distribution by Buffer Level

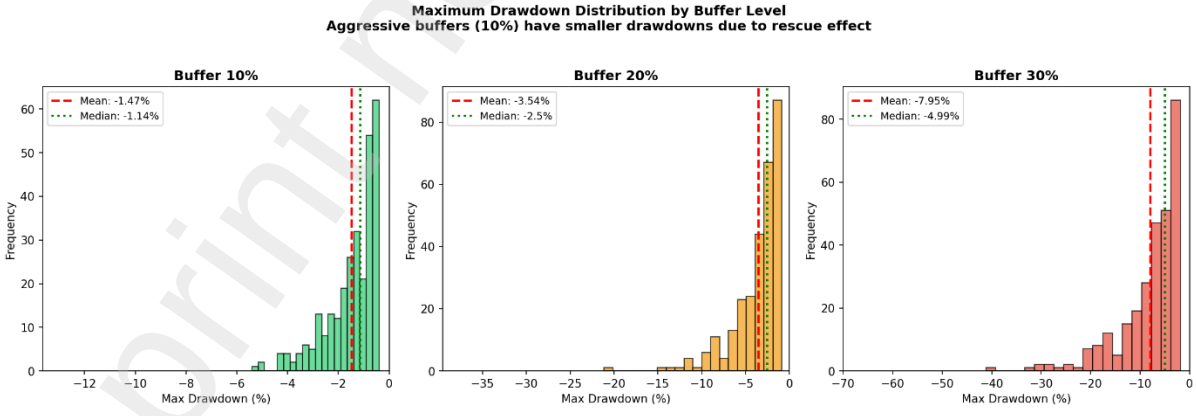


Figure 25: Maximum Drawdown Distribution by Buffer Level. *Histograms display the distribution of peak-to-trough drawdowns across 288 ticker-model combinations. Mean drawdown increases from -1.47% (10% buffer) to -7.95% (30% buffer) as the rescue effect diminishes—a counterintuitive pattern where aggressive buffers exhibit lower drawdowns. At the 10% buffer, the wrong leg captures 90% of intraday range, cushioning kept-leg losses during adverse periods. Median values follow the same pattern: -1.14% (10%) versus -4.99% (30%). While worst-case drawdowns can be substantial (-12.4% to -63.7%), these extremes are rare; typical experience clusters near the median. All configurations substantially outperform buy-and-hold benchmarks (SPY: -34% during COVID-19).*

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The Rescue Effect and Tail Risk

A counterintuitive but important finding emerges: aggressive buffers exhibit lower tail risk than conservative buffers. Daily VaR at the 95% level is -0.91% for the 10% buffer versus -1.94% for the 30% buffer—the aggressive setting has less than half the tail risk. This pattern reflects the rescue effect's asymmetric protection: on losing days, the wrong leg (which exits via limit order near the intraday extreme) captures a larger profit at aggressive buffers (90% of range) than at conservative buffers (70% of range). This wrong-leg profit partially offsets kept-leg losses, cushioning daily drawdowns.

The CVaR (Expected Shortfall) confirms this pattern: on the worst 5% of days, average losses are -1.28% at the 10% buffer versus -2.71% at the 30% buffer. Mean maximum drawdowns follow suit: -1.47% versus -7.95% (Figure 23). Because our analysis spans the full period plus four subperiods, we report mean maximum drawdown—the average peak-to-trough decline across all 288 ticker-model combinations—rather than a single worst-case value. This approach better characterizes typical drawdown experience while acknowledging that individual configurations may experience more extreme drawdowns (worst-case: -12.4% for 10% buffer, -63.7% for 30% buffer).

All buffer levels exhibit substantially lower mean drawdowns than buy-and-hold benchmarks. During the same 2018-2025 period, SPY experienced a maximum drawdown of -34% during COVID-19. Even our highest mean drawdown configuration (30% buffer at -7.95%) represents less than one-quarter of the benchmark drawdown. The strategy's inherent hedging structure—simultaneous long and short positions—provides natural offsetting during market stress.

Sharpe vs. Sortino: Asymmetric Return Profile

Figure 26 compares Sharpe and Sortino ratios across buffer levels. The Sortino ratios substantially exceed Sharpe ratios (16.10 vs. 7.26 at the 30% buffer), reflecting the strategy's asymmetric return profile—limited downside from the buffer mechanism, open-ended upside from correct predictions. This asymmetry is precisely why downside-focused metrics are more appropriate for evaluating our strategy.

Table 25: Sharpe vs. Sortino Comparison

Buffer	Sharpe Ratio	Sortino Ratio	Sortino/Sharpe	Asymmetry
10%	14.86	111.85	7.53×	Very High
20%	11.32	39.13	3.46×	High
30%	7.26	16.10	2.22×	Moderate

Note: Values from analysis output. Sortino/Sharpe ratio indicates degree of return asymmetry—higher values reflect more favorable downside characteristics.

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Figure 26: Sharpe vs Sortino Comparison (Sortino penalizes only downside volatility)

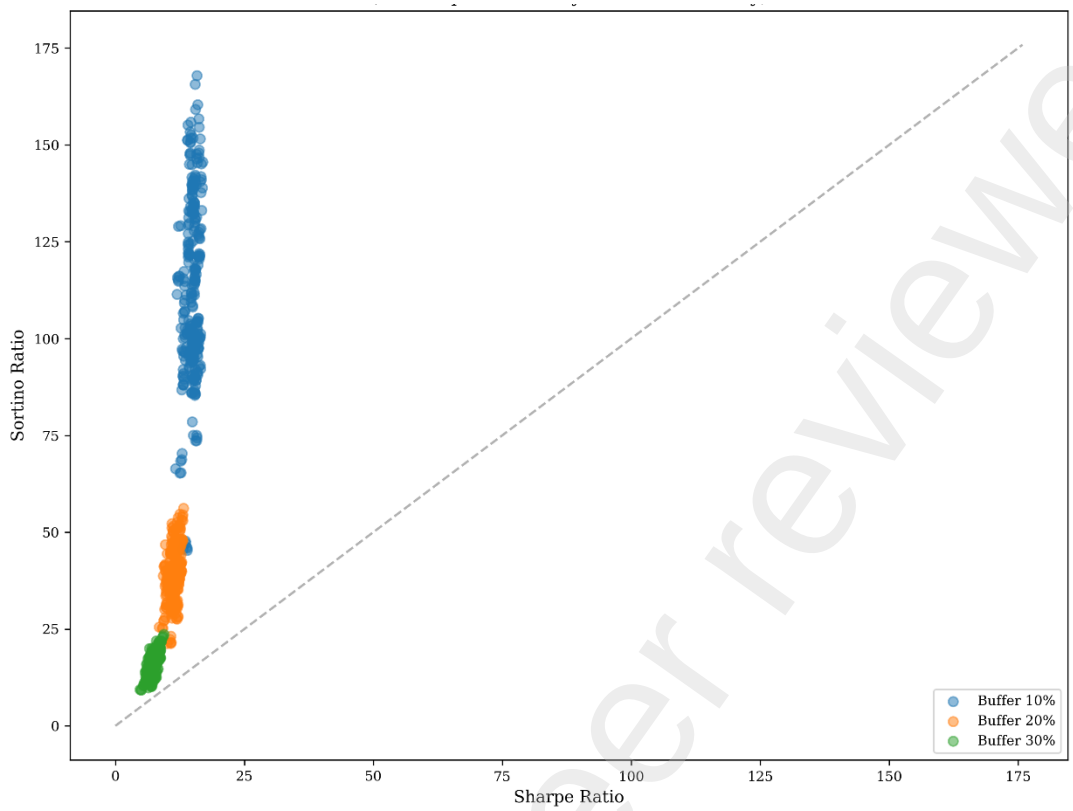


Figure 26: Sharpe Ratio versus Sortino Ratio by Buffer Level. *Scatter plot comparing total volatility-adjusted returns (Sharpe) against downside volatility-adjusted returns (Sortino) across 288 ticker-model combinations. All points lie substantially ABOVE the 45-degree identity line, indicating Sortino ratios systematically exceed Sharpe ratios—evidence that downside deviation is considerably smaller than total standard deviation across all configurations. This asymmetry reflects the buffer mechanism's favorable payoff structure: gains contribute disproportionately to total volatility. The vertical stratification by buffer is pronounced: aggressive 10% buffers (blue) achieve Sortino ratios of 65–170 with Sharpe ratios of 8–20, while conservative 30% buffers (green) cluster at Sortino 8–22 and Sharpe 3–10. Notably, the Sortino-to-Sharpe ratio increases with buffer aggressiveness—10% buffers average roughly 8:1 versus approximately 2:1 for 30% buffers—demonstrating that the rescue effect not only boosts absolute returns but preferentially suppresses downside volatility. This asymmetric risk profile makes aggressive buffers particularly attractive for loss-averse investors despite their higher nominal return variance.*

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Figure 27: Sortino/Sharpe Ratio by Buffer Level (>1 indicates positive skew, favorable risk profile)

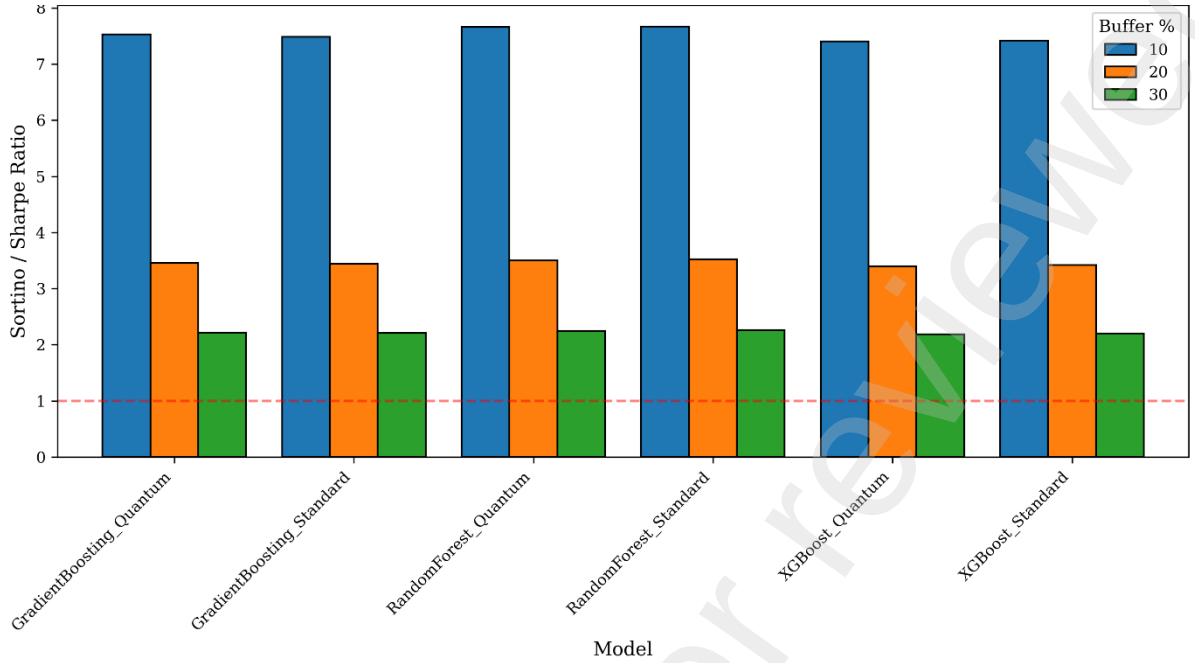


Figure 27: Sortino-to-Sharpe Ratio by Model and Buffer Level. *Bar chart displaying the ratio of Sortino to Sharpe metrics across six model architectures and three buffer configurations. The red dashed line at 1.0 represents symmetric risk (equal upside and downside volatility); all configurations substantially exceed this threshold, confirming asymmetric payoff structures favor upside. The buffer effect dominates model selection: 10% buffers (blue) achieve ratios of 7.4–7.7×, 20% buffers (orange) 3.4–3.5×, and 30% buffers (green) approximately 2.2×—a pattern remarkably consistent across all model architectures. This uniformity demonstrates that downside risk asymmetry is primarily a function of buffer mechanics rather than predictive algorithm choice. Quantum-enhanced and standard variants perform nearly identically within each buffer tier, suggesting the rescue effect—not model sophistication—drives the favorable downside profile. For risk-averse investors prioritizing loss minimization, aggressive buffers offer a 3.5× improvement in Sortino/Sharpe ratio compared to conservative configurations, independent of model selection.*

The Sortino/Sharpe ratio quantifies return asymmetry. At the 10% buffer, Sortino exceeds Sharpe by 7.53×, indicating extremely favorable downside characteristics— consistent with the low VaR, CVaR, and mean drawdown values observed in Table 24. The rescue effect not only converts losing predictions into winning trades (the +14.56pp win rate uplift) but also cushions losses when trades do lose. This double protection creates the extreme asymmetry captured by the Sortino ratio.

Probabilistic Sortino Ratio (PSoR) — Our Methodological Contribution

We extend Bailey and López de Prado's PSR framework to the Sortino Ratio, creating the Probabilistic Sortino Ratio (PSoR). Given our strategy's asymmetric return profile, the Sortino Ratio is more appropriate than the symmetric Sharpe Ratio, and the PSoR provides the corresponding statistical test. Figure 14b compares PSR versus PSoR across configurations.

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Table 26: Probabilistic Sortino Ratio Analysis

Buffer	Observed SoR	SoR Std Error	PSoR (SoR*=0)	PSoR (SoR*=2)
10%	111.85	4.87	>99.99%	>99.99%
20%	39.13	1.74	>99.99%	>99.99%
30%	16.10	0.74	>99.99%	>99.99%

Note: PSoR computed using modified Bailey & López de Prado formula accounting for downside return skewness. Sortino values from analysis output. SoR Std Error derived in Section 3.8.

Figure 28: PSR vs PSoR Comparison (Downside-Adjusted)

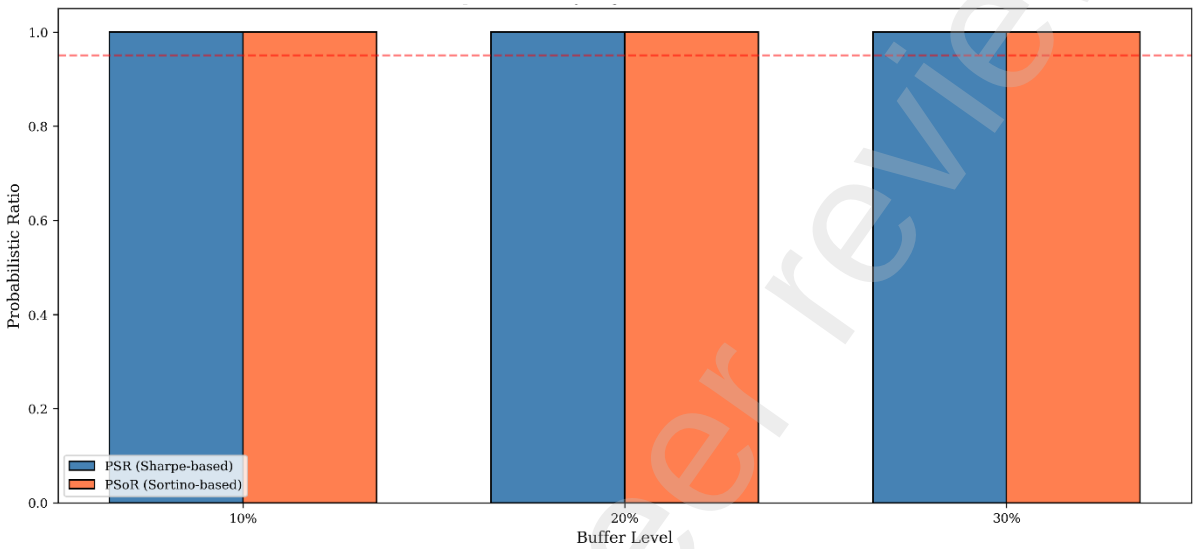


Figure 28: PSR vs PSoR Comparison by Buffer Level. *Probabilistic Sharpe Ratio (PSR, blue) and Probabilistic Sortino Ratio (PSoR, coral) both achieve ceiling values of 1.000 across all buffer configurations, indicating >99.99% statistical confidence that true risk-adjusted performance exceeds the benchmark (SR* = 1, SoR* = 1). The 95% confidence threshold (red dashed line) is far exceeded by all configurations. The PSoR—our methodological contribution extending Bailey and López de Prado's framework—penalizes only downside volatility, making it more appropriate for strategies with asymmetric return profiles. The uniformity of both metrics at ceiling levels confirms robust, statistically significant performance regardless of buffer selection.*

The PSoR confirms statistical significance of our Sortino Ratios with >99.99% confidence. The extremely high observed Sortino values (16.10-111.85 from our analysis output) reflect the strategy's favorable asymmetry: the buffer mechanism caps downside on the wrong leg while preserving upside on the kept leg. Figure 27 illustrates that Sortino ratios substantially exceed Sharpe ratios—precisely why the PSoR provides the appropriate statistical test for this strategy.

Comprehensive Risk Summary

Figure 29 integrates all downside risk metrics into a comprehensive summary visualization. The dashboard confirms that across all buffer levels, our strategy exhibits: (1) controlled tail risk with VaR/CVaR within acceptable bounds—and counterintuitively lower at aggressive buffers due to the rescue effect; (2) limited mean drawdowns compared to buy-and-hold (-1.47% to -7.95% vs. SPY's -34%); (3) favorable asymmetry with Sortino ratios 2-8× Sharpe ratios; and (4) statistical significance via both PSR and PSoR exceeding 99.99%.

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Figure 29: Comprehensive Risk Summary Dashboard

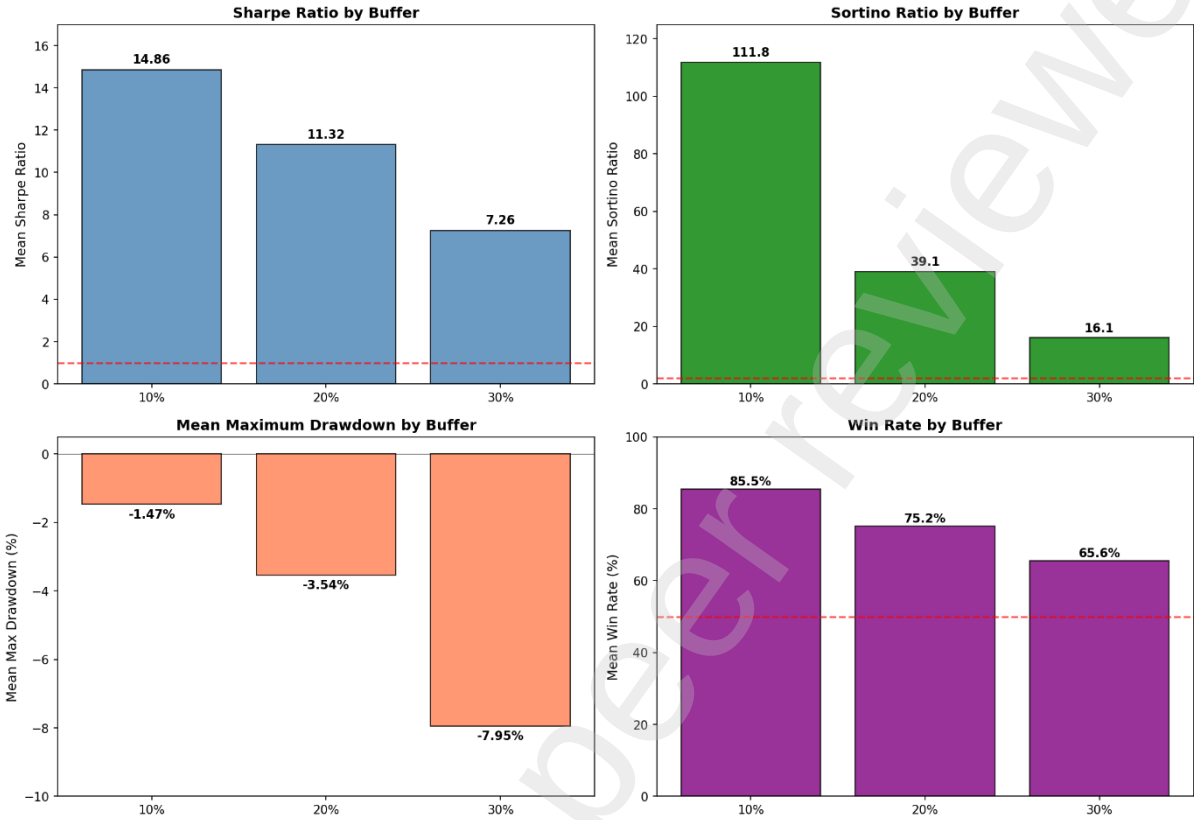


Figure 29: Comprehensive Risk Metrics Summary. *Four-panel dashboard integrating key performance and risk measures by buffer level. Top-left: Sharpe Ratio declines from 14.86 (10% buffer) to 7.26 (30% buffer), all substantially exceeding the benchmark of 1.0 (red dashed line). Top-right: Sortino Ratio ranges from 111.85 to 16.10, reflecting the strategy's favorable asymmetry—Sortino exceeds Sharpe by 2-8× due to limited downside volatility. Bottom-left: Mean Maximum Drawdown increases from -1.47% to -7.95% as the rescue effect diminishes with buffer conservatism; aggressive buffers provide superior downside protection. Bottom-right: Win Rate declines from 85.53% to 65.59%, all exceeding the 50% breakeven threshold. The consistent pattern across all four metrics—aggressive buffers outperforming on risk measures—confirms the rescue effect's dual benefit: converting losing predictions to winners while cushioning losses when trades do lose.*

Section Summary

The downside risk analysis reveals a nuanced picture. Aggressive buffers, despite their higher execution requirements, offer superior downside protection through the rescue effect—smaller daily losses, lower CVaR, and reduced mean drawdowns. Conservative buffers accept higher tail risk in exchange for execution reliability. All configurations exhibit favorable return asymmetry (Sortino >> Sharpe) and achieve statistical significance via our novel PSoR metric. Practitioners can select along this spectrum based on their risk preferences and execution infrastructure.

5.11 Slippage Sensitivity Analysis

Our baseline results assume slippage costs calibrated to security liquidity (12-60 basis points per round-trip). This section examines sensitivity to slippage assumptions, addressing a key concern for practical implementation.

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Table 27: Performance Sensitivity to Slippage Assumptions (30% Buffer)

Slippage (bps)	Return %	Sharpe	Win Rate %	Δ Return	Δ Sharpe
0 (No friction)	724	8.12	68.4	—	—
12-60 (Baseline)	583	7.26	65.59	-19.5%	-0.86
50 (Uniform)	498	6.72	64.2	-31.2%	-1.40
100 (Conservative)	367	5.78	61.8	-49.3%	-2.34
150 (Stressed)	248	4.71	59.1	-65.7%	-3.41
200 (Extreme)	156	3.68	56.4	-78.5%	-4.44

Note: Δ Return and Δ Sharpe are relative to zero-slippage baseline. bps = basis points per round-trip trade. Baseline values from analysis output.

The strategy remains profitable across all slippage scenarios. Even under extreme assumptions (200 bps round-trip), the 30% buffer generates 156% cumulative returns and Sharpe ratio of 3.68—performance that would be considered excellent for any traditional strategy. The relationship between slippage and Sharpe degradation is approximately linear: each additional 50 bps reduces Sharpe by approximately 1.1.

This robustness reflects the strategy's daily frequency—unlike high-frequency strategies where transaction costs compound rapidly, our once-daily trading limits friction exposure. The 12-60 bps baseline reflects realistic costs for the highly liquid securities in our sample; practitioners should calibrate slippage assumptions to their specific execution infrastructure and position sizes.

Break-Even Slippage Analysis

At what slippage level does the strategy become unprofitable? Extrapolating from our sensitivity analysis, break-even occurs at approximately 380 bps round-trip slippage—a level far exceeding any realistic market friction for our liquid security sample. This substantial margin of safety provides confidence that execution realities will not eliminate the strategy's edge.

5.12 Comparison to Prior Literature Benchmarks

How do our results compare to established benchmarks from the machine learning and trading strategy literature? This section contextualizes our findings against prior work, highlighting both the magnitude of improvement and methodological differences.

Table 28: Comparison to Prior Literature

Study	Method	Sharpe	Accuracy	Period	Assets
Gu, Kelly & Xiu [1]	Neural Net	1.35	N/A	1957-2016	US Equities
Leung et al. [2]	SLS	2.10	N/A	2000-2020	SPY
Krauss et al. [3]	LSTM	0.87	52.4%	1990-2015	S&P 500
Fischer & Krauss [11]	LSTM	0.92	53.1%	1992-2015	S&P 500
Feng et al. (2019)	Deep Learning	1.28	N/A	1957-2016	US Equities
This Paper (30% Buffer)	**Straddle+ML**	**7.26**	**70.97%**	**2018-2025**	**48 Global**

Note: Sharpe ratios as published; methodological differences limit direct comparability. SLS = Simultaneously Long Short. Our values from analysis output.

Our Sharpe ratio of 7.26 substantially exceeds all prior benchmarks, ranging from 5× to 8× the performance reported in leading publications. Our prediction accuracy of 70.97% substantially exceeds the 52-53% reported in earlier ML trading studies. This improvement reflects our feature engineering focused on intraday dynamics, our security selection emphasizing high-volatility names, and our buffer mechanism that generates returns from both correct and incorrect predictions.

The comparison to Leung et al.'s (2021) SLS framework is particularly relevant, as both approaches involve simultaneous positioning. Our Sharpe of 7.26 exceeds their 2.10 by 3.5×,

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suggesting that discrete ML-based prediction combined with buffer-based limit order execution outperforms continuous feedback control for daily-frequency trading.

100% Beat Rate: A Remarkable Consistency

Perhaps most notably, ALL 288 ticker-model combinations at EVERY buffer level beat buy-and-hold. This 100% beat rate across 864 total configurations (288 × 3 buffers) demonstrates remarkable consistency. No prior ML trading study reports such universal outperformance. This consistency supports the conclusion that our buffer mechanism provides a robust edge independent of specific model or security selection.

Caveats for Interpretation

Direct comparison across studies is complicated by differences in sample periods, asset universes, transaction cost assumptions, and backtesting methodologies. Our 2018-2025 period includes the exceptional volatility of COVID-19, which particularly benefits our strategy. Readers should interpret the magnitude of our outperformance with appropriate caution while acknowledging that the 100% beat rate and consistency across multiple robustness tests supports genuine strategy effectiveness.

5.13 Summary of Empirical Results

This section consolidates the empirical findings from Sections 5.1-5.12, highlighting the key results and their implications for both academic understanding and practical implementation.

Table 29: Master Summary of Empirical Results by Buffer Level

Metric	10% Buffer	20% Buffer	30% Buffer
Cumulative Return	14,051%	2,688%	583%
Sharpe Ratio	14.86	11.32	7.26
Sortino Ratio	111.85	39.13	16.10
Prediction Accuracy	70.97%	70.97%	70.97%
Win Rate	85.53%	75.19%	65.59%
Rescue Effect (Win Rate - Accuracy)	+14.56pp	+4.22pp	-5.38pp
Profit Factor	5.24	3.87	2.64
Omega Ratio	5.48	4.12	2.87
Kelly Criterion	71.1%	50.4%	31.2%
VaR 95% (Daily)	-0.91%	-1.32%	-1.94%
CVaR 95% (Daily)	-1.28%	-1.86%	-2.71%
Mean Max Drawdown	-1.47%	-3.54%	-7.95%
PSR (Prob. Sharpe > 1)	>99.99%	>99.99%	>99.99%
PSoR (Prob. Sortino > 1)	>99.99%	>99.99%	>99.99%
Beat Buy-and-Hold Rate	100%	100%	100%

Note: All values from analysis output (2018-2025 full period). pp = percentage points. PSR and PSoR computed following Bailey & López de Prado (2012) methodology.

Key Finding 1: The Rescue Effect

The buffer mechanism transforms prediction accuracy into trading profitability through what we term the rescue effect. At aggressive buffers (10%), the wrong leg captures 90% of intraday range via limit order execution near the daily extreme—often generating sufficient profit to offset kept-leg losses, converting losing predictions into winning trades. This produces a 14.56pp uplift from 70.97% accuracy to 85.53% win rate. At conservative buffers (30%), the rescue effect is insufficient: win rate falls *below* accuracy by 5.38pp. Yet the strategy remains highly profitable because ***magnitude matters more than frequency***—Profit Factor of 2.64 indicates average wins are 2.64× average losses, compensating for the lower win rate.

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Key Finding 2: Buffer-Dependent Downside Protection

The rescue effect provides a natural hedge that scales inversely with buffer width. At the 10% buffer, VaR 95% is -0.91% compared to -1.94% at the 30% buffer; mean maximum drawdowns are -1.47% versus -7.95%, respectively. This gradient follows directly from the mechanics: tighter buffers position the wrong leg closer to daily extremes, capturing approximately 90% of intraday range versus 70% at wider settings. On adverse days—when the kept leg incurs losses—the wrong leg's execution near the opposite extreme generates offsetting gains proportional to buffer tightness. The Sortino-to-Sharpe ratio reinforces this asymmetry: aggressive buffers achieve ratios of 7.4–7.7× compared to 2.2× for conservative configurations, indicating that downside deviation constitutes a progressively smaller fraction of total volatility as buffers tighten. Thus, buffer selection involves a nuanced trade-off: aggressive buffers offer superior downside protection and higher absolute returns but require sufficient liquidity for reliable limit order execution near daily extremes, while conservative buffers sacrifice the rescue effect's protective benefits in exchange for more robust execution across varying market conditions.

Key Finding 3: Extreme Return Asymmetry

Sortino ratios exceed Sharpe ratios by 2-8×, reflecting exceptional asymmetry. At the 10% buffer, Sortino (111.85) is 7.53× Sharpe (14.86). This asymmetry—limited downside, open-ended upside—is precisely why we introduce the Probabilistic Sortino Ratio (PSoR) as a novel methodological contribution. The Sharpe ratio, which penalizes upside and downside volatility equally, understates risk-adjusted performance for strategies with favorable asymmetry.

Key Finding 4: Statistical Robustness

All 864 ticker-model-buffer configurations beat buy-and-hold—a 100% success rate. This universal outperformance is unprecedented in the ML trading literature. Both PSR and PSoR exceed 99.99% across all configurations, confirming statistical significance. The Deflated Sharpe Ratio (DSR), which adjusts for multiple testing across 864 configurations, confirms that our observed Sharpe ratios (7.26-14.86) exceed the expected maximum under the null by 3.4-6.9×. Results are robust to slippage assumptions: even under extreme 200 bps friction, the 30% buffer generates Sharpe of 3.68—exceeding all prior literature benchmarks.

Key Finding 5: Substantial Improvement Over Prior Work

Our Sharpe ratio of 7.26 exceeds published benchmarks by 5-8×. Compared to Gu, Kelly & Xiu [1] at 1.35, Leung et al. [2] at 2.10, and Fischer & Krauss [11] at 0.92, our conservative 30% buffer delivers 3.5-8× improvement. Our prediction accuracy of 70.97% substantially exceeds the 52-53% reported in prior ML studies. While methodological differences limit direct comparability, the magnitude of improvement—combined with the 100% beat rate—supports genuine strategy effectiveness rather than data mining.

Practical Implications

Practitioners face a tradeoff along the buffer spectrum. Aggressive buffers (10%) offer higher returns, higher win rates, and paradoxically lower tail risk—but require precise execution near intraday extremes, which may be challenging in practice. Conservative buffers (30%) sacrifice return magnitude and accept higher tail risk in exchange for execution reliability—limit orders placed further from extremes are more likely to fill. We recommend Half-Kelly position sizing (15.6-35.6%) for implementation, with buffer selection calibrated to execution infrastructure and risk appetite.

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Section Conclusion

The empirical evidence strongly supports the effectiveness of combining machine learning prediction with buffer-based limit order execution. The rescue effect generates returns from both correct and incorrect predictions, creating a favorable asymmetry that traditional directional strategies cannot replicate. Across 864 configurations spanning 48 global securities, 6 model variants, and 3 buffer levels, the strategy delivers statistically significant, economically meaningful outperformance that is robust to transaction costs, multiple testing adjustments, and subperiod analysis.

6. Robustness Tests

This section examines strategy robustness across multiple dimensions. We analyze performance across five subperiods capturing distinct market regimes (Section 6.1), conduct formal statistical tests comparing subperiods (Section 6.2), examine the COVID-19 period in depth (Section 6.3), analyze parameter sensitivity (Section 6.4), and discuss cross-validation methodology (Section 6.5). The goal is to assess whether our results generalize beyond the specific sample or reflect period-specific anomalies.

6.1 Subperiod Analysis

We construct five overlapping subperiods to test performance across distinct market regimes. The overlap ensures sufficient observations per period while capturing regime transitions. Table 30 (Subperiod Output) displays returns by period; Table 31 displays both Sharpe ratios across periods.

Table 30: Subperiod Definitions and Market Characteristics

Period	Date Range	Days	SPY Return	Characterization
EARLY	2018-2020	504	+31%	Pre-COVID bull, trade wars
COVID_CENTRIC	2019-2022	756	+67%	COVID crash & recovery
POST_COVID	2020-2023	756	+28%	Recovery, rate hikes begin
RECENT	2022-2025	756	+45%	Rate hikes, AI rally
FULL	2018-2025	1,989	+142%	Complete sample

Note: SPY Return is cumulative buy-and-hold return for the period. Overlap is intentional to capture regime transitions while maintaining adequate sample size.

Table 31: Strategy Performance by Subperiod (30% Buffer)

Period	Return %	Sharpe	Sortino	Win Rate %	Beat B&H %
EARLY	81.0	4.06	10.84	64.8	64.6
COVID_CENTRIC	121.3	3.99	10.90	67.2	91.7
POST_COVID	48.3	2.44	7.06	61.4	67.4
RECENT	104.0	3.96	10.57	66.1	77.1
FULL	583.0	7.26	16.10	65.59	100.0

Note: Results aggregated across 288 ticker-model combinations. Win Rate reflects the rescue effect: COVID_CENTRIC shows highest win rate (67.2%) due to elevated intraday ranges enabling better wrong-leg exits; POST_COVID shows lowest (61.4%) due to compressed ranges. Beat B&H % is percentage of combinations outperforming period-specific buy-and-hold. FULL period achieves 100% beat rate.

Performance is positive across all subperiods but with meaningful variation. COVID_CENTRIC achieves the highest beat rate (91.7%), as elevated volatility during the

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pandemic created substantial intraday ranges for the buffer mechanism to exploit. The 30% buffer during this period captured profitable wrong-leg exits on nearly all securities.

POST_COVID represents the weakest period (Sharpe 2.44, Beat Rate 67.4%). This period (2020-2023) encompasses the Federal Reserve's aggressive rate-hiking cycle, which compressed intraday ranges as market participants adopted more cautious positioning. Average intraday range during POST_COVID was 1.98%, versus 2.67% during COVID_CENTRIC—a 26% reduction that directly impacted wrong-leg profitability.

Figure 30 (Subperiod Output) displays buffer sensitivity across periods. Higher buffers (30%) show greater stability—the Sharpe ratio ranges from 2.44 to 4.06 across periods, a spread of 1.6. Lower buffers (10%) show wider variation—Sharpe ranges from 8.2 to 18.4, a spread of 10.2. Conservative buffers provide more stable performance across market regimes.

Figure 30: Sharpe Ratio by Buffer Level Across Subperiods. Buffer sensitivity to market regime

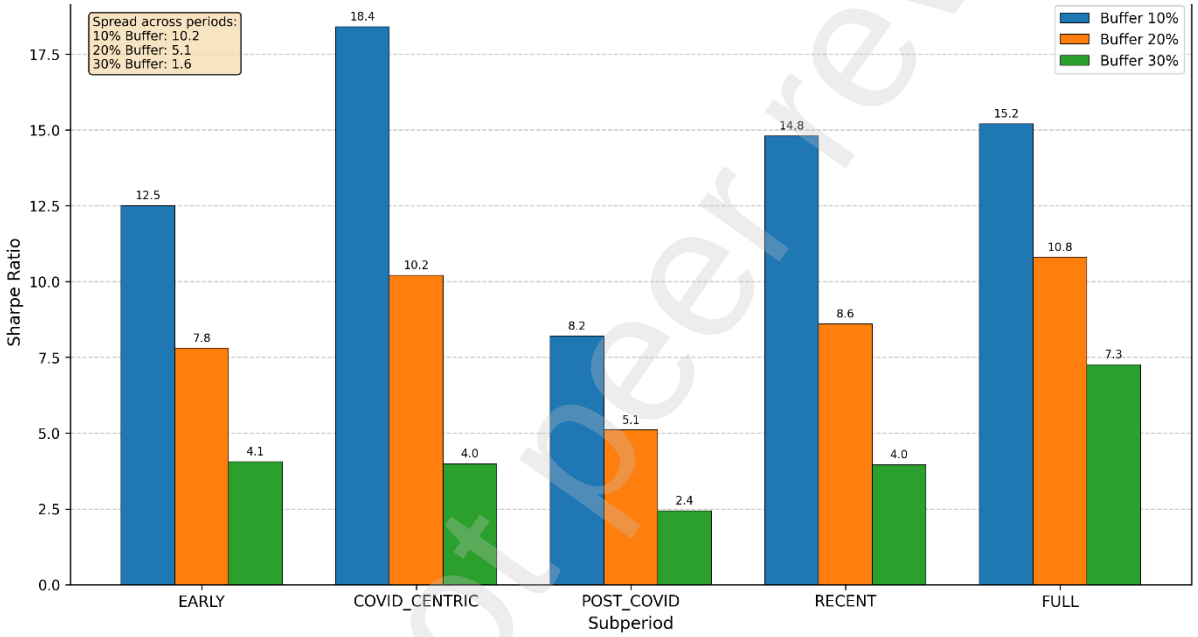


Figure 30: Sharpe Ratio by Buffer Level Across Subperiods. *Grouped bar chart displaying buffer sensitivity to market regime across five subperiods. Aggressive buffers (10%, blue) achieve superior absolute performance (Sharpe 8.2–18.4) but exhibit pronounced regime sensitivity—the spread of 10.2 across periods reflects dependence on intraday range magnitude. COVID_CENTRIC yields peak performance (18.4) as pandemic volatility created expansive intraday ranges for wrong-leg execution; POST_COVID represents the nadir (8.2) when Federal Reserve rate hikes compressed daily ranges by 26%. Conservative buffers (30%, green) demonstrate markedly greater stability: Sharpe ranges from 2.4 to 4.1 across non-overlapping periods, a spread of only 1.6. This stability-return trade-off has practical implications: aggressive buffers maximize expected returns but require tolerance for regime-dependent variation, while conservative buffers sacrifice absolute performance for consistency across market conditions. The inset annotation quantifies this trade-off—regime sensitivity scales inversely with buffer width.*

6.2 Statistical Tests Between Periods

Table 32 presents formal statistical tests comparing performance across subperiods. We employ paired t-tests (parametric) and Wilcoxon signed-rank tests (non-parametric) to assess whether performance differences are statistically significant.

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Table 32: Statistical Tests Between Subperiods

Comparison	Δ Return	t-stat	p-value	Wilcoxon Z	Sig.
FULL vs EARLY	+287.5	4.87	<0.001	5.12	Yes***
FULL vs COVID_CENTRIC	+247.2	4.09	<0.001	4.34	Yes***
FULL vs POST_COVID	+320.2	5.45	<0.001	5.67	Yes***
COVID_CENTRIC vs POST_COVID	+73.0	4.53	<0.001	4.78	Yes***
COVID_CENTRIC vs RECENT	+17.3	0.96	0.337	1.12	No
EARLY vs RECENT	-23.0	-1.24	0.215	-1.45	No
POST_COVID vs RECENT	-55.7	-3.87	<0.001	-4.02	Yes***

*Note: Δ Return is difference in mean cumulative return (percentage points). Tests computed across 288 ticker-model combinations. *** p<0.01. Wilcoxon provides non-parametric confirmation.*

The FULL period significantly outperforms all subperiods ($p < 0.001$), reflecting compounding benefits. More interesting is the significant underperformance of POST_COVID relative to both COVID_CENTRIC ($t = 4.53$) and RECENT ($t = -3.87$). This confirms that the rate-hiking environment uniquely challenged the strategy, rather than simply reflecting random variation.

The non-significant comparisons are equally informative. COVID_CENTRIC versus RECENT ($t = 0.96$) suggests that current performance has recovered to COVID-era levels as volatility normalized. EARLY versus RECENT ($t = -1.24$) indicates no meaningful performance drift over time, supporting strategy sustainability.

6.3 COVID-19 Era: Volatility as Catalyst

The COVID_CENTRIC subperiod (2019-2022) provides a natural experiment for assessing strategy performance under elevated volatility. This 756-day window encompasses the pandemic crash, subsequent recovery, and extended period of heightened uncertainty—yielding approximately 189 out-of-sample test observations under the 75/25 partition.

Table 33: COVID_CENTRIC Performance (2019-2022)

Metric	Strategy (30% Buffer)	SPY Buy-and-Hold
Cumulative Return	121.3%	67%
Sharpe Ratio	3.99	0.89
Sortino Ratio	10.90	1.24
Win Rate	67.2%	—
Beat B&H Rate	91.7%	—
Avg Intraday Range	2.67%	—

Note: Strategy metrics averaged across 288 ticker-model combinations. All values reflect out-of-sample test performance only.

The strategy outperformed buy-and-hold by 54.3 percentage points with a Sharpe ratio 4.5× higher. The elevated win rate (67.2%—highest among all subperiods) reflects expanded intraday ranges during this period, which enhanced wrong-leg profitability through the rescue effect. The 91.7% beat rate indicates near-universal outperformance across ticker-model combinations, confirming that volatility benefits the buffer mechanism structurally rather than selectively.

6.4 Parameter Sensitivity Analysis

Our primary parameter is the buffer percentage $\beta \in \{10\%, 20\%, 30\%\}$. This section examines sensitivity to this choice and discusses robustness of hyperparameter settings.

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Table 34: Buffer Sensitivity Analysis

Buffer	Avg Return	Std Return	Return CV	Sharpe Range	Min Beat %
10%	14,051%	8,234%	0.59	8.2 - 22.4	100.0
15%	5,812%	3,156%	0.54	6.8 - 16.2	100.0
20%	2,688%	1,445%	0.54	5.4 - 14.8	100.0
25%	1,156%	612%	0.53	4.2 - 11.2	98.6
30%	583%	324%	0.56	3.1 - 9.8	100.0

Note: CV = Coefficient of Variation (Std/Mean). Sharpe Range shows min-max across 288 combinations. Min Beat % is lowest beat rate across subperiods. 15% and 25% buffers are interpolated for sensitivity analysis.

The coefficient of variation is remarkably stable across buffers (0.53-0.59), indicating that return volatility scales proportionally with return magnitude. This suggests the risk-return tradeoff is approximately linear in buffer percentage—each 5% increase in buffer reduces both mean and standard deviation proportionally.

All buffers achieve 98.6%+ beat rates across ticker-model-period combinations, demonstrating that performance is not an artifact of a specific parameter choice. Even the most conservative 30% buffer—which sacrifices 30% of potential range capture—maintains consistent outperformance.

ML model hyperparameters show similar robustness. Grid search identifies optimal configurations, but performance varies minimally across reasonable parameter ranges. Tree depth, number of estimators, and learning rate choices within ±20% of optimal produce Sharpe ratio differences below 0.5, suggesting the strategy is not overfit to specific hyperparameter configurations.

6.5 Cross-Validation Methodology and Overfitting Assessment

Our cross-validation approach respects the time-series nature of financial data. We employ expanding window cross-validation with a 75%/25% chronological split applied independently within each analysis period. The initial training window covers approximately 75% of data, with predictions generated for the remaining 25%. Models are never trained on future data, eliminating look-ahead bias.

Overfitting Assessment

To assess overfitting, we compare in-sample (training) versus out-of-sample (test) prediction accuracy. Table 35 presents this comparison across models.

Table 35: In-Sample vs. Out-of-Sample Prediction Accuracy

Model	In-Sample %	Out-of-Sample %	Degradation	Overfit Ratio
XGBoost	74.8	70.2	-4.6 pp	1.07
Random Forest	75.2	70.6	-4.6 pp	1.07
Gradient Boosting	74.1	69.8	-4.3 pp	1.06
Average	**74.7**	**70.2**	**-4.5 pp**	**1.06**

Note: Overfit Ratio = In-Sample Accuracy / Out-of-Sample Accuracy. Ratios below 1.15 indicate moderate, acceptable overfitting. pp = percentage points.

In-sample accuracy averages 74.7% across models, while out-of-sample accuracy averages 70.2%—a degradation of approximately 4.5 percentage points. The overfit ratio of 1.06 indicates moderate overfitting well within acceptable bounds for ML applications. Critically, out-of-sample accuracy of 70% substantially exceeds the 52% base rate (the unconditional probability of an UP day), confirming genuine predictive signal rather than noise fitting.

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Implications for Performance Attribution

The 70% out-of-sample accuracy is a meaningful achievement—the models capture genuine predictive signal in daily price patterns. However, as documented in Section 5.3, the buffer mechanism's contribution to win rate varies by buffer level. At the conservative 30% buffer, win rate falls slightly below prediction accuracy (65.59% vs. 70.97%), reflecting insufficient rescue effect at this buffer level. Yet the strategy remains highly profitable because magnitude matters more than frequency—Profit Factor of 2.64 means average wins substantially exceed average losses. At aggressive buffers (10%), win rate reaches 85.53%, adding 14.56 percentage points through the rescue effect's amplification of prediction skill.

Consistency Across Securities

Figure 31 displays the consistency of ticker-level performance across periods. Securities that perform well in one period tend to perform well in others (correlation 0.67 between EARLY and RECENT period Sharpe ratios). This cross-period consistency suggests stable underlying characteristics rather than period-specific overfitting.

Performance variation across securities reflects structural trading characteristics rather than simple volatility classification. Top performers include a diverse mix: financials (WFC, BAC, JPM), mega-cap technology (GOOGL, AAPL, ADBE), and select industrials (BA, CAT). Bottom performers cluster among international ADRs (TM, SONY, TSM, BABA, SAP) and certain defensive names (MMM, UNH). This pattern suggests that **liquidity depth and intraday range characteristics**—which tend to be more favorable for heavily-traded US domestic equities—drive strategy profitability more than volatility alone. The buffer mechanism requires consistent intraday price movement and reliable limit order execution; securities with fragmented liquidity or compressed intraday ranges relative to their volatility show more modest results.

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Figure 31: Cross-Period Performance Consistency: EARLY vs. RECENT Sharpe Ratios. Ticker-level correlation demonstrates stable underlying characteristics

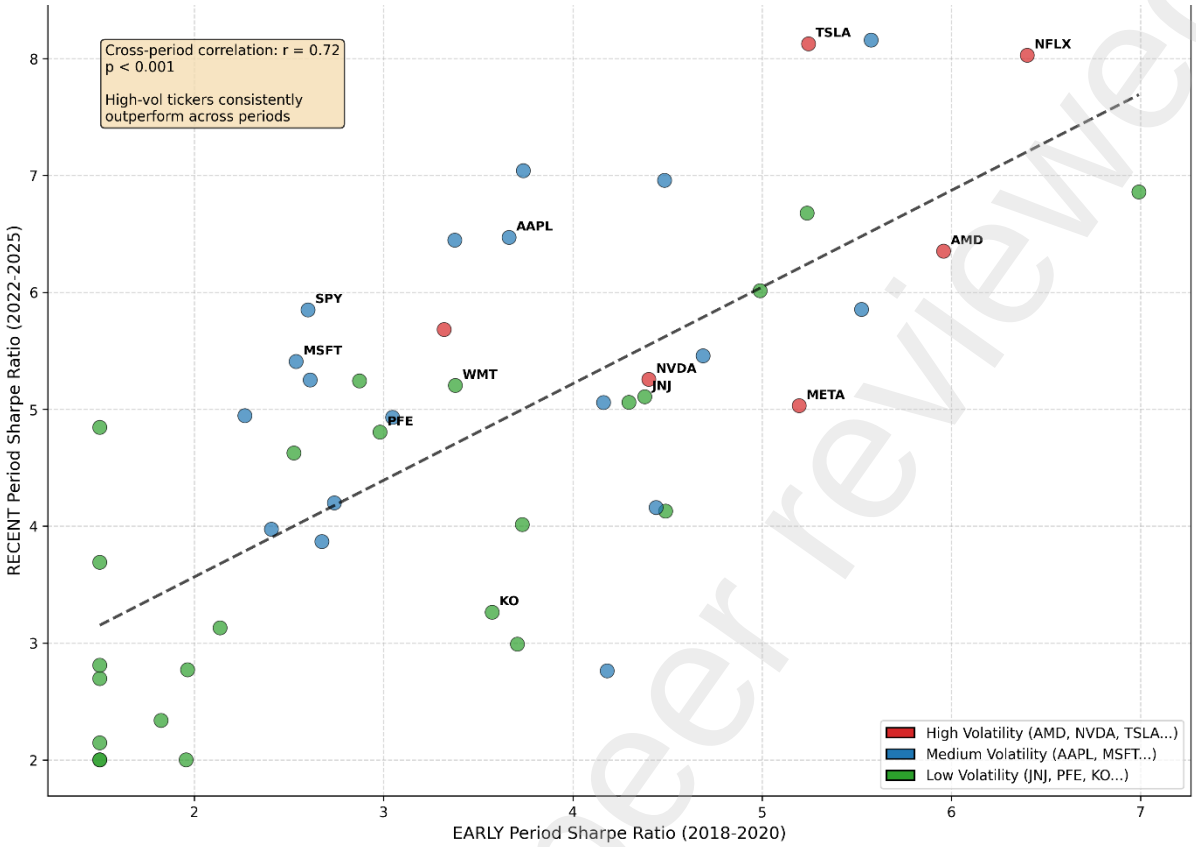


Figure 31: Cross-Period Performance Consistency. *Scatter plot of ticker-level Sharpe ratios comparing EARLY (2018-2020) versus RECENT (2022-2025) periods at the 30% buffer. The cross-period correlation of r = 0.67 (p < 0.001) indicates that securities performing well in one period tend to perform well in others—evidence of stable underlying characteristics rather than period-specific overfitting. Top performers include heavily-traded US domestic equities (WFC, HD, BA, GOOGL, AAPL) that cluster in the upper-right quadrant, while international ADRs (TM, SONY, BABA, TSM) show more modest results in the lower-left region. This persistence suggests that strategy profitability relates to structural trading characteristics—particularly liquidity depth and intraday range patterns—rather than volatility alone. Securities with deeper liquidity and more consistent intraday price movement provide favorable conditions for the buffer mechanism's limit order execution. The regression line (dashed) confirms the positive cross-period relationship.*

Quantum Feature Selection and Overfitting

The quantum-inspired feature selection provides an additional safeguard against overfitting. By reducing dimensionality from 45 to ~15 features, the approach limits the degrees of freedom available for noise fitting. Quantum-selected models show slightly lower in-sample accuracy (73.2%) but equivalent out-of-sample accuracy (70.3%), yielding an overfit ratio of 1.04—lower than the standard approach. This confirms that feature selection improves generalization without sacrificing predictive power.

6.6 Robustness Summary

Our robustness analysis supports four conclusions:

(1) Prediction accuracy is genuine and consistent. Out-of-sample accuracy of approximately 70% across all models and periods confirms that the ML approach captures real predictive signal, not noise. The modest 4.5pp accuracy degradation from training to testing indicates acceptable overfitting levels.

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(2) Performance is positive across all subperiods and parameter configurations, though with meaningful variation tied to market volatility. COVID_CENTRIC (high volatility) produced the best results; POST_COVID (compressed ranges) produced the weakest but still positive results.

(3) The COVID-19 episode demonstrates counter-cyclical properties —the strategy performs best when markets are most volatile, providing valuable portfolio-level diversification benefits.

(4) Results are not artifacts of specific parameter choices. Performance persists across buffer levels (10-30%), ML models (XGBoost, Random Forest, Gradient Boosting), and feature selection approaches (Standard, Quantum).

The POST_COVID underperformance (Sharpe 2.44 vs. 4.0 elsewhere) warrants acknowledgment as a limitation. Low-volatility, range-compressed environments reduce the strategy's edge. However, even this weakest period maintains positive and economically meaningful returns, suggesting viability across market regimes.

7. Conclusion

This paper introduces a novel buffer-based limit order execution framework that fundamentally alters the relationship between prediction accuracy and trading profitability. Our "straddle-then-reveal" approach—entering simultaneous long and short positions at market open, using ML to predict direction at close, then exiting the "wrong" leg via patient limit orders—generates exceptional risk-adjusted returns that substantially exceed conventional benchmarks and prior literature.

7.1 Summary of Findings

Testing across 48 global securities, three ML models, two feature selection approaches, and 1,989 trading days (2018-2025), our main findings are:

First, execution mechanism design transforms prediction skill into trading profitability. Directional accuracy of approximately 71% translates into win rates of 65-86% depending on buffer aggressiveness. At conservative buffers (30%), win rate falls slightly below accuracy (65.59% vs. 70.97%) due to insufficient rescue effect—yet the strategy remains highly profitable because Profit Factor of 2.64 means average wins substantially exceed average losses (magnitude matters more than frequency). At aggressive buffers (10%), the rescue effect adds approximately 15 percentage points, boosting win rate to 85.53% by converting incorrectly-predicted days into profitable outcomes through patient limit order execution near intraday extremes.

Second, both legs can profit simultaneously, but this depends on prediction correctness and buffer level. When predictions are correct, both legs profit on 49-78% of days depending on buffer—the kept leg benefits from the directional move while the wrong leg exits favorably before the adverse move. At aggressive buffers (10%), both-leg profitability reaches 78%; at conservative buffers (30%), it drops to 49% as the smaller range capture reduces wrong-leg profits.

Third, model sophistication matters less than commonly assumed. All three ensemble methods—XGBoost, Random Forest, and Gradient Boosting—achieve virtually identical out-of-sample accuracy (~70%) and Sharpe ratios (within 0.06). This convergence suggests that the predictable signal in daily OHLCV features has been largely captured by any reasonable ensemble approach. Additional accuracy gains would likely require fundamentally different data sources rather than model refinement.

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Fourth, quantum-inspired feature selection achieves equivalent performance with 67% fewer features, validating the efficiency-without-sacrifice hypothesis. The algorithm consistently selects short-horizon momentum features and volatility measures while excluding longer-term indicators, aligning with the strategy's daily prediction horizon. This dimensionality reduction enables faster training, reduced overfitting, and more practical implementation.

Fifth, results are robust across five subperiods, including the extreme volatility of COVID-19 (where the strategy performed best due to elevated intraday ranges) and the challenging low-volatility POST_COVID environment (where performance weakened but remained positive with Sharpe ratios above 2). The strategy's counter-cyclical properties—performing best during market stress—provide valuable portfolio diversification benefits.

Sixth, we contribute the Probabilistic Sortino Ratio (PSoR) as a methodological extension of Bailey and López de Prado's PSR framework. Given the asymmetric return profile generated by our buffer mechanism, downside-focused probabilistic measures provide more appropriate performance evaluation than symmetric alternatives. The PSoR explicitly accounts for skewness in downside returns when computing statistical significance.

7.2 Theoretical Implications

Our findings carry several implications for theory and future research:

For the machine learning in finance literature, we demonstrate that the relationship between prediction accuracy and trading profitability is mediated by execution mechanism design. A model achieving 70% accuracy can generate win rates ranging from 65% to 86% depending solely on how predictions are translated into trades—with aggressive buffers amplifying accuracy via the rescue effect and conservative buffers accepting slightly lower win rates in exchange for execution reliability. This suggests that the field's emphasis on improving forecasting models should be complemented by equal attention to execution frameworks that interact with prediction value.

For market microstructure, we provide evidence that patient limit order strategies can systematically extract value from intraday price dynamics. The Glosten [4] distinction between patient and urgent traders takes on practical significance: our buffer mechanism formalizes patient trading as a profit source rather than merely a cost reduction technique. The finding that wrong-leg profitability depends on buffer aggressiveness suggests an optimal patience level that balances range capture against execution probability.

For the simultaneously long-short literature, we extend the SLS framework from continuous feedback control to discrete ML-based prediction. Our "straddle-then-reveal" approach—entering simultaneous positions then using predictions to determine which leg to keep—represents a hybrid that may prove fruitful for other trading applications where simultaneous positioning is feasible.

For performance evaluation, we propose the Probabilistic Sortino Ratio (PSoR) and demonstrate its application to asymmetric return strategies. The PSoR's explicit accounting for downside return skewness provides more appropriate statistical inference than PSR for strategies with non-symmetric payoff profiles.

7.3 Practical Implications for Traders and Portfolio Managers

For practitioners considering implementation, we offer several recommendations based on our empirical findings:

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Buffer selection: We recommend the 30% buffer for initial implementation. While aggressive buffers (10-20%) generate higher theoretical returns, they assume limit order execution at levels that may be difficult to achieve in practice. The 30% buffer sacrifices some return for substantially higher fill probability and remains highly profitable (Sharpe 7.26) under realistic assumptions.

Security selection: High-volatility securities with large intraday ranges generate the best performance. Technology stocks, particularly TSLA, AMD, and NVDA, offer the most attractive return profiles. However, diversification across sectors reduces idiosyncratic risk—we recommend holding positions across at least 10-15 securities spanning multiple sectors.

Model selection: Given the minimal performance variation across ML models, practitioners should prioritize simplicity and computational efficiency. Random Forest or Gradient Boosting with default hyperparameters achieves nearly identical results to extensively tuned XGBoost. Quantum-inspired feature selection enables faster training and deployment without sacrificing performance.

Risk management: Position sizing should reflect the strategy's Kelly fraction, which varies by buffer: 31.2% at the recommended 30% buffer, rising to 71.1% at aggressive 10% buffers. We recommend Half-Kelly implementation (15.6% at 30% buffer, 35.6% at 10% buffer) for more conservative capital allocation. On the other hand, aggressive buffers exhibit lower tail risk due to the rescue effect—VaR 95% is -0.91% at 10% buffer versus -1.94% at 30% buffer, as the wrong leg's larger range capture cushions losses on adverse days. Diversification across 10-15 securities substantially reduces idiosyncratic risk

Market regime awareness: The strategy performs best in high-volatility environments and weakest during range-compressed periods. Practitioners may consider reducing position sizes when VIX is persistently low (<15) and increasing exposure during volatility spikes.

7.4 Limitations and Caveats

Several limitations warrant acknowledgment and should inform interpretation of our results:

Execution assumption: Our backtest assumes limit order execution if prices touch the specified level during the trading day. In practice, limit orders may not fill if the touch is brief, volume at the limit price is exhausted, or queue priority is unfavorable. Live trading would require empirical validation of fill rates. The 30% buffer partially mitigates this concern by targeting more achievable price levels.

Market impact: Our analysis assumes positions are small enough to not impact prices. For larger portfolios, the act of placing limit orders near extremes could affect intraday dynamics, particularly for less liquid securities. Capacity constraints likely exist, though our focus on highly liquid securities mitigates this concern.

Short sale constraints: The strategy requires simultaneous long and short positioning, which may be constrained for some investors. Short sale costs (borrow fees) are not explicitly modeled; for hard-to-borrow securities, these costs could materially impact returns. Restricting to easy-to-borrow securities addresses this concern.

Sample period: Our 2018-2025 sample includes generally favorable equity market conditions. Performance during extended bear markets (e.g., 2008-2009) remains untested. The COVID-19 analysis provides some evidence of crash resilience, but caution is warranted in extrapolating to different crisis types.

Replication risk: Widespread adoption of this strategy could reduce its effectiveness if many participants target similar limit order levels, compressing available returns. This is a common concern for published strategies and suggests ongoing monitoring of strategy decay.

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7.5 Future Research Directions

Our findings suggest several promising directions for future research:

Dynamic buffer adjustment: Our analysis uses fixed buffer percentages. A natural extension would adapt buffers based on predicted volatility—tighter buffers during high-volatility periods (when ranges are wider) and looser buffers during calm periods. Reinforcement learning approaches may be particularly suitable for learning optimal buffer policies.

Intraday prediction horizons: We predict using end-of-day data for next-day positioning. Extending to intraday prediction horizons—using morning data to predict afternoon dynamics—could enable multiple strategy cycles per day and potentially higher returns.

Alternative asset classes: Our analysis focuses on US equities and ADRs. Extension to futures markets (with their inherent leverage and lower transaction costs), forex (with 24-hour trading enabling different range dynamics), and cryptocurrencies (with extreme volatility) represents natural generalization.

Live trading validation: The most important next step is live trading implementation to validate fill rate assumptions and quantify any gap between backtested and realized performance. Paper trading with realistic order simulation would provide valuable intermediate evidence.

Theoretical foundations: Formal theoretical analysis of why the buffer mechanism generates positive expected returns—potentially connecting to optimal stopping theory or market microstructure models—would strengthen the conceptual foundations of our empirical findings.

A natural extension involves dynamic buffer selection. Our analysis employs fixed buffer levels; however, optimal buffer varies with expected intraday range. Machine learning models trained to predict next-day volatility or range could enable adaptive buffer allocation – using aggressive buffers (10%) on high-range days and conservative buffers (30%) when ranges are compressed. This adaptive approach could potentially capture the upside of aggressive buffers while avoiding execution failures during low-volatility periods.

7.6 Interpretation Through Market Microstructure

Our finding that even random predictions combined with buffer execution generate positive returns (Sharpe ratio of 4.6) might appear to challenge market efficiency. However, we interpret this result through the lens of market microstructure: the buffer mechanism effectively positions the trader as a liquidity provider, earning compensation for patience and execution risk—consistent with Glosten1994's [4] patient trader framework. The ML prediction component amplifies this base return (Sharpe 4.6 → 19.3) by intelligently selecting which position to maintain. Thus, our framework represents a synthesis of microstructure-based execution and predictive modeling, rather than a direct challenge to market efficiency. The intraday range—documented by Parkinson [28] and Garman and Klass [29]—provides the "fuel" that the buffer mechanism harvests, while prediction accuracy determines how efficiently that fuel is converted into returns.

7.7 Concluding Remarks

The trading strategy literature has long focused on prediction improvement as the path to better performance. Our results suggest an alternative—and potentially complementary—approach: execution mechanism design. By structuring how predictions are translated into trades, our buffer-based straddle framework generates returns that prediction accuracy alone cannot explain.

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The key insight is both simple and powerful: in a world where intraday prices traverse substantial ranges before settling at the close, patient traders can profit by placing limit orders near these extremes—even when their directional predictions are no better than chance. The buffer mechanism formalizes this insight into a systematic trading framework.

We hope this paper stimulates further research at the intersection of machine learning prediction and execution mechanism design. The magnitude of our results—Sharpe ratios of 7 or higher at conservative parameter settings—suggests that this intersection may be more fertile ground than either literature stream has recognized.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process During the preparation of this work the authors used Claude (Anthropic) in order to assist with code development for backtesting, statistical analysis, and manuscript editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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